INTERNATIONAL LOGISTICS

INTRODUCTION

The environment of any organization is the aggregate of all conditions, events and influences that surround and affect it. The Global business environment imposes several constraints on an international enterprise and has considerable impact and influence on the scope and direction of its activities. The enterprise, on the other hand, has very little control over its international environment. The basic job of the enterprise, therefore, is to identify with the environment in which it operates, and to formulate its policies in accordance with the forces, which operates in its environment.

International business is different from domestic business because the environment changes when a firm crosses international borders. Typically, a firm understands its domestic environment quite well, but is less familiar with the environment in other countries and must invest more time and resources into understanding the new environment. The environment of international business is regarded as the sum total of all the external forces working upon the firm as it goes about its affairs in foreign and domestic markets.

The environment can be classified in terms of domestic, foreign, and international spheres of impact. The domestic environment is familiar to managers and consists of those uncontrollable external forces that affect the firm in its home market.

GLOBAL ENVIRONMENTAL FACTORS AFFECTING LOGISTICS

With increase in global business the environmental complexity, uncertainty and asymmetry also increased. It affects the supply chain decision also. These factors are as follows:

1. Complexity

It is a reflection of the fact that political, economic and cultural factor differ across countries, resulting in a complex set of variables to be considered when operating across multiple countries. This global diversity is reflected in difference in government regulations and legal requirements, different state of economic and infrastructure development and difference in consumer demand among different countries. It has a great impact on supply chain decision.

2. Uncertainty

It is increased by the fact that in some countries and region of the world economic and political instability may result in dynamic exchange rate shifts, unpredictable inflation rate and unexpected change in political climate, government regulation etc. The participant in global supply chain has no control over conditions so they make a decision while considering the uncertainty of environment.

3. Asymmetry

With increase in global diversity, the asymmetry shall be managed when doing business between countries with very different economic status or very different culture. Asymmetry deals with the relationship. An information asymmetry is relates to openness, the greater the openness, and the smaller the information asymmetry.

INTERLINKED INTERNATIONAL ECONOMY

Trade is increasingly taking place within blocs, such as the European Union, the North American Free Trade Area, and other regional groups. These blocs are established to promote regional development, which often leads to tariff and non-tariff barriers to imports from outside of the bloc. Global companies wishing to supply markets within these blocs will eventually have to set up production inside the bloc if they wish to serve their markets with significant volume. The result accelerates a phenomenon already taking place that emphasizes the growth of interregional economies and trade, at the expense of trade from outside of these blocs.

REGIONAL INTEGRATION

Another important trend in this regard is that the bulk of international trade takes place in materials and intermediate components. Global corporations will ship semi-manufactured products between their own facilities or from suppliers to assembly plants and distribution centers. The trend toward global supply chains has been fueled by needs for centralized research and development, the development of homogeneous markets and global products and global market segments for many products, the need for economies of scale in specific industries. Wage differentials between advanced countries and other less developed countries, and political efforts to maintain low tariff barriers. Global networks have also been further aided by lower transportation and communication costs, the explosion of the World Wide Web and Internet and the expansion of services that connect remote parts of the world with each other and with major markets. Finally, the development of organizational capacity to manage complex supply chains has improved.

Stages of Regional Integration

1. Free trade Agreement

The first stage, a free trade agreement, eliminates tariffs on trade between countries in a region. A free trade agreement may either stimulate or reduce interregional trade. Such agreements can also reduce access of the firms to more efficient producers or markets outside their region.

2. Customs Unions

The second stage, a customs union, eliminates tariffs between member countries and establishes a common external tariff structure toward other regions and non-member countries. Under this and the remaining two stages, member countries are required to give some control over economic policies to the group.

3. Common Market

The third integration stage, a common market, is characterized by the same tariff policy as the customs union. In addition, a common market allows factors of production such as labor and capital, as well as goods and people, to move freely between member countries as dictated by market conditions.

4. Economic Union

The economic union is the fourth and most advanced stage of development because it implies harmonization of economic policies beyond a common market. Economic union standardizes monetary and fiscal policy among member countries.

Economic structures will have a major effect on how global supply chains are designed and developed, including both technical separation in production and distribution and in production process and ownership of functions. A supply chain design will also be impacted by risk, trade restrictions, and long lead times. For instance, transportation systems required to manage supply chains in European markets are very unique. The movement toward a single market has led to an impending set of bottlenecks that will occur in the already strained transportation infrastructure. This structure is being challenged by the deregulation of the transport market and liberalization on cabotage (i.e., carrying goods in domestic commerce by a foreign carrier). (These trends are currently being disputed and resisted by the German trucking industry).

INTERNATIONAL OPERATING LEVELS

It traces the levels of enterprise evolution from domestic logistics Operations to becoming a global competitor the duration of each level is a reflection of managerial philosophy rather than elapsed time. The five levels are my length relationship, internal export, internal Operations, insider business, and denationalized operations. Figure Illustrates the risk and return associated with each global operating Stage.

The five levels are arm's-length relationship, internal export, internal operations, insider business, and denationalized operations.

1. Arm's-Length Relationship (Level 1)

Arm's-Length Relationship is characterized by an arm's-length relationship between an enterprise and an international distributor that serves a country or region. The enterprise, which probably has limited international experience, either sells or consigns its goods to the international specialist, which accepts responsibility for ordering, providing international transportation, completing documentation, as well as coordinating marketing, inventory management, invoicing, and product support. On the positive side, the arm's-length relationship significantly lowers the enterprise's risk by providing staff and expertise to establish and manage necessary relationships and operations. On the negative side, the arm's-length relationship reduces the firm's contribution margin and decreases product and logistical control.

2. Internal Export (Level 2)

Internal Export enterprises develop the expertise to coordinate and manage international transportation and documentation. However, the local agent or distributor is retained to provide marketing, inventory management, invoicing, and product support. Thus, the enterprise might

increase its contribution margin by internalizing some of the export activities but still uses specialized assistance for marketing, sales and customer service activities once the product exported. The length export operations business operations the international market also reduce the enterprise's sensitivity to local requirements.

3. Internal Operations (Level 3)

Internal Operations is characterized by development of operations in the local or foreign country. Internal operations include combinations of marketing, sales, production, and distribution. Establishment of local facilities and operations increases market awareness and sensitivity. This is typically referred to as local market presence. However, the internal operations stage often uses punt company practices, management, and personnel, rather than employing locals. The rationale for home country staffing is that headquarters can rely on international operations to better execute policy if proven company managers are in charge. Since managers are assigned from the home country, this first foothold of internal operations strongly reflects home country operations. The internal operations stage increases enterprise control and sensitivity in comparison to the initial two stages but still relies heavily on home country values, procedures, and operations.

4. Insider Business Practices (Level 4)

Insider Business Practices the fourth level further initializes international operations and institutes local business practices. This level of sophistication typically involves hiring host country management, marketing, and sales organizations and may include the use of local business systems. This stage requires that headquarters management support local management decisions regarding practices and procedures. As insider operations develop, a separate host country philosophy emerges. However, the home country philosophy is clearly dominant. Individual international operations are still measured against home country expectations and standards.

5. Denationalized Operations (Level 5)

Denationalized Operations the denationalized operations level maintains foreign country operations and develops a regionalized headquarters to oversee the coordination of operations in the area. At this point, the enterprise is stateless in the sense that no specific home or parent country dominates policy. Senior management likely represents a combination of nationalities. Denationalized operations function on the basis of local marketing and sales organizations and are typically supported by world-class manufacturing and logistics operations. Product sourcing and marketing decisions can be made across a wide range of global alternatives. Systems and procedures are designed to meet individual country requirements and are aggregated as necessary to share knowledge and for financial reporting. The sidebar discussion of the 3M Company describes many characteristics of a firm with denationalized operations.

While most enterprises engaged in global logistics are operating at levels 2, 3, and 4, a truly international firm must focus on denationalizing operations. Enterprises in any of the other stages will retain a home country perspective with its implicit pecking order. However, denationalized operations require a significant level of management trust across countries and cultures. Such trust can grow only as managers increasingly live and work in other cultures. While

this cross culturalization is beginning to occur, it will be some time before it transcends a broad range of logistics managers

LOGISTICS IN A INTERNATIONAL ECONOMY

INTRODUCTION

The current definition of the Council of Logistics Management can be logically extended to include the elements of international environment. Global or International logistics is the process of planning, implementing and controlling the flow and storage of goods, services, and related information from a point of origin to a point of consumption located in a different country. This definition accounts for all the intricacies inherent to crossing borders.

The global logistics management function is naturally more complex than the logistics function managed within one particular country. This is primarily because of the large distances involved and the customs and other procedural formalities required to be completed.

International logistics is the process of planning, implementing and controlling the flow and storage of goods, services and related information from a point of origin to a point of consumption located in a different country. The dominant theme in the logistics literature is the increased complexity of the logistics processes in a global environment. Wood used a channels framework to categories types of complexity as:

1. Transactions Channel

It deals with when and where to transfer title and payment of goods, how the process is managed to deal with the uncertainties of exchange rate fluctuations and inflation rates, and the increased risk associated with fraud and piracy.

2. Documentation/Communication Channel

It involves the complex and multiple legal requirements involved in cross-border movements as well as the increased difficulty of communicating across spatial and cultural barriers.

3. Distribution Channel

It deals with the physical flow of goods and incorporates the international complexities of packaging and labeling requirements in different countries, as well as mode choice.

The new demands of the global marketplace have caused incredible pressures on antiquated logistical systems of companies. The improvements needed in logistical systems in order to compete in the more competitive global marketplace are numerous.

COMPONENTS OF GLOBAL LOGISTICS

There are few activities that are exclusively specific to international or global logistics; however, the traditional logistical activities are managed differently in an international environment than they are in domestic environment.

1. International Transportation

It is eminently more complicated, involving different modes of transportation, different carriers, different transportation documents, and much greater transit times. It! Inherent risks and hazards are also much more Significant.

2. International Insurance

It is also much more complex and honestly somewhat treacherous, as the contracts are written using archaic language and terminology that varies in meaning depending on the country in which the insurance company is located

3. Packaging needs

They are different, as the goods are exposed to a number of risks rarely encountered in domestic shipments.

4. International means of Payment

They are more involved, with the risks of non-payment and currency fluctuations calling for specific strategies that are never used in domestic transactions.

5. Terms of Trade

These are much more involved, as the greater number of nodes and links increases the number of possible alternatives for transfer of responsibility and ownership.

6. The Crossing of Borders

It also represents specific challenges. Products sold abroad or purchased from abroad have to go through customs, a complicated and paper-intensive process in most countries.

7. Inventory

It is managed differently as the risk of delays and variations in shipping times are increasing the challenges of Just-in-Time production.

8. Environment of International Logistics

Finally, the environment of international logistics is quite different; while there is obviously the issue of language and culture.

GLOBAL LOGISTICAL DECISIONS

The major decisions in Global logistics that are needed to manage in order to get success in logistics operations are as follows:

1. Warehouse Management

Global warehousing serves the same purposes as its domestic counterparts; that is, receiving, transferring, picking, and shipping. However, rapid movement of products through the warehouse receives more emphasis than in domestic warehousing. The global warehouse mirrors a distribution centre, with little long-term storage. Third party warehousing is also quite common globally.

2. Packaging

For overseas shipments packaging is more critical as compared to domestic shipments because of the nature and number of hazards the packaging has to face or undergo during its journey to its destination. The logistical packaging needs to withstand varying storage, transit, and

handling conditions during transit, and protect the packed material. The logistical packaging has to comply with the shipping regulations of the countries of origin and destination.

3. Inventory Management

Inventory is the life of integrated logistics. Without it, there is nothing to do in physical supply-nothing to store, nothing to carry, and nothing to package. Inventory management on a global scale becomes even more difficult and more important. Distance, port delays, custom delays, and transit times encourage higher inventories. This, of course, increases the costs of inventory. Political decisions such as, closing borders to trade or instituting or increasing tariffs and duties only add to the inventory problems.

4. Material Handling

Material handling systems vary globally. In countries like the United States, Canada, Australia, New Zealand, Hong Kong, Singapore, and most of Western Europe, these systems will be among the world's best, usually highly mechanized or automated. In developing countries, the majority of systems are manual. Product throughput in warehouses and plants tends to be much slower in manual systems. Also, some products require more sophisticated handling equipment, so some countries or ports may not be able to receive them. For example, some ports can handle only twenty-foot containers, while others cannot handle containers at all.

5. Information Systems

While developed countries have very sophisticated integrated logistics information systems, not all countries do. Many third world countries use pencil and paper. Technologies such as EDI, automated data collection, and radio frequency systems are useless. Communications may be poor, and available information may be inaccurate.

6. Transportation

Transportation mode plays a vital role in the movement of cargo within or between countries. Normally cargo is moved using three modes of transportation, i.e., road, sea, or air depending on the cost, urgency, and destination. However, for cross border cargo movement, mostly sea and air are the preferred modes of transportation as most countries are connected well by air and sea. Transportation confers 'time and place utility' to the product; it determines the company's customer service; it also has a crucial bearing on the other elements of physical distribution and marketing, like warehousing, inventory control and channel management. Main modes of transportation are as follows:

a) Railway Transport

Indian Railways is one of the largest systems in the world. It is also one of the very few railway systems in the world generating operating surpluses. With a modest beginning in India on April 16, 1853, when the first wheels rolled on rails from Bombay to Thane.

b) Road Transport

India has one of the largest road networks in the world, aggregating to about 3.34 million kilometers at present. However, this network is not adequate for speedy and efficient transportation.

c) Water Transport

It is a very ancient mode of transport was found in India. In ancient time shipping was quite popular in India. Water transport plays an important role in carrying heavy and bulky goods over long distances on the transport.

d) Air Transport

Compared to all other modes of transport, air transport is the least hazardous in nature. Air transportation is costly and is used only for highly perishable commodities and items, whose life is short and value is high.

e) Insurance

To protect against damage or loss of cargo during transit in air or sea, shippers insure their goods. In domestic cargo movement, carriers take the responsibility for transit loss or damage. However, in sea transportation carrier do not take this responsibility because of the high degree of risk due to unavoidable perils during sea journeys. The purpose of marine insurance is to protect sea cargo against loss or damage in transit. The coverage of marine insurance is much broader as compared to domestic cargo insurance.

IMPORTANCE OF INTERNATIONAL LOGISTICS

Unlike the old approach to logistics, which was restricted to the army, today, it has gained recognition in business organizations as one of the important business functions and a tool for developing competitiveness. It is of critical importance to the organization, how it delivers products and services to the customers, whether the product is tangible or intangible. Effective and efficient physical movement of the tangible product will speak about the intangible services associated with the product and the organization, which is delivering it. In case of intangible products, the delivery of tangibles at the right place, at the right time, will speak about its quality.

Logistics is one of the major enablers of growth of trade and commerce activity in a country. On the macro level international logistical infrastructure such as the various modes of transportation, transportation equipment, storage facilities, connectivity and information processing are contributing to a large extent in the physical movement of goods produced in manufacturing, mining, and agricultural sectors. The speed and reliability in distribution from the place of production to the place of consumption will contribute to a great. Extent in the growth of a country's domestic and international trade.

At the micro level international logistics plays a critical role in the value delivery system of the business organization to provide superior customer service. Hence, the mission of the international logistical system is to achieve the desired level of delivered service and quality at the lowest possible cost. The importance of logistics ranges from the management of material procurement and manufacturing to the delivery of finished products to the consumer. In a nutshell, any productivity improvement that could be achieved in any part of the international logistics system, at the macro or micro level, would help in cost reduction and proper deployment of scarce resources to productive purposes.

Generally, production activities are credited with providing form utility; logistics activities with time, place, and quantity utilities; and marketing activities with possession utility. Each will be discussed briefly:

1. Form Utility

It refers to the value added to goods through a manufacturing or assembly process. For example, form utility results when raw materials or components are combined in some predetermined manner to make a finished product. This is the case, e.g., when Dell combines components along with software to produce a computer to a customer's specifications. The process of combining these different components represents a change in the product form that adds value to a product. In today's business environment, certain logistics activities can also provide form utility.

For example, breaking bulk and mixing products, which typically take place at distribution centers, change a product's form by changing its shipment size and packaging characteristics. Thus, unpacking a pallet of Kraft macaroni and cheese into individual consumer-size boxes adds form utility to the produce. However, the three principal ways in which logistics adds value are place, time, and quantity utilities.

2. Place Utility

Logistics provides place utility by moving goods from production points to market points where demand exists. Logistics extends the physical boundaries of the market area, thus adding economic value to the goods. Logistics creates place utility primarily through transportation; For example, moving Haggis diapers from a Kimberly-Clark manufacturing facility by motor carrier to markets where consumers need these diapers creates place utility. The same is true when steel is moved by rail to an automotive supplier to stamp automobile parts. The market boundary extension added by place utility increases competition, which usually leads to lower prices and increased product availability.

3. Time Utility

Not only must goods and services be available where customers need them but also at the point when customers demand them. This is called time utility, or the economic value added to a good or service by having it at a demand point at a specific time when it is needed. Logistics creates time utility through proper inventory maintenance, the strategic location of goods and services, and transportation. For example, logistics creates time utility by having heavily advertised products and sale merchandise available in retail stores at the time promised in the advertisement. This can be done by having the right products in inventory, having them stored close to the point of demand, or using a premium (faster) mode of transportation. Time min. y is much more important today because of the emphasis on reducing lead time and minimizing inventory levels through logistics-related strategies such as Just-in-Time (JIT) inventory control to improve cash flow.

5. Quantity Utility

Today's business environment requires that products not only be delivered on time to the correct destination but also be delivered in the correct quantities to minimize inventory cost and prevent stock outs. So, the utilities of when and where must be accompanied by how much. Delivering the proper} quantities of an item to where it is demanded are creating quantity utility.

Logistics creates quantity through production forecasting, production scheduling, and inventory control, for example, the importance of quantity utility in the automobile industry. Assume that General Motors will be assembling 1,000 automobiles in one day and is using a JIT inventory strategy. This will require that 4,000 tires be delivered to support the automobile production schedule. Assume that tire supplier delivers 1,900 tires in time at the correct location. Even though the when and where utilities are created, the how much utility is not. Thus, GM will not be able to assemble the 1,000 cars as planned. So, logistics must deliver products at the right time, to the right place, and in the right quantities to add utility and economic value to a product.

6. Possession Utility

It is primarily created through the basic marketing activities related to the promotion and sale of products and services. Promotion can be defined as the effort, through direct and indirect contact with the customer, to increase the desire to possess a good or to benefit from a service. The role of logistics in the economy depends on the existence of possession utility, for time, place, and quantity utilities make sense only if demand for the product or service exists. Marketing also depends on logistics, since possession utility cannot be accomplished unless time, place, and quantity utilities are provided.

BARRIERS TO INTERNATIONAL LOGISTICS

While many forces facilitate borderless operations, some significant barriers continue to impede global logistics. Three barriers are significant: markets and competition, financial barriers, and distribution channels. Global logistics management must balance me cost of overcoming these barriers with the potential benefits of international trade to achieve the actual benefits of successful international operations. Each barrier is described.

1. Markets and Competition

Perceived and real market and competitive barriers include entry restrictions, information availability, pricing, and competition. Entry restrictions limit market access by placing legal or physical barriers on importing. An example of a physical barrier is the European practice of local presence, which requires that market-based manufacturing or distribution facilities be established prior to market access. An example of a legal entry barrier is the Japanese practice of allowing local retailers to “vote” on acceptance of new retailers, particularly foreign ones, into the market. Poor information availability is another global logistics barrier. In addition to limited information availability regarding market size, demographics, and competition, little coordinated information is available defining import and documentation requirements. Typical requirements differ by government and even by specific case.

2. Financial Barriers

The financial barriers to global logistics result from forecasting and the institutions infrastructures. While it is not easy to forecast in any situation, it is particularly difficult in global environments. The domestic forecasting challenge is to predict unit or dollar sales based on customer trends, competitive actions, and seasonality. In a global environment, these challenges are compounded by enhance rates, customs actions, and government policy complexities. Institutional

infrastructure barriers result from major differences in how facilitating intermediaries such as banks, insurance firms, legal counselors, and transportation carriers operate.

3. Distribution Channels

Distribution channel differences such as, infrastructure standardization and trade agreements are a third barrier confronting logistics managers. Infrastructure standardization refers to differences in transportation and material-handling equipment, warehouse and port facilities, and communication systems. While there have been recent efforts to improve standardization with respect to containerization, there are still major differences in global transportation equipment such as vehicle dimensions, capacity, weight, and rail gauge. When infrastructure is not standardized, it is necessary for products to be unloaded and reloaded into different vehicles or containers as they cross national boundaries, resulting in increased cost and time.

4. The Global Challenge

Firms desiring to expand globally need to assess the balance of forces that en- courage such activity and the barriers they must overcome. Increasing international trade requires logistics managers to develop both a global awareness and a global perspective. Managers must be aware of the aforementioned logistics barriers. Consider alternative solutions, and have the insight to apply them in non-traditional environments. While logistics principles are the same domestically and globally, operating environments are more complex and costly. Cost and complexity are represented by the four D's-distance, documents, diversity in culture and demands of customers. Developing strategies and tactics to respond to the' 'four D" environment is the global challenge for logistics management. Global logistics development requires creation of an international operating philosophy and vision. The vision must result in operating strategies, performance expectations, measurement, and decision alternatives.

SUMMARY

In a rapidly integrating and globalizing world market, efficient and effective logistics support structures relating to international goods transactions are indispensable. Logistics managers involved in international distribution must acquaint themselves with distribution channels because they need to understand how products flow to foreign consumers. The longer the channel, the greater the number of intermediaries. International goods transport is provided by all modes. However, the vast majority of goods is moved by sea and air transport.

International logistics is the process of planning, implementing and controlling the flow and storage of goods, services and related information from a point of origin to a point of consumption located in a different country. The dominant theme in the logistics literature is the increased complexity of the logistics processes in a global environment.

Logistics is one of the major enablers of growth of trade and commerce activity in a country. On the macro level international logistical infrastructure such as the various modes of transportation, transportation equipment, storage facilities, connectivity and information processing are contributing to a large extent in the physical movement of goods produced in manufacturing, mining, and agricultural sectors.

REVIEW QUESTIONS

1. What is international logistics?
2. Elaborate the views of international logistics.
3. Explain the importance and barriers to international logistics.
4. What are the major channels used in international logistics?
5. In what an aspect is international logistics different from domestic logistics function?
6. Discuss the global environmental factors affecting logistics and SCM decisions along with global operating levels.
7. Discuss the components and decisions of international logistics. What are various issues in international trade and logistics?

INTRODUCTION TO AIRLINE INDUSTRY

Learning Objectives

After reading this chapter, you should be able to:

- Understand the concept of airline industry.
- Know the different types of airlines.
- Know the history of airline industry.
- Understand the major players in airline industry.
- Know the SWOT analysis of airline industry.

INTRODUCTION

The airline industry encompasses a wide range of businesses, called airlines, which offer air transport services for paying customers or business partners. These air transport services are provided for both human travellers and cargo, and are most commonly offered via jets, although some airlines also use helicopters.

Airlines may offer scheduled and/or chartered services and the airline industry forms a key part of the wider travel industry, providing customers with the ability to purchase seats on flights and travel to different parts of the world. The airline industry offers a variety of career paths, including pilots, flight attendants and ground crew.

AIRLINE INDUSTRY AND THE AVIATION INDUSTRY

The terms 'airline industry' and 'aviation industry' are sometimes thought of as being synonymous, but in reality, they have different meanings. As stated, the airline industry refers to companies that offer air transport services to paying customers, whereas the aviation industry includes all aviation-related businesses.

With this in mind, the airline industry can be classed as just one sector of the wider aviation industry. In addition to this sector, aviation companies would also include businesses like aircraft manufacturers, companies offering non-commercial flights, aerospace companies, regulation authorities and those involved with research.

DIFFERENT TYPES OF AIRLINES

Airlines are often grouped into different categories and precise definitions can vary in different parts of the world. However, broadly speaking, airlines can be separated into three main types, which are as follows:

1. International Airlines

International airlines are a group of the largest, most high-profile and most successful airlines. They make billions in revenue each year and operate large passenger jets. These airlines also tend to focus their efforts on offering global services, carrying passengers and cargo over large distances.

Additionally, international airlines usually employ tens of thousands of people, often have multiple hubs and will provide access to hundreds of destinations. Examples would include Delta Air Lines and American Airlines.

2. National Airlines

National airlines represent the next step down from the largest international airlines. They will typically offer both medium-sized and large-sized jets and will often focus on offering services to areas within their home country, but many will offer access to international destinations too.

A national airline is still likely to employ thousands of people, but will have a smaller fleet size. In many cases, the destinations they offer flights to are influenced by seasonal fluctuations in demand.

3. Regional Airlines

Finally, as the name suggests, regional airlines are the smallest of the three main types and focus on offering services within specific regions. In many cases, they provide passenger services to parts of the world with lower levels of demand and where services are not offered by either national or international airlines.

Some regional airlines also function as an affiliate for a national or international airline. Within this context, they will typically provide connection flights from the region they cover to the airline's main hubs.

HISTORY OF AIRLINE INDUSTRY

Since the birth of flight in 1903, air travel has emerged as a crucial means of transportation for people and products. The hundred-plus years following the invention of the first aircraft have brought about a revolution in the way people travel. The airline business is a major industry, relied upon by millions not only for transportation but also as a way of making a living.

1. EARLY 20TH CENTURY

Airplanes were around the first few years of the 20th century, but flying was a risky endeavor not commonplace until 1925. In this year, the Air Mail Act facilitated the development of the airline industry by allowing the postmaster to contract with private airlines to deliver mail. Shortly thereafter, the Air Commerce Act gave the Secretary of Commerce power to establish airways, certify aircraft, license pilots, and issue and enforce air traffic regulations. The first commercial airlines included Pan American,

Western Air Express and Ford Transport Service. Within 10 years, many modern-day airlines, such as United and American, had emerged as major players.

2. MID-20TH CENTURY

In 1938, the Civil Aeronautics Act established the Civil Aeronautics Board. This board served numerous functions, the two most significant being determining airlines' routes of travel and regulating prices for passenger fares. The CAB based airfares on average costs, so because airlines couldn't compete with each other by offering lower fares, they competed by striving to offer the best quality service. If the CAB found an airline's service quality was lacking on a certain route, it would allow other carriers to begin operating on that route. In this environment, established airlines enjoyed an advantage over startups, as new carriers found it difficult to break into existing routes. The Federal Aviation Agency, now known as the Federal Aviation Administration, was created in 1958 to manage safety operations.

3. DEREGULATION

In the mid-1970s, Alfred Kahn, an economist and deregulation advocate, became chairman of the CAB. Around the same time, a British airline began offering exceptionally inexpensive transatlantic flights, awakening a desire for U.S.-based airlines to lower their fares. These influences led to Congress passing the Airline Deregulation Act of 1978, ushering in an era of unencumbered free market competition. The CAB disbanded a few years thereafter.

4. LATE 20TH CENTURY

Post-deregulation, new carriers rushed into the market, and new routes directly connected cities previously accessible only via a string of layovers. Fares dropped as competition and the number of customers increased. A 1981 air traffic controllers strike brought a temporary setback to the growth, which continued throughout the 1980s. Some of the major carriers who had dominated the skies during the middle portion of the century, such as Pan American and TWA, began to collapse in the wake of competition. Such carriers disappeared completely following the Gulf War and subsequent recession of the early 1990s. Surviving airlines rode out the recession and returned to record profitability by the late 1990s.

5. 21ST CENTURY

In 2001, the industry dealt with the effects of another economic downturn, as business travel decreased substantially while labor and fuel costs increased. The events 9/11 greatly magnified the airlines' issues, leading to a sharp decline in customers and significantly higher operating costs. Losses continued for years; the industry as a whole didn't return to profitability until 2006. A relatively stable period followed, although controversies arose over service quality and passenger treatment in terms of flight delays, particularly those involving planes waiting on the runway. In 2010 and 2011, the U.S. Department of Transportation issued a series of rules mandating that the airlines provide adequate modifications for passengers in extenuating circumstances.

BIGGEST AIRLINE COMPANIES IN THE WORLD

When it comes to identifying the largest airlines in the world, one of the most widely utilized metrics is revenue passenger-kilometers (RPKs) and this information is published each year by the International Air Transport Association (IATA). As the name indicates, this metric is the sum total of the distances flown by each of the airline's paying passengers over the course of a year. With this in mind, the seven biggest airline companies in the world today are:

1. American airline

The biggest airline company in the airline industry is American Airlines. Founded in 1926, American Airlines is the largest airline in the world, not only according to the aforementioned RPK metric, but also based on fleet size, total passengers and number of destinations served. The company has its main headquarters in Fort Worth, Texas, flies to around 350 destinations, and handles 200 million annual passengers.

American Airlines' largest hub is Dallas/Fort Worth International Airport, with more than 50 million passengers being served via this hub alone each year. The business also employs more than 100,000 people.

2. Delta Air Lines

The second-largest airline company in the airline industry is Delta Airlines. Based in Atlanta, Georgia, in the USA, Delta Air Lines was established in 1925 and commenced passenger operations in 1929. The company, which employs more than 80,000 people, operates in excess of 5,000 daily flights, covers more than 300 destinations and handles in the region of 200 million passengers each year.

In total, Delta Air Lines has nine hubs, including Atlanta Airport, New York-JFK and Detroit Metropolitan Airport. It is also the second-largest airline in the world by fleet size, with more than 900 aircraft.

3. United Airlines

The third-largest airline company in the airline industry is United Airlines. United Airlines was founded in 1926 and started offering passenger services in 1931. The company is based in Chicago, Illinois and offers flights to more than 300 destinations. United Airlines is ranked as the third-largest airline, not only by RPK, but also by fleet size, with more than 750 Boeing or Airbus aircraft.

In total, the airline has eight hubs, with Chicago–O'Hare Airport serving as the main one. United Airlines serves more than 150 million passengers each year and is also a key member of the Star Alliance.

5. United Arab Emirates

Another main airline company within the airline industry is Emirates, based in Dubai, in the United Arab Emirates. A state-owned airline, it started operations in 1985 and is also recognized as one of the largest airlines in the world for cargo activities. Emirates currently has a fleet size of more than 250 aircraft and is notable for not having joined any of the major airline alliances.

It is estimated that Emirates provides flights for more than 50 million annual passengers and offers services to more than 150 destinations in total. The company's sole hub is Dubai International Airport.

6. Southwest Airlines

The fifth biggest airline company in the airline industry is Southwest Airlines. Established in 1967 and commencing with passenger airline services in 1971, Southwest Airlines is recognized as the largest low-cost carrier in the world, according to most metrics. It has more than 60,000 staff, operates with a fleet of more than 750 aircraft and carries more domestic passengers than any other US airline.

Unlike most of the other airlines on this list, Southwest Airlines does not have a main hub and instead relies on a point-to-point system. Most of its destinations are in North America and Central America.

7. China Southern Airlines

The sixth biggest airline company in the airline industry is China Southern Airlines. China Southern Airlines was founded in 1988 and is the largest airline in Asia by fleet size, delivering its services through more than 600 aircraft. The company provides services to in excess of 200 destinations, carries around 150 million passengers each year and employs somewhere in the region of 100,000 staff.

The airline's two main hubs are Beijing Capital International Airport and Guangzhou Baiyun International Airport.

8. Ryanair

One more large airline company within the airline industry is Ryanair. Ryanair is Founded in 1984, Ryanair is based in Dublin, Ireland and is recognized as one of the largest budget airlines in the world, providing flights for more than 100 million

passengers every year. These flights cover 225 destinations and the company has a fleet size of more than 300 aircraft.

The airline makes use of a point-to-point model, rather than having a central hub, but its main operational bases can be found at Dublin Airport and London Stanstead Airport. Its route network centres on Europe and the Middle East.

MAJOR PLAYERS IN AIRLINE INDUSTRY IN INDIA

This aviation sector contributed close to $72 billion to Indian GDP and is en-route to also become the 3rd largest air passenger market by 2024. These bright prospects are mainly due to the untapped potential considering that air transport is still considered expensive for the majority of the country's population which will change with the country's economic growth.

The Indian government realizing this potential has allowed 100% FDI in these sectors. Investments over 49% will, however, require government approval. The Indian government has also planned to invest up to $1.82 billion for the development of airport infrastructure and aviation navigation services by 2026.

As of March 2019, India had 103 operational airports, this number is expected to increase to 190-200 by FY40. The rising demand expects the number of airplanes to reach 1,100 by 2027 and by 2038 will require 2,380 new commercial airplanes.

1. Indigo

Indigo is the leader in the Indian aviation industry with the current market cap of Rs 49,923.47 Cr. This company trades on Indian stock exchanges with the latest share price of Rs 1,364.95 per share.

Indigo is India's largest airline by passengers carried and fleet size and boasts a 60.4% domestic market share as of July 2020. The airline was founded by Rahul Bhatia of Interglobe Enterprises and Rakesh Gangwal in 2006 after it took delivery of its first aircraft in July 2006.

The company operates in Indian domestic markets as a low-cost carrier. The company has grown its fleet to 276 aircrafts today and has been one of the few airlines that have been profitable for 10 years.

2. Spice Jet

Spice Jet is India's 2nd largest airline in terms of domestic passengers carried and has a market share of 13.6% as of July 2020. It has the current market cap of Rs 3,085.41 Cr.

The airline was established as ModiLuft in 1994 with backing from Lufthansa but ceased operations in 1996. In 2004 Indian entrepreneur Ajay Singh acquired the company and renamed it as SpiceJet. SpiceJet started its operations with 2 aircraft on lease and currently has a fleet of over 90 aircraft.

3. Jet Airways

JET AIRWAYS

Jet Airways Ltd is an Indian international airline. In March 2019 it was reported that nearly a fourth of Jet Airways' aircraft was grounded from operations due to unpaid lease rates. It is a smallcap company trading in the Indian stock market with a market cap of Rs 311.82 Cr. The stocks of Jet Airways used to trade at a share price above Rs 1,300 per share in 2005. However, as of Oct 2020, this stock is hovering at a share price of Rs 31.

4. Air India

एअर इंडिया
AIR INDIA

Air India is India's flag carrier owned by the Indian government. The airline is India's largest international carrier with a market share of 18.6%. Domestically the airline falls behind market leaders Indigo and SpiceJet with a market share of 9.1% as of July 2020. The airline was founded in 1932 by JRD Tata and made a profit of Rs 60,000 in its first year carrying weekly mail and 155 passengers.

After WW2 Tata airlines became a public ltd company under the name Air India. Unfortunately, years of loss-making operations have turned the airline into a debt-stricken

entity. The government has time and again tried to sell the airline but unsuccessfully. The new owner would have to take on a debt of US$4.7 billion.

The airline operates flights domestically and to its Asian destinations through its subsidiaries Alliance Air and Air India Express. The airline currently has a fleet of 173 aircraft.

5. Air Asia

Air Asia India was established in 2013 as an Indian airline through a joint venture between Tata Sons and Air Asia Investment Ltd. (Malaysia).

The airline also marked the return of the Tata Group into the aviation industry after 60 years. Air Asia holds a 6.2% market share as of July 2020. The airline has a fleet of over 30 aircraft.

6. Vistara

Vistara is another airline that includes a Joint Venture between Tata and a global airline. The airline was founded in 2013 as a joint venture with Singapore airlines. Vistara is a name taken from the Sanskrit word meaning "Limitless expanse". Vistara had a market share of 4.2% as of July 2020. The airline has a fleet of 42 aircraft.

7. GoAir

GoAir, part of the Wadia Grp. is a low-cost airline. It launched its operations in 2005 and as of July 2005 holds a market share of 3.8%. The airline has a fleet size of 55 aircraft.

The airline had earlier looked for a merger with SpiceJet and later appointed JPMorgan to scout for potential investors. In its attempt to raise capital the airline had planned to go for an IPO this year but these plans been delayed.

The takeoff of the Indian Airline industry has been delayed unfortunately due to COVID-19. Industry experts CAPA expects the Indian aviation industry to shrink to 2-3 players if they do not receive additional funding. This according to them would result in sustainable damage in connectivity throughout India.

Ajay Awtaney, Founder, LiveFromALounge.com said that he expects only IndiGo, Vistara, and Air India will be able to survive post-COVID-19. "IndiGo is cash-rich. They are doing the right thing by raising liquidity and have also taken cost-cutting measures. Vistara too is in a good position right now. They are bleeding financially but have the support of two strong backers. Air India, on the other hand, has the backing of the government". This calls for added government focus on the industry in order to ensure that the industry does not suffer irreparable damage.

AIRLINE INDUSTRY SWOT ANALYSIS

The airline is a company that offers an air source of transport to passengers and freight. The airline companies use airplanes and aircraft to operate their services. These companies also have partnerships and alliances with one another to complete the same flight. The airline companies usually require an air certificate and license by the Government aviation authority to run and operate their flights.

The history of the Airline industry started with the German company DELAG that launched its first aircraft on November 16, 1909. In the beginning, airlines were the personal ownership until the 1930s. But from 1940 to 1980, the government took control over the ownership of major airlines across the world. However, the period of mass-scale privatization and alliances started in the 1980s.

Now we'll discuss the SWOT (strengths, weaknesses, opportunities, threats) analysis of the airline industry. What the internal strength and external opportunities companies could use to face the internal weaknesses and external threats.

I STRENGTHS OF AIRLINE INDUSTRY

The four major strengths of the airline industry.

1. High Income

One thing we all know that airlines are one of the profitable companies, and their income level must be very high. Why? It is because the fare of airlines is very high, and many people who haven't traveled on an airplane in their entire lives. Traveling on an airplane is a dream to them, but they can't afford the fare.

2. Growing Tourism Industry

The tourism industry has been rising for the past decade and a half. TV, media, and films have made people adventurous, and they would like to travel and explore new places across the world. When it comes to traveling inter-continent, then airplanes are the only viable and easy mode of transportation. According to an estimate, there has been a growth of 50% among domestic and local tourists, and a 25% increase in international travelers.

3. Continued Growth

If we look at the graph of airline growth, then except for a few downturns, it has been moving upward consistently. It is due to the number of factors like growth in population, and increased businesses. When more people engaging in businesses, then they will fly from one country to another. If the aviation management uses effective marketing strategies, then they increase the profit rate more than ever.

4. Safe & Speedy

One thing that we all agree on that airplane are the fastest mode of transportation. But when it comes to safety, aviation engineers are working on the research and development to make it the safest transportation mode. Even though, it's a safe mode if we compare it with others. The airline's security and other staff is highly trained and well equipped, they're ready and capable to face all kinds of challenges.

II WEAKNESS OF AIRLINE INDUSTRY

1. Slow Rate of Infrastructure

Slow infrastructure is the major problem in the airline industry. If you look at the technological development, the engineers are building new big and sophisticated airplanes. But on ground infrastructure is the same old airports. You would see a huge contrast between the airports and airplanes if you visit an airport. Some of the airports in the developed countries have a good latest technological infrastructure. But it's not the case everywhere. There should be a balance of investment in both of these sectors.

2. High Spoilage Rate

Spoilage rate means that the passengers miss their flights. If the passengers miss their flight, then they would have to refund their fare, and the company would losses the income. If you compare the spoilage rates with other transportation means, then it's very high among airlines.

The airlines companies send a reminder message or call to all of their customers before the flight, the purpose is to make sure that every passenger should be on board, and lower the loss rate.

3. Huge Investment

The airline is one of those industries that require a huge early investment to start. Therefore, a group of shareholders come together and start an airline company. It is nearly impossible to be a sole proprietor in this industry, and the company and shareholder is the only solution for the formation of airlines.

4. Competitive Market

The airline industry is also getting very competitive. In the off-seasons, small airline companies have to reduce the traveling fare. Therefore, it becomes necessary for the bigger companies to reduce the fare as well, and keep attracting the customers. If they don't lower the price, customers would prefer the smaller airlines with lower prices, and the bigger airline company would losses the market share.

5. Huge Workforce

Airline companies require a huge workforce for the smooth running of various operations at one time. Multiple departments are performing multiple tasks at the same time. If a company has a less workforce, then it cannot perform various functions like control tower, runway staff, security entrance personals, ticketing staff, and many others. When all of them perform their duties, then airplanes can safely land, take off, customers come and go, and the whole process continues.

III OPPORTUNITIES OF AIRLINE INDUSTRY

1. Inviting New Stakeholders

As we know that the airline industry is a very profitable business and it has been growing consistently over some time, and hopefully it continues to improve so in the future. However, the management can use its background to attract and invite new investors. They would bring more capital, and the company can improve various functions with it.

2. Worldwide Mailing Services

Since the aircraft of the airline companies go worldwide. Therefore, such companies should launch the new delivery service by partnering up with the local shipment services of different countries. This delivery service won't cost the company much, but it can bring a lot of profit. All the airline company has to do is to advertise it, and partner up with the local courier service. Receive packages from the courier of one country and deliver it to the courier service of the other country, and they would receive their profit out of it.

3. Use Technology to Minimize Cost

The airline companies should invest in the use of technological development at different levels. The cost may be high at one time, but it would reduce the long-term

miscellaneous expenses. For instance, the website and a smartphone by connecting the company with customers, it would reduce the spoilage rate. A centralized application connecting all the employees, departments and management, where everyone knows one's roles and how it would affect the performance of others. That's how investment in the technology would remove delays and spoilage rates.

4. Market Expansion

The airline industry already works beyond borders, the management should take this as an opportunity and expand their geographical market into the new airports and target more people. If the marketers advertise it well, then they surely can attract new customers to new places.

5. Better Deals with Supplier

The airline companies have direct and contact with the oil companies, and they are very costly suppliers. If the companies manage to lower the operational cost, then they can get better with them.

6. Timely Service

If the markers and the customer service center of the company effectively communicate with the customers and send a good reminder. Then they can remove the delays, spoilage rate, and refunding. Customer would get a better timely service, and it would be good for the reputation of the company.

IV THREATS TO AIRLINE INDUSTRY

1. Global Economic Crisis

The economy of all the countries is on the downturn. Jobs are less and people are working at the least minimum wages. The tourism industry is declining, and people are using the airlines only for the business purposes. Traveling for adventure and pleasure is getting less and less because people don't have sufficient savings. In such circumstances, the airline industry has to reduce the expenses to the minimum level. This is the only way to survive under the global crisis.

2. Government Intervention

Government policies can intervene in the airline industry and have a very bad impact. For instance, government pressure to lower or increase the prices, delays in the flight due to the security officials. Such interventions can result in the form of the loss to the company.

3. Environment Threats

The airline industry is helpless to the environment. For instance, tornados, heavy storms, and the strong wind could result in the form of delays. When it happens, some

customers have to cancel their flight and the company has to refund the fare. In other words, bad weather and the environment is a loss to the company.

4. Seasonal Threats

It is a seasonal business because people aren't usually free the whole year. They have jobs and work to do the whole years; they only get free in the holidays and festivals. When few customers travel in the offseason, then the fare would barely cover the fuel expenses.

5. Pandemic

The current pandemic of Coronavirus (COVID-19) has caused the lockdown of all the airports and airlines. People are stuck in the country and spending their savings. Tourism and the airlines are one of the few industries that are badly affected by the lockdown. It's still uncertain to estimate the loss that it would cost after the lockdown. One thing is certain that things won't go back to normal after the lockdown immediately.

The airline industry is a very profitable business, but the lockdown is becoming very detrimental to the life of this industry. The longer it goes, the more it will be damaging.

REVIEW QUESTIONS

1. Explain the concept of airline industry.
2. Write the different types of airlines.
3. Discuss the history of airline industry.
4. What are the major players in airline industry?
5. Explain the SWOT analysis of airline industry.

CHALLENGES IN AIRLINE, AIR CARGO INDUSTRY, IATA, FIATA

Learning Objectives

After reading this chapter, you should be able to:

- Understand the challenges in aviation industry.
- To know the challenges in air cargo industry.
- Understand the functions and objectives of IATA.
- To know the FIATA mission and objectives.
- Understand the FIATA functions.

INTRODUCTION

Energy is definitely the center of human life – without it there would be no modern life. Ever since the 19th century, the role of fossil fuels has expanded with every passing year. Much has been made out about the negative impacts on the climate and the resultant effects on health, but the fact remains that it is still impossible to imagine a world without fossil fuels and the energy that they have to offer. The airline industry offers the perfect example of a sector that is almost entirely dependent on fossil fuels and its availability. In fact, about 30% of the costs of the industry goes towards kerosene. This article tries to offer insights into the biggest challenges faced by the airlines today.

Auxiliary change in the Indian flying area has profited customers with lower normal costs, progressively broad and visit administration and proceeded with innovative advancement in both aero plane and air terminals. The division keeps on pulling in extensive consideration from financial experts and policymakers, in any case, to a limited extent since its strategic approaches have been so dynamic in nature and separated while carrier profit has been enormously unpredictable. In any case, the noteworthy issue that speaks to a test for the avionics segment in India is framework impediment which is required to be quickly expanded.

CHALLENGES OF THE AVIATION INDUSTRY

The aviation industry is a world in itself, encompassing a massive workforce and contributing a significant percentage to the global economy. The sector was set for major growth, challenges notwithstanding, until the pandemic hit, post which it plummeted to a considerable extent.

According to research released by the Air Transport Action Group (ATAG) in 2018, the global aviation market outlook had been quite positive over a couple of years earlier. The report stated that at the time, the global air transport sector supported more than 65.5 million jobs and was responsible for a mammoth USD 2.7 trillion in economic activity worldwide. Additionally, the report claimed that a free-trade approach will help further the growth in air transport, and it will come to record USD 5.7 trillion in economic activities by 2036, while supporting around 97.8 million jobs.

The airline sector does, therefore, play a fundamental role in today's society, but it is also important to highlight that it has its own fair share of challenges. From battling recessions to government regulations and terrorism to labor shortage, there are innumerable issues this sector has been facing. The paragraphs below enlist some of the challenges of the aviation industry.

1. Fuel Efficiency

Aviation fuel availability and costs have remained one of the major economic factors affecting the airline industry since nearly three decades. High jet fuel prices have a

direct impact on the airline's financial portfolio. With the number of airline companies rising year-on-year, fuel prices were on an all-time high, creating a vicious circle. Alternative fuels haven't really been that impactful, thereby, maintaining fuel efficiency falls among a battery of key challenges of the aviation industry.

2. Global Economy

The state of the world economy is one of primary challenges of the airline market. The 2008 economic recession for instance, had a disastrous impact on the aviation industry size. As global economy collapses, travel and fuel costs increase, while passenger numbers decrease. The impact of the recession on the tourism sector is also one of the key economic factors affecting the airline industry.

As air travel companies seek expansion across various nations and explore new routes, they must keep in mind the market conditions and volatility in the regional zones. Also, various nations have different economic conditions for growth, a factor which airline companies must remember when they seek international expansion.

3. Passenger Comfort and Experience

The aviation industry is service-driven; its success depends enormously on how satisfied passengers are. The reason passenger comfort forms the crux of challenges faced by the airline industry is that no passenger is alike; at some point, a section of customers may always be dissatisfied. However, this is what has pushed airline companies to ensure seamlessness throughout – the actual air travel, excellent security, less/no hassles in airport lines, convenient baggage claim, customer care, and more.

To achieve these in a flawless manner is a consistent tussle of sorts. Customer surveys and feedback always depict that not all passengers have great travel experiences. For instance, IATA's 2017 Global Passenger Survey depicts that only around 56% of North American customers were happy with their last travel experience. To that end, there's always space for bettering customer comfort, which airline companies will continue to battle with.

4. Airline Infrastructure

Airports must consistently upgrade their infrastructure – the runways, terminals, concourses, hotels, shopping centers, lounges, and more. There is major competition in this area; to maintain the reputation of the airline and remain ahead of competition, onsite amenities such as aircraft ground handling systems will need to be periodically renovated. While doing so occasionally is advantageous no doubt, leading to increasing passenger numbers, repeating the upgrade every now and then will have a significant impact on the airline company's finances.

Aircrafts need to be periodically upgraded and maintained as well – this in fact, is even more crucial, as passenger safety is reliant on the same. Having faulty aircraft doors or aircraft seating is liable to get the airline sued. Airline infrastructure is one of most vital challenges of the aviation market, as carriers need to maintain their current fleet and also ensure to purchase new, modern ones, while ensuring fuel efficiency and lowered costs.

5. Global Congestion

A couple of decades back, air travel had been a luxurious affair; playthings of the rich, especially in the developing countries. The scenario has changed drastically, today, airports the world over are so crowded that it has led to unnecessary flight delays. Most flights seem full, terminals are always congested, and more importantly, the skies are overcrowded with the excessive number of aircraft.

Air traffic and airport congestion are major challenges faced by the airline market, which seem to have no feasible solution, at least in the immediate future. Carriers continue to make trips seamless for flyers, however, this will continue to remain a viable challenge.

6. Technological Advancements

To think that technological developments are one of the challenges of the airline industry may come as a surprise. Technology, however, is a double-edged sword, and despite the revolution it has brought about, rising dependence on the same can make the entire industry vulnerable. For instance, in case of a software issue, the operations of the airline may remain crippled until it is resolved. In case of insufficient funding, upgrading crucial infrastructure such as aircraft communication systems may be impossible, causing the entire system to collapse.

7. Labor Unrest

A number of airlines in Europe have suffered because of issues like pilot walkouts. Just recently, both Air France and KLM were hit by a 14-day pilot strike which hit the earnings by almost 500 million euros. Even Lufthansa has complained of suffering from the pain caused by pilot walkouts, with issues like these outweighing the benefits that lower oil prices had to offer in recent times.

8. The Emergence of Low-Cost Carriers

The problems faced by many of the larger operators have benefited low-cost carriers operating in the region tremendously. At a time when Air France employees were on strike, low-cost carriers like EasyJet managed to increase their pretax profit guidance. Europe's largest low-cost carrier, Ryanair, has also managed to show strength when its full-service competitors suffered from strikes.

9. Terrorism

Tragic events from the past have created ripples of fear not only among the masses, but also airport personnel. Although terrorist activities have dimmed down in recent times, this is still very much a threat, as airline companies must remain consistently vigilant. Increasing fear of terrorism leads to stringent check-ins and consequently longer lines and delays. It also pressurizes airline companies to come up with highly secure state-of-the-art screening procedures and equipment.

On a related note, even political scuffles between two nations can pose as one of the major challenges of the aviation industry. After all, airlines are the connection between two countries, and unreasonable government regulations due to strife between two nations can make it very difficult for carriers to conveniently transport passengers and cargo in and out of international borders.

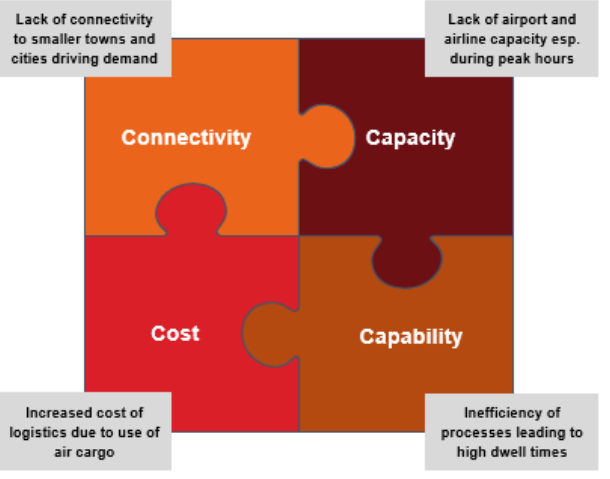
10. Climate Change

Without a shred of doubt, climate change and environmental issues have remained among the key challenges faced by the airline market since a very long time. Since commercial aviation is responsible for a considerable percentage of carbon emissions, the airline industry is under tremendous pressure to undertake measures that will reduce environmental impact of air travel.

As part of a response to this challenge, the IATA had suggested a multi-faceted, four-pillar strategy, including enhanced technology, efficient aircraft operations, improved infrastructure, and a single global market-based measure. Until these are completely and seamlessly met, climate change will continue to be one of the current challenges of the airline industry.

The aviation industry forms an important part of the global economy. Pre-COVID, airline companies had to deal with considerable challenges, the solutions of which were being thought out and assessed by industry experts periodically. The advent of the pandemic however, brought forth a set of groundbreaking challenges for the airline industry that it hadn't encountered in any of the previous global disasters such as the 9/11 attack or the economic recession of 2008. It has not only brought about an entire 180-degree change in the way previous challenges were perceived, but a new set of risks that will now form the base of how the industry will function in the forthcoming decades.

CHALLENGES IN AIR CARGO INDUSTRY



I Capacity and Infrastructure Constraint

a) Airline Capacity Constraint

One of the biggest challenges faced by the retailers has been insufficient cargo capacity for shipments via air which is leading to delayed deliveries in distant markets. This is mainly due to lack of freighter jets and lower priority to belly cargo by passenger airlines. There is need for airlines to relook at their freight strategy and develop plans to optimize belly capacity.

b) Airport Capacity Constraint

Historically, airports have always been developed from passenger stand point of view, and thus requirement of cargo facility development has been relegated and considered not important. Therefore, the entire logistics of cargo is woefully inadequate and poorly managed area of the airport.

Cargo infrastructure at any airport is just not the cargo terminal building that houses the warehouse but also the related facilities including special facilities for express freight, frozen foods, airmail, and hazardous goods. It is essential to develop capabilities to handle all types of cargo.

Further, quick throughput of cargo requires specialized equipment, fast processes, truck parking and loading zones and separate access roads for cargo vehicles. Today, lack of capacity is a major reason Indian air cargo industry is suffering from high dwell time at international airports

c) Manpower Capacity Constraint

The logistics industry is facing a shortage of skilled workers able to help meet complex supply chain challenges and sustain the industry's growth rate, according to a new report commissioned by the World Bank. Not only do logistics employers find it tough to recruit good candidates for positions, but many existing employees are not sufficiently qualified, and this skills shortage is likely to worsen in the absence of new initiatives one reason for the shortage is the difficulty of competing with other industries for talented staff, in part because the pay is low, working conditions are poor, and the industry's reputation is not good. There needs to be a substantial increase in the amount of time and resources devoted to logistics training at all levels.

II Capability and Efficiency

a) Inefficient processing and lack of automation

Inefficiencies are created in system due to involvement of multiple procedures with lengthy processing times and long queues at the airports. Further, lack of automation resulting in high manual intervention, inordinate dwell time, missing and untraceable cargo.

As per IATA, each cargo shipment on an average today carries around 30 types of documents and well over 100 copies thus resulting into significant documentation overheads, increased dwell times and supply chain opaqueness. Therefore, certain systems/ processes such as automation and digital platform are essential to capitalize on the technological advances to promote a hurdle free system for movement of cargo.

b) Lack of transparency and governance

While the e-retailers and logistics players can provide transparency at all other parts of the value chain, this transparency is completely lost once the shipments are handed over to the airlines in cargo terminal. There is lack of capability in the airline/ airport system to record data providing a status regarding movement of goods. This furthers makes it difficult to measure service quality at the airport.

III Connectivity for cargo

a) Connectivity across the country

E-commerce is an important driver of air cargo. Today, 60-70% of the total orders for retailers are from Tier-II and Tier-III cities. Most of these cities do not lie within the serving radius of a fulfilment centre. With faster customer delivery as the focus and top priority, air shipment connectivity to these locations is a key requirement.

However, India currently does not have the connectivity required for this industry, and challenges are faced on three fronts:

- Lack of airport facilities.

- Lower frequency of flights on connections.
- Inconvenient time schedule of flights as e-commerce players prefer to deliver shipments at the end of the day/ later evening, restricting availability of connections.

b) Connectivity with the city

An airport with poor access and limited cargo commute alternatives acts as a bottleneck in the growth of cargo volumes. Most airports do not have separate connectivity for cargo transport to airport. Combining passenger and cargo traffic on the same highway leads to inconvenience to both.

IV Costs of air transport

Logistics capture a high share of the available margins to the players; representing 10 to 20% of the overall value of the order, where gross margins range from 15 to 40%. Thus, optimization of logistics and delivery cost is one of the key financial success parameters for the players Mode mix optimization and modal shift from air to less expensive or perceived more environmentally friendly waterway or rail transport has been occurring for some years.

Freight forwarders offer more air/sea, air/road or air/rail products that combine to create price and total shipment time flexibility. Today its railways, tomorrow it will be waterways and emerging technologies such as hyperloop. With the introduction of GST, road travel has become more attractive as local taxes have been abolished. But it also presents a unique opportunity for air cargo as ease of doing business increases and so does the potential for digitalization. The key factors driving the air freight rates are:

a) Economic Conditions

Economic conditions directly impact supply and demand - factors at the heart of all pricing decisions, regardless of industry. That is certainly true for air cargo. With air cargo demand rising due to e-commerce boost, the prices of air cargo have been rising in the past few years.

b) Operating Expenses

Operating expenditure can vary widely and change rapidly. The price of fuel is a prime example. Fuel prices are rising in the past few months further adding to the pricing. Apart from fuel there are various other costs such as landing fees, parking fees, taxes and financing costs that effect the overall pricing of cargo services at airports.

c) Cargo Specialization

Time and temperature sensitive cargo have historically been sent via sea freight. But with increasing availability of temperature-controlled ULDs, the dynamic is changing. Carriers can charge higher prices for these shipments, and even offer temperature-controlled ULDs as a service upgrade.

d) Premium / Value added Service

One of the primary competitive advantages of air freight over sea freight is the ability to offer premium services at a higher price based on things services like packing, door-2-door service, express delivery, shipment tracking, and shipments that require special handling, like food with a short shelf life, hazardous materials, or human remains. In addition, while cargo-only carriers have the advantage of knowing exactly what their capacity is, combined carriers don't. The freight capacity for passenger flights that carry belly cargo fluctuates depending on the number of passengers and how much luggage they have. Therefore, combined carriers can offer shipment prioritization as a premium service.

Customers can pay more to ensure that their shipments jump to the front of the line in the event of a shortfall in capacity. All these have led to increasing in air cargo prices which make them less affordable and impede the growth of the industry. Airlines need to develop pricing optimization programs to capture this opportunity as cargo revenues can add on to airlines profitability in a significant manner.

REGULATORY BODIES IN AIR CARGO

The air cargo industry incorporates an industrial supply chain, which includes airlines, customs, ground services, air cargo forwarders, brokers, domestic transportation, air cargo terminals, distribution centers and integrated international express services. Of these, air cargo terminals are critical in the air cargo supply chain. A typical air cargo terminal has three main users – airlines, air cargo terminal operators and forwarders/cargo-agents who are the principal contributors to the revenue of air cargo terminals.

The demand for air freight is limited by cost, typically priced 4–5 times that of road transport and 12–16 times that of sea transport. These values differ from country to country, season to season and from product to product and for different volumes also. Cargo shipped by air thus have high values per unit or are very time-sensitive, such as documents, pharmaceuticals, fashion garments, production samples, electronics consumer goods, and perishable agricultural and seafood products. They also include some inputs to meet just-in-time production and emergency shipments of spare parts. As the volume of air freight grows, there is a natural progression from passenger aircraft to chartered cargo planes of increasing size and ultimately to scheduled cargo services.

IATA (INTERNATIONAL AIR TRANSPORT ASSOCIATION)

The International Air Transport Association (IATA) is an international industry trade group of airlines headquartered in Montreal, Canada, where the International Civil Aviation Organization is also headquartered. IATA was formed on 19 April 1945, in Havana, Cuba. It is the successor to the International Air Traffic Association, founded in The Hague in 1919, the year of the world's first international scheduled services. At its founding, IATA had 57 members from 31 nations, mostly in Europe and North America. Today it has about 230 members from more than 140 nations in every part of the world.

At its founding, IATA had 57 members from 31 nations, mostly in Europe and North America. Today it has some 290 members from 120 nations in every part of the globe. The modern IATA is the successor to the International Air Traffic Association founded in the Hague in 1919 – the year of the world's first international scheduled services.

The old IATA was able to start small and grow gradually. It was also limited to a European dimension until 1939 when Pan American joined. The post-1945 IATA immediately had to handle worldwide responsibilities with a more systematic organization and a larger infrastructure. This was reflected in the 1945 Articles of Association and a much more precise definition of IATA's aims than had existed before 1939.

The most important tasks of IATA during its earliest days were technical, because safety and reliability are fundamental to airline operations. These require the highest standards in air navigation, airport infrastructure and flight operations. The IATA airlines provided vital input to the work of ICAO, as that organization drafted its Standards and commended Practices. By 1949, the drafting process was largely complete and reflected in "Annexes" to the Chicago convention, the treaty which still governs the conduct of international civil aviation.

The Chicago Conference of 1944 which gave birth to the Chicago Convention tried to achieve a multilateral answer to the first two questions, but failed to do so. The questions of who flies, and where, were resolved on a bilateral basis. The benchmark Bermuda Agreement of 1946 between the US and the UK was the first of almost 4,000 bilateral air transport agreements so far signed and registered with ICAO.

In the early days, governments insisted on the right to oversee the prices charged by international airlines but could not, in practical terms, develop those prices for themselves. IATA was delegated to hold Traffic Conferences for this purpose, with all fares and rates subject to final government approval. The aim was twofold: ensuring that fares and rates would not involve cut-throat competition, while ensuring that they could be set as low as possible, in the interests of consumers.

A coherent pattern of fares and rates pattern was established, avoiding inconsistencies between tariffs affecting neighboring countries - and thereby avoiding traffic diversion. The predictability of fares and rates in this pattern also enabled airlines to accept each other's' tickets on multi-sector journeys and thus gave birth to interlining. Today, 50 million international air passengers a year pay for their ticket in one place, in one currency, but complete their journey using at least two, and sometimes five or more, airlines from different countries using different currencies.

IATA's legal efforts anticipated the effects of new technology associated with the period of rapid growth. It was able to advise the industry on new aircraft and systems, electronic data processing and advances in sales and marketing techniques. Since acts of hijacking and sabotage became more frequent during this period, IATA assisted in the development of the Tokyo, Hague and Montreal Conventions, the first international legal counter-measures.

Automation became commonplace in airline operations during the 1960s. Standardization in its use was less commonplace and IATA began its involvement in perfecting standard message formats for inter-company data exchange, which continues to this day. The goal has been to save money for the airlines, while enhancing airline service. This philosophy was carried over too many activities during the 1950s and 1960s. Development of the Clearing House was followed by the establishment of Billing and Settlement Plans and Cargo Accounts Settlement Systems - in effect, one-way clearing houses to speed the flow of revenue from agents to airlines.

Sales Agents were given the opportunity to prove their professional status under an accreditation process, and training - in conjunction with the Universal Federation of Travel Agents' Associations and the Federation of Freight Forwarders' Associations - was introduced. The pattern for airline-agent relations was set with the introduction of the Standard Agency Agreement in 1952. There are now nearly 81,000 IATA agents worldwide and 135,000 students have enrolled for IATA-sponsored agency training courses since they were introduced.

ROLE OF IATA

1. IATA's mission is to represent, lead, and serve the airline industry.

2. IATA represents some 230 airlines comprising 93% of scheduled international air traffic. The Director General and Chief Executive Officer is Giovanni Bisignani.
3. Currently, IATA is present in over 150 countries covered through 101 offices around the globe. All the Airline rules and regulations are defined by IATA.
4. The main aim of IATA is to provide safe and secure transportation to its passengers.
5. The Organization work benefits all industry stakeholders, such as simplifying the travel and shipping process for consumers, and keeping governments informed about aviation industry complexities.
6. The IATA Head Office are located at 800 Place Victoria (Montreal Stock Exchange Tower) in Montreal since 1977 (having been located at Central Station (Montreal) since its founding) and the executive offices are at the Geneva Airport in Switzerland.
7. The IATA's mission includes three types of actions:
    a) Representing the industry and lobbying governments and regulators;
    b) Leading the industry by working with airlines to increase passenger convenience;
    c) Serving the industry by providing support with products and expert services.

OBJECTIVES OF IATA

1. To promote safe, regular and economical air transport for the benefit of the people of the world,
2. To foster air commerce and study the problems connected therewith
3. To promote means for collaboration among air transport enterprises engaged directly or indirectly in international air transport service
4. To cooperate with ICAO and other international organizations there is always close association and dialogue between ICAO and IATA.

FUNCTIONS OF IATA

1. IATA was involved in establishing a sound technical and legal frame work for the operation of the world's airlines.
2. It organized the first worldwide Traffic Conference in 1947 which worked out agreements related to fare construction rules for multi-sector trips, revenue allocation-prorating rules, baggage allowance, ticket and airway bill design and agency appointment procedures.

3. IATA is also the coordinator of Multilateral Interline Traffic Agreements that allow nearly 300 airlines around the world to accept each other's' ticket and airway bills-and thus their passenger and cargo traffic-on a reciprocal basis.
4. In 1952, the pattern for airline-travel agent relations was set with the introduction of Standard Passenger Sales Agency Agreement.
5. There are now nearly 81,000 IATA sponsored agency training courses since they were introduced.
6. One of the IATA's major issues today is taxation and user charges (payment for using airports and air navigation services).
7. IATA makes specific challenges to the legality of certain taxes and point out to governments the counter-productive effect of excessive aviation taxation.
8. Higher user charges mushroomed during the 1960's and 70's. IATA's task is to minimize their impact by ensuring that the charges are for facilities actually required, that charges are cost-related and that productivity improvements are built into cost projection

FIATA (INTERNATIONAL FEDERATION OF FREIGHT FORWARDERS ASSOCIATIONS)

FIATA, is a non-governmental, membership-based organization representing freight forwarders in some 150 countries. FIATA's membership is composed of 109 Association Members and over 5,500 Individual Members, overall representing an industry of 40,000 freight forwarding and logistics firms worldwide.

FIATA is a reference source on international policies and regulations governing the freight forwarding and logistics industry. FIATA works at the international level to represent service providers who operate in trade logistics and supply chain management. Through its FIATA documents and forms, congress, training and publications, it promotes trade facilitation and best practices among the freight forwarding community.

FIATA is committed to representing the interest of its members by actively engaging with the World Trade Organization, United Nations agencies and other

international organizations, transport organizations, global partners and governments to promote and protect the interest of the industry.

Founded in 1926 in Vienna, Austria, FIATA owes its acronym to its French name (Fédération Internationale des Associations de Transitaires et Assimilés) and is known as 'the global voice of freight logistics'. FIATA is headquartered in Geneva, Switzerland.

MISSION OF FIATA

FIATA's objectives are fivefold:

1. To unite the freight forwarding industry worldwide;
2. To represent, promote and protect the interests of the industry by participating as advisors or experts in meetings of international bodies dealing with transportation;
3. To familiarize trade and industry and the public at large with the services rendered by freight forwarders through the dissemination of information, distribution of publications, etc.;
4. To standardize and improve the quality of services rendered by freight forwarders by developing and promoting uniform forwarding documents, standard trading conditions, etc.;
5. To assist with vocational training for freight forwarders, liability insurance problems, tools for electronic commerce including electronic data interchange (EDI) and barcode.

FIATA MEMBERSHIP

FIATA has different categories of members:

1. Association Members

There are organizations representing the entire (or part of) the forwarding sector in a country/territory and/or a region. Each Association Member holds the right to present motions, as well as nomination and voting rights.

2. Individual Members

There are forwarding and logistic companies, as well as legal entities closely associated with the forwarding and logistic branch.

3. Group Members

Group members can be:

a) International organizations representing the forwarding branch of a group of countries if all members are national associations, which in turn are FIATA Association Members
b) International forwarding groups representing the same or similar interests to FIATA
c) International associations whose members are active only in a sub-sector of the forwarding branch

4. Honorary Members

There are persons who have been of particular service to FIATA or the forwarding industry.

FUNCTIONS OF FIATA

I Advocacy

As the global voice of freight logistics and the only international organization truly focusing on all modes of transport, FIATA advocates trade facilitation policies and collaborates with all industry stakeholders to further develop approaches for the well-functioning of the supply chain. Through its institutes and bodies, FIATA leverages the global expertise of its delegates to tackle today's key challenges and ensure the resilience and adaptability of the supply chain. Combined with a collaborative approach with partner organizations, FIATA ensures representation of freight forwarders around the world as well as the effective movement of international trade.

1. International Trade and Law

The International Trade and Law unit oversees FIATA's advocacy work and supports the activities of the Customs Affairs Institute, Airfreight Institute, Multimodal Transport Institute and the Advisory Body on Legal Matters. Within the ambit of each institute and body, FIATA develops crucial tools and resources for its membership and represents the industry at the international level.

2. FIATA documents

Already 50 years ago, the freight forwarding community felt that there was a need for certain harmonization and guidance in freight forwarding transport documents. FIATA has developed a corpus of documents and forms available for members to use in their daily business. These documents, which include the FIATA Multimodal Transport Bill of Lading, enjoy an excellent reputation and are highly regarded for their integrity and traceability.

These documents are distributed by licensed FIATA Association Members in their respective territories. Where this is not possible, FIATA documents may be obtained

directly from FIATA by individual members and members of FIATA Association Members.

II FIATA Digital Strategy

In the global supply chain, freight forwarders own and control much more data than airlines or shipping lines do. Yet, they are currently struggling to communicate and connect with their partners who aren't necessarily using the same transport management system (TMS). The lack of interoperability between systems and inconsistency in data quality and data standards is a real challenge for the freight forwarding industry.

The digitalization of the supply chain is an industrywide challenge, where collaboration between all stakeholders is critical. FIATA is committed to playing its role, representing the interests of freight forwarders by leading projects that will help them embrace the digital world and ensure they stay operationally up to date with new business models.

FIATA's digital strategy will allow its members to exchange secured and authenticated data, access/use/sign up-to-date FIATA standard trade data/documents, and leverage industry intelligence through their everyday tools (TMS and other software). The strategy focuses on four main axes:

a) Creating a network of trust

Defining a digital identity format and registry, which will allow freight forwarders to exchange authenticated and secured data.

b) Enforcing the usage of standards

- Making FIATA digital standard templates available through TMS and other relevant logistics platforms, in a controlled way;
- Allowing freight forwarders to digitally sign documents so that the issuer of the document can be easily identified by their partners.

c) Facilitating interoperability

To connecting existing software platforms through the definition of a data exchange model and API specifications

d) Promoting business intelligence

To ensuring FIATA members keep control over their data and can easily collect and analyse data from software suppliers to optimize their business.

III Training

Education is central to the development of the freight forwarding industry and to the improvement of the lives of freight forwarders and logistics professionals. Education

is important in promoting sustained, inclusive, and equitable economic growth and sustainable development. Increased efforts towards accessibility, quality, and affordability of education in our industry are part of the FIATA presidency's priorities.

At FIATA, training and education activities take place under the umbrella of the FIATA Logistics Institute (FLI). FLI equips the industry workforces with the necessary skills and competences through initiatives such as the FIATA Diplomas, the ICAO-FIATA Dangerous Goods by Air Training Programme, and the Young International Freight Forwarder of the Year Award.

FIATA TRANSPORT DOCUMENTS

As FIATA documents are internationally recognized and enjoy significant reliability, integrity, and traceability directly linked to their reputation, their uniformity is of utmost importance. Due to copyright protection, these documents can only be introduced upon conclusion of a Licensing Agreement with FIATA and within the national association's territory. FIATA offers the possibility for Association Members to distribute the following documents:

1. FIATA Forwarders Certificate of Receipt (FIATA FCR)
2. FIATA Forwarders Certificate of Transport (FIATA FCT)
3. FIATA Warehouse Receipt (FWR)
4. Negotiable FIATA Multimodal Transport Bill of Lading (FBL)
5. Non-Negotiable FIATA Multimodal Transport Waybill (FWB)
6. Shippers Declaration for the Transport of Dangerous Goods (FIATA SDT)
7. Shippers Intermodal Weight Certification (FIATA SIC)
8. FIATA Forwarding Instruction (FFI)

REVIEW QUESTIONS

1. Write the challenges in aviation industry.
2. Discuss the challenges in air cargo industry.
3. Explain the functions of IATA.
4. What are the objectives of IATA? Explain
5. Discuss the FIATA mission and objectives.
6. Write the FIATA functions.

AIR CARGO IMPORT / EXPORT RATES

TYPES OF AIR CARGO

People who are not trying the air freight service very often may have two main assumptions; the first one is that air freight is very expensive, and the second one is that air freight is a simple one type of service from one airport to another for cargo with special handling or short shelf life, such as gold, human tissue, blood samples, … etc. Well, the answer is Yes and NO. Airfreight can be expensive in some cases but in other circumstances, it may even save you money. You can also ship a lot of things by air, even though it is usually needed for FMCG (Fast-Moving Consumer Goods) but we can also find other types of cargo as well.

1. General Cargo

This type of service is usually less in price than the Special Cargo service. Transit time may be sometimes longer. For cargo to be categorized as "General Cargo", it shouldn't have any special needs like temperature control or fast-moving for short shelf life. It also should not be oversized or have odd dimensions (like overlength). Example of General Cargo: Clothing (in boxes and palletized), Computers and computer parts (no battery included), books, … etc.

2. Special Cargo

Items that fall under this type of service usually should be handled with care. The air freight invoice for this service is more than the general cargo one. Some airlines are usually skeptical about accepting certain commodities when it comes to special cargo service. Some of the cargo types that fall under the special cargo category are:

a) Temperature Controlled

This service is for the items that require to be stored under a certain temperature due to the special nature of the cargo. Such as fruits, vegetables, medical kits, etc.

b) Dangerous / Hazardous Cargo

Few categories are falling under hazardous cargo. Some of these categories are flammable, poisonous, radioactive, explosive etc. These items should be packed properly and loaded in a certain space in the airplane. For this reason, the rates are expected to be higher than the general cargo. Especially in a passenger aircraft.

c) Live Animals

The service is normally used for pets. Shipping live animals (dogs, cats, turtles, etc.) requires a properly ventilated area and they should be placed in a suitable crate that allows their comfortable stay during the flight. The rate will be based on the chargeable weight of the crate (with the pet inside).

AIR CARGO IMPORT / EXPORT RATES

Learning Objectives

After reading this chapter, you should be able to:

- Understand the air cargo rates.
- To know sample air cargo tariff.
- Understand the import cargo procedures.
- To know the procedure for clearness of import and export goods.
- Understand the export cargo procedures.

INTRODUCTION

Air cargo is used by global importers and exporters when they need to get goods somewhere rapidly and reliably. While 90% of everything is shipped by ocean freight, air freight connects the world faster, cutting China-US freight shipping time from 20-30 days by ocean to only three days by air cargo.

India's customs authorities have launched the Indian Customs Compliance Information Portal detailing customs procedures and regulatory compliances for import-export trade. This article briefly highlights relevant import-export procedures in India as well as various promotional schemes and initiatives being implemented to benefit exporters.

Businesses planning to set up a trading company, or start importing or exporting from India, must understand the stages and stakeholders involved in the process, as well as the regulatory framework and documentation required.

In India, the imports and exports are regulated by the Foreign Trade (Development and Regulation) Act, 1992, which empowers the federal government to make provisions for development and regulation of foreign trade. The current provisions relating to exports and imports in India are available under the Foreign Trade Policy, 2015-20.

SAMPLE CARGO TARIFF

Air India Air Transport Services Ltd.

EXPORT CARGO RATES EFFECTIVE FROM 1ST December, 2020

1. TERMINAL CHARGES

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	General	Rs. 0.70	Rs. 120/-
2	Special (Hazardous/ Cold Storage)	Rs. 1.40	Rs. 235/-
3	Perishable	For F/V Fish Rs. 0.65 wherever State of Art facility is provided. For Live Stock Rs. 1.30 wherever exclusive facility is not provided. For Vaccines Rs. 2.65 wherever State of Art facility is provided.	Rs. 200/- Rs. 235/- Rs. 235/-

2. WAREHOUSING CHARGES. (Leviable from Shipper)

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	General	Rs. 0.72	Rs. 120/-
2	Special	Rs. 1.44	Rs. 235/-
3	Perishable	Rs. 2.00 wherever State of Art facility is provided.	Rs. 200/-

3. BACK TO TOWN SHIPMENT CHARGES

If the shipment is back to town, carting charges Re 1/- per kg + Demurrage charges will be levied.

NOTES

1. The free period for export cargo shall be 12 Hours for examination/processing by the shippers. After the free period, Late clearance charges will be levied.
2. Consignment of human remains, coffin including baggage of deceased & Human eyes will be exempted from the purview of Terminal and Warehousing Charges.
3. No separate Forklift charges will be levied. Arrangement for consignment of 10 tons and above will be done by consignee/agent and crane charges will be borne by them.
4. Special Cargo consists of live animals, hazardous goods, valuable cargo and cargo stored in cold storage.
5. Charges will be levied on the "gross weight" or the "chargeable weight" of the consignment whichever is higher. Wherever the gross weight and /or volume weight is wrongly indicated on the Airway Bill and is actually found more, charges will be levied on the 'actual gross weight' or 'actual volumetric weight' whichever is higher.
6. Opening/Repacking/Strapping charges will be included in TSP/DEMURRAGE.
7. All the bills shall be rounded off to the next of Rs 5/- 0r Rs. 10/-.
8. Service Tax and any other statutory taxes shall be levied extra as per government notifications.
9. Opening/Repacking/Strapping charges will be included in TSP/DEMURRAGE/MISCELLANEOUS receipts, as

Type of Service	Charges per packages in Rs.
Opening/Repacking/Stitching (Bags/Packages)	Rs. 5.00/-

Strapping with Plastic Strip	Rs. 19.00/-
Strapping with Steel Strips	Rs. 28.00/-

IMPORT CARGO RATES

1. TERMINAL CHARGES

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	General	Rs. 4.00	Rs. 100/-
2	Special (Hazardous/ Cold Storage)	Rs. 8.00	Rs. 200/-
3	Valuable	Rs. 8.00	Rs. 500/-

2. WAREHOUSING CHARGES

Free Period for Import Cargo shall be Two Working days (48HOURS) including the Date of Arrival of Flight. For next Two Days i.e., 4th & 5th Day, Warehousing will be charged at 'per kg/per day', non-cumulative basis provided the consignment is taken delivery within 5 days of arrival. If delivery is affected after 5 days from the date of arrival, the Warehousing will accrue for the entire period from the date of arrival of flight as follows,

S. No.	Type of Cargo	Up to 3 days including free Period.	4 days	5 days	Minimum rate per Consignment
A	B	C	D	E	F
1	General	Rs. 1.13 per kg per day	Rs. 2.26 per kg per day	Rs. 6.78 per kg per day	Rs. 250/-
2	Special (Hazardous/ Cold Storage)	Rs. 2.26 per kg per day	Rs. 4.52 per kg per day	Rs. 13.54 per kg per day	Rs. 500/-
3	Valuable	Rs. 4.50 per kg per day	Rs. 9.00 per kg per day	Rs. 27 per kg per day	Rs. 1000/-

3. DELIVERY ORDER CHARGES

Category	Revised D.O. charge (in Rs.)
Per MAWB	2000
Per HAWB	1300
Unaccompanied Baggage	1300

NOTES

1. Consignment of human remains, coffin including baggage of deceased & Human eyes will be exempted from the purview of Terminal and Warehousing Charges.
2. No separate No separate Forklift Charges will be levied.
3. Charges will be levied on the "gross weight" or the "chargeable weight" of the consignment whichever is higher. Wherever the "gross weight and /or volume weight is wrongly indicated on the Airway Bill and is actually found more, charges will be levied on the 'actual gross weight' or 'actual volumetric weight' whichever is higher.
4. Cargo consists of gold, bullion, currency notes, securities, shares, share coupon, traveler's cheques, diamonds (including diamonds for industrial use), Diamond jewelry, jewelry and watches made of silver, gold platinum and items valued at USD 1000 per kg and above.
5. Opening/Repacking/Strapping charges will be included in TSP/DEMURRAGE.
6. Packing charges shall be levied as per existing rates.
7. Service Tax and any other statutory taxes shall be levied extra as per government notifications.
8. Due to persistent increase in cases of dishonouring of cheques received from the CHAs / Consignees resulting into delivery of cargo without collection of dues besides delaying availability of legitimate dues to MIAL, it has been decided to levy penal charge of 10% of payable dues or Rs. 1,000/- whichever is higher, on each case of dishonouring of cheques.
9. Opening/Repacking/Strapping charges will be included in TSP/DEMURRAGE/MISCELLANEOUS receipts, as

Type of Service	Charges per packages in Rs.
Opening/Repacking/Stitching (Bags/Packages)	Rs. 5.00/-
Strapping with Plastic Strip	Rs. 19.00/-

Strapping with Steel Strips	Rs. 28.00/-

SCHEDULE OF AGENT CHARGES FOR DOMESTIC CARGO (INBOUND AND OUTBOUND)

1. TERMINAL CHARGES

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	General Cargo	Rs. 0.90	Rs. 100/-
2	Special (AVI)	Rs. 1.80	Rs. 200/-
3	PER/DGR/VAL	Rs. 1.80	Rs. 200/-

2. WAREHOUSING CHARGES

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	General Cargo	Rs. 0.90	Rs. 125/-
2	Special (AVI)	Rs. 1.80	Rs. 200/-
3	PER/DGR/VAL	Rs. 1.80	Rs. 200/-

3. OTHER CHARGES

S. No	Type of Cargo	Rate per Kilogram	Minimum rate per Consignment
A	B	C	D
1	Courier Handling	Rs. 1.00	Rs. 100/-
2	Amendment of Airway Bill (Outbound)	-	Rs. 100/-
3	Return Cargo Charges (Outbound)	-	Rs. 100/-
4	Strapping Charges	-	Rs. 10/per Pkg.

5	In addition to the above, in the event of Mis-declaration of weight, following charges	2% to 5% Variation Rs. 1.30 More than 5% Rs. 3.25	-

NOTES (INBOUND & OUTBOUND CARGO)

1. Consignment of Human Remain, Coffins including unaccompanied Baggage of the deceased & Human eyes will be exempted from the purview of Terminal and Warehousing Charges.
2. Free Period for Outbound Cargo/Inbound Cargo shall be one day (24hrs) for Shipper's/ Consignee.
3. All bills will be rounded off to the next INR 5 as per rules.
4. All charges above are excluding any taxes and levies and the same will be charged extra as per rules.
5. The Charges will be levied on "Gross Weight" or "Chargeable Weight" whichever is higher. Wherever the Gross Weight or Volume Weight is indicated on Airwaybill and found more, Charges would be levied on "Actual Volumetric Weight" whichever is higher.

IMPORT CARGO PROCEDURE

All goods imported into India have to pass through the procedure of customs for proper examination, appraisal, assessment and evaluation. This helps the custom authorities to charge the proper tax and also check the goods against the illegal import. Also, it is important to note that no import is allowed in India if the importer doesn't have the IEC number issued by the DFGT. There is no requirement of IEC number if the goods are imported for the personal use.

STEPS IN PROCEDURE

1. Arrival of goods and procedures prior to lodgment of goods
2. Conveyances to call only at Notified Customs Ports / Airports
3. Power to board conveyance, to question and to demand documents
4. Delivery of Import Manifest
5. General Conditions
6. Amendments
7. Penal Liability
8. Entry Inwards
9. Enclosures to Import General Manifest
10. Procedure for filing IGM at Custom Houses operating EDI service centres

11. Filing of Stores List
12. Unloading and Loading of Goods
13. Other liabilities of carriers

PROCEDURE FOR CLEARANCE OF IMPORTED GOODS

1. Bill of Entry

A Bill of Entry also known as Shipment Bill is a statement of the nature and value of goods to be imported or exported, prepared by the shipper and presented to a customhouse. The importer clearing the goods for domestic consumption has to file bill of entry in four copies; original and duplicate are meant for customs, third copy for the importer and the fourth copy is meant for the bank for making remittances.

If the goods are cleared through the EDI system, no formal Bill of Entry is filed as it is generated in the computer system, but the importer is required to file a cargo declaration having prescribed particulars required for processing of the entry for customs clearance. In the non-EDI system along with the bill of entry filed by the importer or his representative the following documents are also generally required,

- Signed invoice
- Packing list
- Bill of Lading or Delivery Order/Airway Bill
- GATT declaration form duly filled in
- Importers/ CHA's declaration
- License wherever necessary
- Letter of Credit/Bank Draft/wherever necessary
- Insurance document
- Import license
- Industrial License, if required
- Test report in case of chemicals
- Adhoc exemption order
- DEEC Book/DEPB in original
- Catalogue, Technical write up, Literature in case of machineries, spares or chemicals as may be applicable
- Separately split up value of spares, components machineries
- Certificate of Origin, if preferential rate of duty is claimed
- No Commission declaration

2. Amendment of Bill of Entry

Whenever mistakes are noticed after submission of documents, amendments to the bill of entry are carried out with the approval of Deputy/Assistant Commissioner.

3. Green Channel facility

Some major importers have been given the green channel clearance facility. It means clearance of goods is done without routine examination of the goods. They have to make a declaration in the declaration form at the time of filing of bill of entry. The appraisement is done as per normal procedure except that there would be no physical examination of the goods.

4. Payment of Duty

Import duty may be paid in the designated banks or through TR-6 challans. Different Custom Houses have authorized different banks for payment of duty and is necessary to check the name of the bank and the branch before depositing the duty.

5. Prior Entry for Shipping Bill or Bill of Entry

For faster clearance of the goods, provision has been made in section 46 of the Act, to allow filing of bill of entry prior to arrival of goods. This bill of entry is valid if vessel/aircraft carrying the goods arrive within 30 days from the date of presentation of bill of entry.

6. Specialized Schemes

Import of goods under specialized scheme such as DEEC and EOU etc is required to execute bonds with the custom authorities. In case failure of bond, importer is required to pay the duty livable on those goods. The amount of bond would be equal to the amount of duty livable on the imported goods. The bank guarantee is also required along with the bond. However, the amount of bank guarantee depends upon the status of the importer like Super Star Trading House/Trading House etc.

7. Bill of Entry for Bond / Warehousing

A separate form of bill of entry is used for clearance of goods for warehousing. Assessment of this bill of entry is done in the same manner as the normal bill of entry and then the duty payable is determined.

OTHER PROCEDURES

1. Conveyances to call only at Notified Customs Ports / Airports

Customs Act, 1962, envisages that only places notified by the Government shall be Customs ports or Customs airports for the unloading of imported goods and loading of export goods. At each such customs ports, the Commissioner of Customs is empowered to approve proper places for the unloading and loading of goods and he also specifies the

limits of any Customs area. The law further provides that the person in charge of the vessel or an aircraft shall not call or land at any place other than the Customs port/airport, except in cases of emergencies.

2. Power to board conveyance, to question and to demand documents

Section 37 empowers the proper officer to board any conveyance carrying imported goods or export goods. He may remain on board as long as he decides to remain. The proper officer may question the person in charge of the vessel or aircraft. He may demand production of documents and also ask questions, to be answered by such person. The person in charge of the conveyance is bound to comply with these requirements [Section 38].

3. Delivery of Import Manifest

The Master / Agent of the vessel or an aircraft has to deliver an import manifest (an import report in case of a vehicle), within 24 hours after arrival in the case of a vessel and 12 hours after arrival in the case of an aircraft or a vehicle in the prescribed form. The time limit for filing the manifest is extendable on showing sufficient cause. In the case of a vessel or an aircraft, a manifest may also be filed even before arrival of the vessel or aircraft (known as Prior Entry Manifest). In the case of vessels, for administrative convenience, such advance manifests are accepted on any day within 14 days before the expected arrival of the vessel.

Obtain Delivery Order from concerned Airline(s) / Break Bulk Agent
Location: Respective Airline / Break Bulk Agent office

↓

Submit Bill of Entry along with Delivery Order, Letter of Authority , Airway Bill, Invoice & Import License to Customs for noting
Location: Customs Service Centre

↓

Assessment / Payment of duty and issue of Examination Order by Customs Appraiser, Air Cargo Complex
Location: Custom Offices / Air Cargo Terminal (ACT)

↓

Obtain Location Slip of Cargo from AAI
Location: Import Public Dealing Counter, Air Cargo Terminal (ACT)

↓

Obtain forwarding of packages for Customs examination from AAI
Location: Import Examination Area,Air Cargo Terminal (ACT)

↓

Complete custom examination by Customs Examiners
Location: Customs Examination Hall,Air Cargo Terminal (ACT)

↓

Obtain Out-of-Charge from concerned Customs Officer
Location: Customs Examination Hall,Air Cargo Terminal (ACT)

↓

Obtain AAI dues Challan
Location: Import Public Dealing Counter, Air Cargo Terminal (ACT)

↓

Payment of AAI dues in Bank
Location: Bank Counter, Air Cargo Terminal (ACT)

↓

Obtain Gate Pass against proof of charges paid
Location: Import Public Dealing Counter, Air Cargo Terminal (ACT)

↓

Present AAI Gate Pass at Bonded (Pre Delivery) Gate for delivery of pkgs.
Location: Import wing shed, Air Cargo Terminal (ACT)

↓

Present AAI Gate Pass at Delivery Gate to AAI staff duly endorsed by Customs and obtain delivery
Location: Delivery Gate – Import, Air Cargo Terminal (ACT)

If the vessel does not arrive within the stipulated time of 14 days or such extended time as may be granted by the Assistant Commissioner (Imports), the manifest accepted provisionally is cancelled and the fact circulated through public notices. All the Bills of Entry filed against the cancelled manifest, become void. The importers have to return those Bills of Entry to the Import Department and to claim refund of duty, if paid on any such Bills of Entry. If the same vessel enters the port after the cancellation of the original manifests, it will be treated as a fresh entry and a fresh manifest is insisted upon.

4. General conditions

A person filing declarations under this section has to declare the truthfulness of contents. This declaration has legal consequences, which bind the carrier. [Section 30(2)].

5. Amendments

If for any reason, the carrier desires to amend or supplement the IGM, it will be permitted by the proper officer on payment of prescribed fees, if he is satisfied that there is no fraudulent intention behind the move. [Section 30(3)].

6. Penal Liability

Any mis-declaration in this document will attract the penal provisions of Section 111(f) and Section 112.

a) Excision from IGMs of items originally manifested:

b) Excision from IGMs of items originally manifested are permitted only:

- On application in writing from the ship's Agents;
- On production of the documentary evidence of short shipment of goods;
- On payment of a fee as prescribed.

c) Excisions or amendments of items in the Import Manifest involving reduction in number of Packages:

An application from the steamer agents for excisions or amendments in the Import Manifest involving reduction in the number of packages or quantity or weight thereof, should be submitted with the connected documentary evidence. Such excisions or amendments will only be allowed; if after due investigation, it is proved that the excess quantity was originally shown in the import manifest as a result of an error. In the absence of such proof, the application will be kept over for being dealt with by the Manifest Clearance Section at the time of closure of the ship's file.

d) Applications for the excision or amendments of items for which Bills of Entry have been noted will be dealt with by the Manifest Clearance Section if made two months after the arrival of the vessel.

Matters such as the number of copies of manifests to be filed, nature of forms, manner of declaring cargo etc. are governed by the following Regulations:

- Import Report (Form) Regulation, 1976;

- Import Manifest (Aircraft) Regulation, 1976; and
- Import Manifest (Vessels) Regulation, 1971;

Generally speaking, the above Regulations stipulate declaring separately cargo to be landed, unaccompanied Baggage, goods to be transported and same bottom or retention cargo. Separate declarations are also to be filed in respect of dangerous/prohibited/ sensitive goods such as Arms and Ammunitions, Narcotics, Gold etc. The prime condition in the Regulations is that the manifest shall cover all the goods carried in the conveyance. In respect of a vessel, an import manifest shall, in addition, consist of an application for entry inwards.

7. Entry Inwards

The Master of the vessel is not to permit the unloading of any imported goods until an order has been given by the proper officer granting Entry Inwards of such vessel. Normally, Entry Inwards is granted only after the import manifest has been delivered. This entry inward date is crucial for determining the rate of duty, as provided in section 15 of the Customs Act, 1962. Unloading of certain items like accompanied baggage, mail bags, animals, perishables and hazardous goods are exempted from this stipulation.

8. Enclosures to Import General Manifest

The amendment made in 1995 (w.e.f. 1-7-1995) introduces a new form for obtaining entry inwards. The forms are designed according to IMO-FAL Convention. The forms have to be filed in prescribed sizes only. Host of enclosures are sought along with these forms. This practice has its origin in other statutes such as Merchant Shipping Act, 1880. However, keeping the said convention in view, Board has issued instructions dispensing with submission of various documents. The following declarations have, however, to be filed along-with IGM:

a) Deck Cargo Declaration / Certificate.
b) Last port clearance copy.
c) Amendment application (when relevant).
d) Income Tax Certificate in case of Export Cargo.
e) Nil export cargo certificate.
f) Port Trust "No Demand" certificate.
g) Immigration certificate.
h) Application for sign on/sign off of crew (when relevant).
i) Application for crew baggage chceking when they sign on (When relevant).

9. Procedure for filing IGM at Custom Houses operating EDI service centres

a) IGM by Shipping lines

The shipping line/steamer agent needs to submit the manifest in prescribed form at the Service Centre. The shipping lines are required to submit the electronic version of the Import General manifest in floppies, containing all the details and particulars. It is to be ensured by the Shipping Lines that all the particulars and details of the Import general manifest submitted either manually or through floppies are correct. The shipping agents who do not have arrangement for data in floppy, may approach the Service Center of Customs and get the data of IGM submitted in system. They are also to ensure that details of House Bill of Lading are also incorporated in the IGM in case of console cargo.

On arrival of the vessel, the shipping line needs to approach the Preventive officer for granting entry inwards. Before making the application, the shipping line has to make payment of the Light House dues.

In case the shipping line is filing an IGM after arrival of the vessel, the procedure as mentioned above for prior IGM, is to be followed except that the date of arrival of vessel is also indicated. After submission, the shipping line has to approach the proper officer for grant of entry inwards in the system.

b) IGM by Air

The airlines are required to file IGM in prescribed format. In case of Air Cargo Complexes having EDI, the IGMs may be filed through electronic mode. The IGMs to be submitted need to contain all details and particulars. In other words, the airlines would not only be furnishing the details of the Master Airway Bills but also the House Airway bills in the case of console cargo. The airlines are also to furnish the additional information, namely, the ULD Nos. for use by the custodians.

10. Filing of Stores List

When entering any port, all ships are required to furnish to the Commissioner of Customs, a list (or nil return) of ships stores intended for landing (excluding any consumable stores issued from any deified shops in. Retention on board of imported stores is governed by Import Store (Retention on board) Regulations, 1963. The consumable stores can remain on board without payment of import duties during the period the vessel/Aircraft remains foreign going. Otherwise, such consumable stores are to be kept under Customs seal. Even in respect of foreign going vessels, only the stores required for immediate use of the personnel may be left unsealed. Excessive stocks of stores such as liquor, tobacco, cigarettes, etc. are kept under Customs seal.

11. Unloading and Loading of Goods

Imported goods are not to be unloaded from the vessel until Entry Inwards is granted. No imported goods are to be unloaded unless they are specified in the import manifest/report for being unloaded at that Customs station. No imported goods shall be unloaded at any place other than the places provided for such unloading. Further,

imported goods shall not be unloaded from any conveyance except under the supervision of the proper officer. Similarly, for unloading imported goods on any Sunday or on any holiday, prior notice shall be given and fees prescribed in this regard shall be paid.

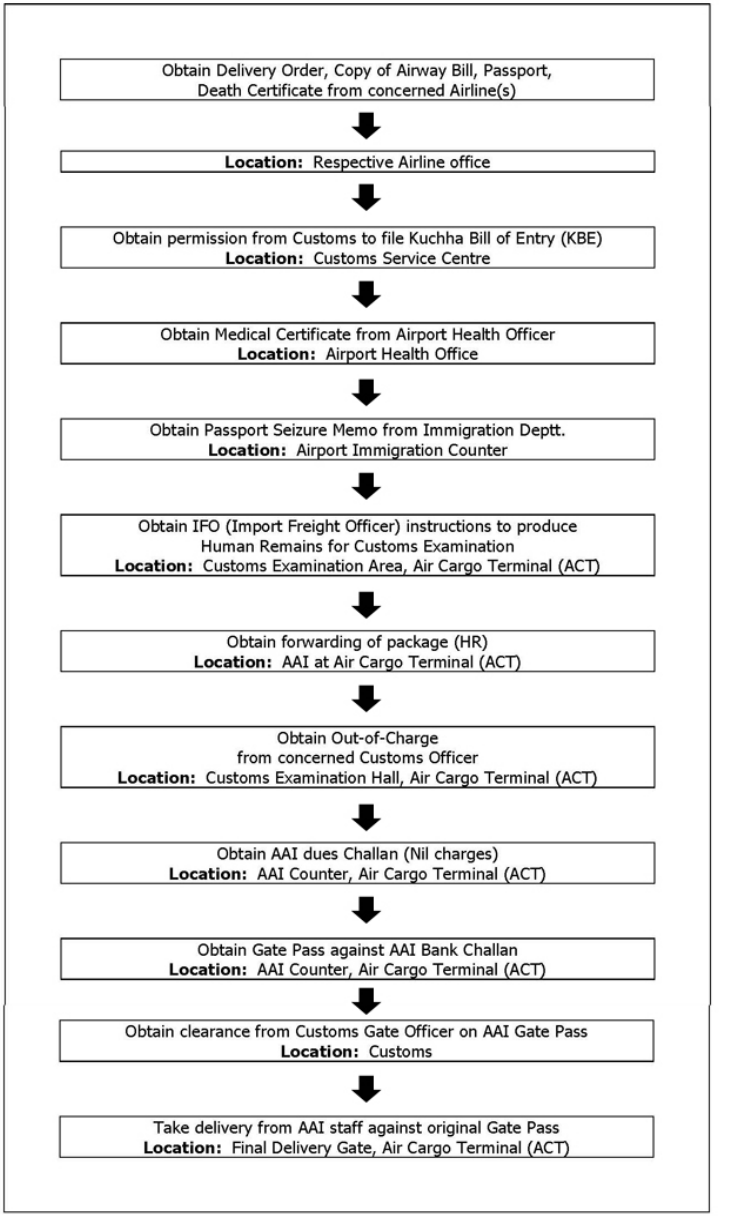
Under Section 115 and 116, the persons in charge of vessel or aircraft have other liabilities, which are important and noteworthy. Section 115 provides for confiscation of vessel or other conveyance under the following circumstances:

a) A conveyance within Indian waters or port or customs area which is adopted, fitted, modified or altered for concealing goods.
b) A conveyance from which goods are thrown overboard, staved or destroyed so as to prevent seizure by customs officers.
c) A conveyance which disobeys any order under Section 106 to stop or land, without sufficient cause.
d) A conveyance from which goods under drawback claim are unloaded without proper officer's permission.
e) A conveyance which has entered with goods, from which substantial portion of goods are missing and failure of the master to account therefor.

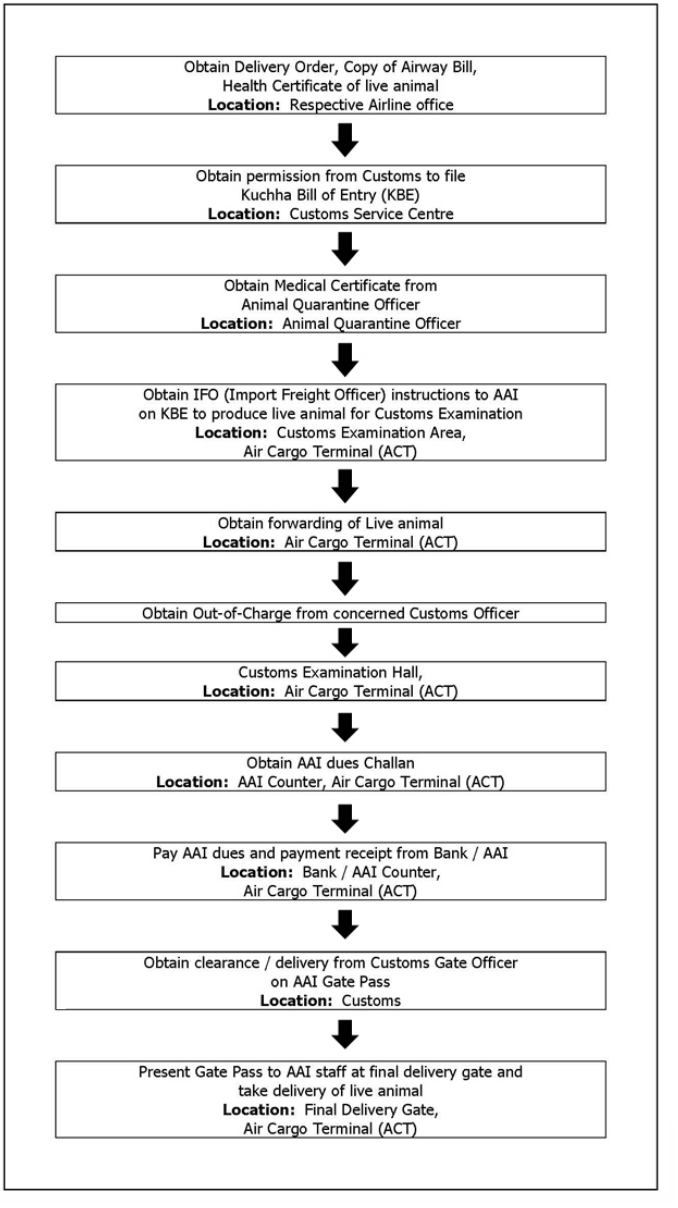
Any vessel when used as means of transport for smuggling of any goods or in the carriage of any smuggled goods, is liable to confiscation, unless the owner establishes that it was used without the knowledge or connivance of the owner, his agent and the person in-charge of the vessel. When any such conveyance is confiscated, an option to pay redemption fine has to be given to the owner of the conveyance. The upper limit for imposing the redemption fine is the market value of impugned goods.

Under Section 116, penalty may be imposed on the person in charge of vessel if there is failure to account for all goods loaded in the vessel for importation into India or transshipped under the provisions of Customs Act and these are not unloaded at the place of destination in India or if the quantity unloaded is short of the quantity to be unloaded at particular destination. Penalty may be waived if failure to unload or deficiency in unloading is accounted for to the satisfaction of competent officer. Thus, if there is any shortage which is not satisfactorily accounted for, the person in charge of the vessel will be liable to penalty, which may be twice the duty payable on the import goods not accounted for.

PROCEDURE FOR CLEARANCE OF HUMAN REMAINS IN IMPORT



PROCEDURE FOR CLEARANCE OF LIVE ANIMAL IN IMPORT



CARGO EXPORT PROCEDURE

1. Registration

The exporters have to obtain PAN based Business Identification Number (BIN) from the Directorate General of Foreign Trade prior to filing of shipping bill for clearance of export goods. Under the EDI System, PAN based BIN is received by the Customs System from the DGFT online. The exporters are also required to register authorized foreign exchange dealer code (through which export proceeds are expected to be realized) and open a current account in the designated bank for credit of any drawback incentive's required to register authorized foreign exchange dealer code (through which export proceeds are expected to be realized) and open a current account in the designated bank for credit of any drawback incentive.

Whenever a new Airline, Shipping Line, Steamer Agent, port or airport comes into operation, they are required to be registered into the Customs System. Whenever, electronic processing of shipping bill etc. is held up on account of non-registration of these entities, the same is to be brought to the notice of Assistant/Deputy Commissioner in-charge of EDI System for registering the new entity in the system.

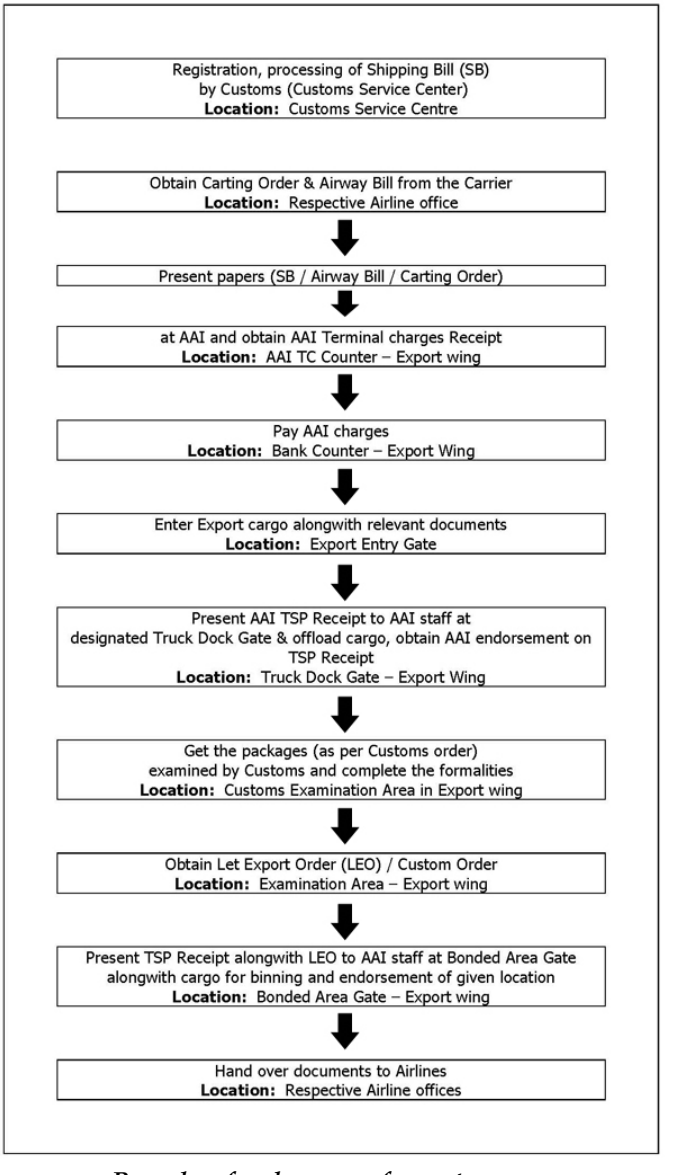
All the exporters intending to export under the export promotion scheme need to get their licenses/DEEC book etc. registered at the Customs Station. For such registration, original documents are required.

2. Processing of Shipping Bill - Non-EDI

Under manual system, shipping bills or, as the case may be, bills of export are required to be filed in format as prescribed in the Shipping Bill and Bill of Export (Form) regulations, 1991. The bills of export are being used if clearance of export goods is taken at the Land Customs Stations. Different forms of shipping bill/bill of export have been prescribed for export of duty-free goods, export of dutiable goods and export under drawback etc.

Shipping Bills are required to be filed along with all original documents such as invoice, AR-4, packing list etc. The assessing officer in the Export Department checks the value of the goods, classification under Drawback schedule in case of Drawback Shipping Bills, rate of duty/Cess where applicable, exportability of goods under EXIM policy and other laws in force. The DEEC/DEPB Shipping bills are processed in the DEEC group. In case of DEEC Shipping bills, the assessing officer verifies that the description of the goods declared in the shipping bill and invoice match with the description of the resultant product as given in the DEEC book. If the assessing officer has any doubts regarding value, description of goods, he may call for samples of the goods from the docks. He may also call for any other information required by him for processing of shipping bill. He may

assess the shipping bill after visual inspection of the sample or may send it for test and pass the shipping bill provisionally.



Procedure for clearance of export cargo

Once, the shipping bill is passed by the Export Department, the exporter or his agent present the goods to the shed appraiser (export) in docks for examination. The shed appraiser may mark the document to a Custom officer (usually an examiner) for examining the goods. The examination is carried out under the supervision of the shed appraiser (export). If the description and other particulars of the goods are found to be as declared, the shed appraiser gives export order, after which the exporter may contact the preventive superintendent for supervising the loading of goods on to the vessel.

In case the examining staff in the docks finds some discrepancy in the goods, they may mark the shipping bill back to export department/DEEC group with their observations as well as sample of goods, if needed. The export department re-considers the case and decide whether export can be allowed, or amendment in description, value etc. is required before export and whether any other action is required to be taken under the Customs Act, 1962 for mis-declaration of description of value etc.

3. Processing of Shipping Bill - EDI

Under EDI System, declarations in prescribed format are to be filed through the Service Centers of Customs. A checklist is generated for verification of data by the exporter/CHA. After verification, the data is submitted to the System by the Center operator and the System generates a Shipping Bill Number, which is endorsed on the printed checklist and returned to the exporter/CHA. For export items which are subject to export Cess, the TR-6 challans for Cess is printed and given by the Service Center to the exporter/CHA immediately after submission of shipping bill. The Cess can be paid on the strength of the challan at the designated bank. No copy of shipping bill is made available to exporter/CHA at this stage.

4. Octroi procedure, Quota Allocation and Other certification for Export Goods

The quota allocation label is required to be pasted on the export invoice. The allocation number of AEPC is to be entered in the system at the time of shipping bill entry. The quota certification of export invoice needs to be submitted to Customs along-with other original documents at the time of examination of the export cargo. For determining the validity date of the quota, the relevant date needs to be the date on which the full consignment is presented to the Customs for examination and duly recorded in the Computer System. In EDI System at Delhi Air cargo, the quota information is automatically verified from the AEPC/TEXPROCIL system.

Since the shipping bill is generated only after the 'let export order' is given by Customs, the exporter may make use of export invoice or such other document as required by the Octroi authorities for the purpose of Octroi exemption.

5. Arrival of Goods at Docks

The goods brought for the purpose of examination and subsequent 'let export' is allowed entry to the Dock on the strength of the checklist and other declarations filed by the exporter in the Service Center. The Port authorities have to endorse the quantity of goods actually received on the reverse of the Check List.

6. System Appraisal of Shipping Bills

In many cases the Shipping Bill is processed by the system on the basis of declarations made by the exporters without any human intervention. In other cases where the Shipping Bill is processed on screen by the Customs Officer, he may call for the samples, if required for confirming the declared value or for checking classification under the Drawback Schedule. He may also give any special instructions for examination of goods, if felt necessary.

7. Status of Shipping Bill

The exporter/CHA can check up with the query counter at the Service Center whether the Shipping Bill submitted by them in the system has been cleared or not, before the goods are brought into the Docks for examination and export. In case any query is raised, the same is required to be replied through the service center or in case of CHAs having EDI connectivity through their respective terminals. The Customs officer may pass the Shipping Bill after all the queries have been satisfactorily replied to.

8. Customs Examination of Export Cargo

After the receipt of the goods in the dock, the exporter/CHA may contact the Customs Officer designated for the purpose present the check list with the endorsement of Port Authority and other declarations as aforesaid along with all original documents such as, Invoice and Packing list, AR-4, etc. Customs Officer may verify the quantity of the goods actually received and enter into the system and thereafter mark the Electronic Shipping Bill and also hand over all original documents to the Dock Appraiser of the Dock who many assign a Customs Officer for the examination and intimate the officers? name and the packages to be examined, if any, on the check list and return it to the exporter or his agent.

The Customs Officer may inspect/examine the shipment along with the Dock Appraiser. The Customs Officer enters the examination report in the system. He then marks the Electronic Bill along with all original documents and check list to the Dock Appraiser. If the Dock Appraiser is satisfied that the particulars entered in the system conform to the description given in the original documents and as seen in the physical examination, he may proceed to allow "let export" for the shipment and inform the exporter or his agent.

9. Variation Between the Declaration & Physical Examination

The check list and the declaration along with all original documents is retained by the Appraiser concerned. In case of any variation between the declaration in the Shipping Bill and physical documents/examination report, the Appraiser may mark the Electronic Shipping Bill to the Assistant Commissioner/Deputy Commissioner of Customs (Exports). He may also forward the physical documents to Assistant Commissioner/Deputy Commissioner of Customs (Exports) and instruct the exporter or his agent to meet the Assistant Commissioner/Deputy Commissioner of Customs (Exports) for settlement of dispute. In case the exporter agrees with the views of the Department, the Shipping Bill needs to be processed accordingly. Where, however, the exporter disputes the view of the Department principles of natural justice is required to be followed before finalization of the issue.

10. Stuffing/Loading of Goods in Containers

The exporter or his agent should hand over the exporter copy of the shipping bill duly signed by the Appraiser permitting "Let Export" to the steamer agent who may then approach the proper officer (Preventive Officer) for allowing the shipment. In case of container cargo, the stuffing of container at Dock is dome under Preventive Supervision. Loading of both containerized and bulk cargo is done under Preventive Supervision. The Customs Preventive Superintendent (Docks) may enter the particulars of packages actually stuffed in to the container, the bottle seal number particulars of loading of cargo container on board into the system and endorse these details on the exporter copy of the shipping bill presented to him by the steamer agent.

If there is a difference in the quantity/number of packages stuffed in the containers/goods loaded on vessel the Superintendent (Docks) may put a remark on the shipping bill in the system and that shipping bill requires amendment or changed quantity. Such shipping bill also may not be taken up for the purpose of sanction of Drawback/DEEC logging, till the shipping bill is suitably amended for the changed quantity. The Customs Preventive Officer supervising the loading of container and general cargo in to the vessel may give "Shipped on Board" endorsement on the exporters copy of the shipping bill.

11. Drawl of Samples

Where the Appraiser Dock (export) orders for samples to be drawn and tested, the Customs Officer may proceed to draw two samples from the consignment and enter the particulars thereof along with details of the testing agency in the ICES/E system. There is no separate register for recording dates of samples drawn. Three copies of the test memo are prepared by the Customs Officer and are signed by the Customs Officer and

Appraising Officer on behalf of Customs and the exporter or his agent. The disposal of the three copies of the test memo are as follows: -

a) Original - to be sent along with the sample to the test agency.
b) Duplicate - Customs copy to be retained with the 2nd sample.
c) Triplicate - Exporter's copy.

The Assistant Commissioner/Deputy Commissioner if he considers necessary, may also order for sample to be drawn for purpose other than testing such as visual inspection and verification of description, market value inquiry, etc.

12. Export of Goods Under Claim for Drawback

After actual export of the goods, the Drawback claim is processed through EDI system by the officers of Drawback Branch on first come first served basis. There is no need for filing separate drawback claims. The status of the shipping bills and sanction of DBK claim can be ascertained from the query counter set up at the service center. If any query has been raised or deficiency noticed, the same is shown on the terminal. A print out of the query/deficiency may be obtained by the authorized person of the exporter from the service center. The exporters are required to reply to such queries through the service center. The claim will come in queue of the EDI system only after reply to queries/deficiencies are entered by the \Service Center

All the claims sanctioned on a particular day are enumerated in a scroll and transferred to the Bank through the system. The bank credits the drawback amount in the respective accounts of the exporters. Bank may send a fortnightly statement to the exporters of such credits made in their accounts.

The Steamer Agent/Shipping Line may transfer electronically the EGM to the Customs EDI system so that the physical export of the goods is confirmed, to enable the Customs to sanction the drawback claims.

13. Generation of Shipping Bills

After the "let export" order is given on the system by the Appraiser, the Shipping Bill is generated by the system in two copies i.e., one Customs copy, one exporter's copy (E.P. copy is generated after submission of EGM). After obtaining the print out the appraiser obtains the signatures of the Customs Officer on the examination report and the representative of the CHA on both copies of the shipping bill and examination report. The Appraiser thereafter signs & stamps both the copies of the shipping bill at the specified place.

The Appraiser also signs and stamps the original & duplicate copy of SDF. Customs copy of shipping bill and original copy of the SDF is retained along with the original declarations by the Appraiser and forwarded to Export Department of the

Custom House. He may return the exporter copy and the second copy of the SDF to the exporter or his agent.

As regards the AEPC quota and other certifications, these are retained along with the shipping bill in the dock after the shipping bill is generated by the system. At the time of examination, apart from checking that the goods are covered by the quota certifications, the details of the quota entered into the system needs to be checked.

14. Export General Manifest

All the shipping lines/agents need to furnish the Export General Manifests, Shipping Bill wise, to the Customs electronically within 7 days from the date of sailing of the vessel.

Apart from lodging the EGM electronically the shipping lines need to continue to file manual EGMs along with the exporter copy of the shipping bills as per the present practice in the export department. The manual EGMs need to be entered in the register at the Export Department and the Shipping lines may obtain acknowledgements indicating the date and time at which the EGMs were received by the Export Department.

The above is the general procedure for export under EDI Systems. However special procedures exist for specified schemes, details of which may be obtained from the Public Notice/Standing Orders issued by the respective Commissionerate's.

REVIEW QUESTIONS

1. Explain the air cargo rates.
2. Write the sample air cargo tariff.
3. Discuss the import cargo procedures.
4. Explain the procedure for clearness of import and export goods.
5. What are the export cargo procedures?

LOGISTICS TRANSPORT MANAGEMENT

Learning Objectives

After reading this chapter, you should be able to:

- Understand the scope of transportation.
- Know the relationship of transportation to other business.
- Understand the legal types of transportation.
- Know the transportation mode selection.
- Understand the transport regulations in India.
- Know the interstate movement of goods.

INTRODUCTION

From the beginning of history, human sensitivity has revealed an urge for mobility leading to a measure of Society's progress. The history of this mobility or transport is the history of civilization. For any country to develop with right momentum modern and efficient Transport as a basic infrastructure is a must. It has been seen throughout the history of any nation that a proper, extensive and efficient Road Transport has played a major role. 'Transporters' perform one of the most important activities, at every stage of advanced civilization. Where roads are considered as veins and arteries of a nation, passenger and goods transported are likened to blood in circulation.

Passenger Road Transport Service (PRTS) is an essential connected to the economic development. Transport is the essential convenience with which people not just connect but progress. Throughout history, people's progress has been sustained on the convenience, speed and safety of the modes of transport. Road transport occupies a primary place in to-day's world as it provides a reach unparallel by any other contemporary mode of transport. Transport contributes in Growth of industries whose product requires quick marketing. Perishable articles like fish and green vegetables are carried to various consumers quickly even in distant markets through transport. Transport helps in increase in the demand for goods. Through transport newer customers in newer places can be easily contacted and products can be introduced to them. Today markets have become national or international only because of transport.

SCOPE OF TRANSPORTATION

The two C's i.e. Capacity and Capability of transportation system, determines the boundaries of market system it serves (Wood and James, 1993). The availability in inexpensive, efficient and easily accessed transportation services activates several critical drivers of economic activity.

1. It provides opportunity for growth in market competition. The more nature and efficient the transportation system is in an economy, the more the opportunities it provides to various companies to compete on an equal footing by exploiting economies of scale.

2. Efficient transportation system allows deeper penetration into the market even if they are far from the point of production.
3. The wider the product distribution and greater the demand, the more producers can leverage economies of scale in production and channel transportation costs.
4. The more efficient and lower the transportation cost is, the lower would be the selling price. This is so because transportation cost born by the producers are included in the selling price of the product, thus a decrease in transportation cost, leads to increase in the delivery capabilities of the producers, thereby leading to marketplace advancement.

RELATIONSHIP OF TRANSPORTATION TO OTHER BUSINESS FUNCTIONS

Business ultimately aims at satisfying the customers for a profit. For this purpose it needs to efficiently manage various activities like traffic management, storing, purchasing and product pricing. But these functions are influenced by transportation activity in some way. Thus, a study of the relationship is necessary to manage/manipulate these business functions. The relationship between transportation and other business functions can be described as follows:

1. Traffic management and transportation

More and more businesses now a days are opting for traffic management department, which is responsible for appropriate transportation of products. A traffic manager has to perform the following responsibilities:

- Freight classification
- Rate negotiation
- Equipment scheduling
- Documentation
- Tracing and expediting
- Auditing
- Claims and fleet management

Transportation decisions in terms of vehicles and capacity have direct bearing on these functions.

2. Warehouse receiving and shipping

The effective scheduling of receiving and shipping of products is fundamental to the smooth flow of both warehousing and transportation. Poor scheduling can cause daily periods of expensive idleness. It may cause warehouse to accommodate uneven flow of material, requiring unnecessary material handling equipment and stagging areas.

3. Purchasing

Purchasing decisions have a direct impact on transportation. Purchasing influences

a) Containerization methods for inbound products
b) Cost of transport
c) Capacity of transport required

d) Methods of packaging and unitization so as to decrease damage while increasing carrier capacity utilization.

4. Customer service

The two factors which influence transport and customer service functions are time and cost. Normally, the shorter the order cycle time the more the customer satisfaction level would be.

5. Product pricing

Transportation function influences a firm's operational costs, which in turn has an impact on the selling price. Transportation costs can be as high as 50 per cent or more of the selling price of bulk commodities like coal and wood or it can be as less as 1 per cent for other products.

6. Distribution locations

Where to locate a new plant and where to locate a warehouse? are some of the most difficult decisions which face the manufacturer. Strategic location of plant and warehouse is necessary to reduce order cycle time. But the location is dependent upon the transportation facilities available. For example, if speedy transport is available to the market from the plant, then warehouse can be located near the manufacturing plant.

7. Inventory planning

The level of inventory to be maintained, the economic order quantity and safety stock levels to be maintained are dependent upon the transportation facilities available. For instance, 'just in time' inventory technique is successful, partly because of efficient transportation facilities.

The relationship between transportation and other business functions has elevated the transportation function to a position of strategic importance.

TRANSPORTATION MANAGEMENT

Effective day-to-day management of transportation activities is absolutely necessary for smooth functioning of the enterprise. The principle functions to be performed by transport management group include activities like

- Transport pricing
- Rate negotiation
- Routing and carrier selection
- Regulatory matters, including effect on economy operations and appearances before regulatory agencies
- Operations of co-transportation
- International transportation
- Co-operate with carriers in the development of technological changes that can enhance transportation productivity.
- Analysis of transportation cost and services
- Customer service and transportation selection

In technical terms these functions can be discussed as follows:

1. Transportation pricing

This is one of the most important functions of transport management i.e. securing but it is up to the transport management department to find the rate that is more advantageous to their company.

2. Rate negotiations

This activity entails initiation of requests to carriers, classification committees, or whoever concerned for adjustments in ratings and rates. The proposals are accompanied by justification for the proposal (decrease in rates). Sometimes appearances before regulatory bodies are also important to assure the best interest of the company.

3. Routing

This function of transportation management department involves the choice of the mode of transportation and the selection of carriers within the mode. In order to perform this function, a transport manager must have a thorough knowledge of the services offered by various carriers. He must determine the route through which the shipment should be sent through different carriers in order to reduce inventory cost, warehousing cost and transportation cost. But the primary aim should be delivering of right product, at right time and place in good condition.

4. Transportation operations

This function requires provision of equipment for handling transportation within the plant or warehouse or vice versa. Main options under these are either company owned equipment or leasing, or chartering of equipment. Main duties include ensuring efficient operation of equipment, suggesting improvements, scheduling the routes and supervising the cost of operations.

5. Diversion and re-consignment

It is sometimes necessary to change the routing of a shipment or its final destination after the shipment has been turned over to a carrier, the consignee remaining the same. This is called "Diversion". But if the consignee is changed, the activity is called "re-consignment". The transport manager, many a times, is required to perform these functions so as to ensure customer service, or to protect the interest of the company.

6. Risks and claims

Where the company uses self-owned means of transportation, it is necessary that adequate protection be provided for the risk that exists in the use of company owned equipment. Protection may be acquired through

- Self Insurance fund of the company
- Insurance companies

The responsibility for detecting loss and damage and then handling the claim procedure with the carrier falls in the jurisdiction of transport manager. Procedures are established to expedite handling of such claims.

Although the collection of claims is important, a claim prevention activity is also a part of the claim function. To effectively carry out this activity, a transport manager must do a periodic analysis, aimed at finding the causes of the claims and then taking remedial steps to reduce these causes, in turn improving services rendered to the customers.

7. Export and import

Handling of matters involving export or import shipments is generally the responsibility of the transport management department. Reservation of space, documentation, insurance are some of the details a department must handle.

8. Analysis of transportation cost and services

A continuous and careful evaluation of all the elements of transportation cost and services is necessary to establish a proper control mechanism. The range of analysis may relate to matters of managerial priority or to routine functions.

LEGAL TYPES OF TRANSPORTATION

A transportation services is a set of performance characteristics purchased at a given price. The services are limitless. The six modes of transport (water, rail, road, air, pipeline and ropeways) may be used in combination or exclusively. The user selects the services or combination of services that provide the best balance between the quality of service offered and the cost of that service. Practically the choice is reduced to a few reasonable possibilities by the circumstances of shipment. Before we discuss modes of transport, it is necessary to discuss various legal types by which transportation may be owned and organized.

Four types of transportation

1. Common carriers

A common carrier is the one, which provides transport service to a general public according to a fixed rate.

2. Contract carriers

These are similar to common carriers but instead of servicing to the general public, they provide transport services for a negotiated price to selected customers, defined by contract agreement. Normally, contract carrier rates are less than common carrier rates, in that

(a) They select customer who wish to transport products that optimize their capacities and capabilities.

(b) The customers provide carriers with specific product transit and delivery schedules while the carrier can plan better the budget for equipment resource.

Contract carriers are more popular than other forms as they allow shippers greater control over the transportation resources.

3. Exempt carriers

This type of 'for hire' carrier company differs from the above two in that it is not regulated with respect to routes, areas served and rates. It enjoys exempt status based on product transported and the nature of operation. The exempt status may be based on the type of product like agricultural products, poultry or livestock, or on the area of operation - within a commercial and trade zone.

4. Private carriers

This type of carrier is distinguished by the fact that it is wholly owned or leased by the firm, and is incidental to a company's main line of business. Wholly owned transport provides great deal of flexibility and economy.

In addition to these types of carriers, new forms of transport services have grown dramatically in the last ten years. These companies purchase transport services from primary carriers, which they in turn retail to their customers in small-customized packages. These transportation agencies can be defined as

a) Freight forwarders

It provides general public with transportation services. In the ordinary course of business, freight forwarders provide/assemble and consolidate shipments at the point of origin. They also provide for break-up of bulk and distribution of shipment at destination. They key responsibility of freight forwarders is consolidation of traffic. This has become more significant as, in the first place, railways move to rake load movement and secondly, containerization gathers momentum.

Freight forwarders are divided into two types, such as

i) Surface freight forwarders which utilize surface common carriers for line haul movement.

ii) Air freight forwarders, which utilize certified air carriers.

Freight forwarder assumes responsibility of transportation of shipment from point of receipt to point of destination.

b) Parcel post

Companies engaged in this type of transport focus o rapid shipment of small goods designed for general public use. Parcel post use all types of modes of transport. Examples of parcel post are FedEx and UPD.

c) Shippers' association

These are co-operative organizations operating on non-profit basis. Realization of lower shipping costs is the primary motive behind members belonging to the association. These associations perform services similar to freight forwarders.

d) Brokers

Transport brokers are agents that bring shippers and carriers together by providing timely information about rates, routes and capabilities. They may arrange transportation but assume no liability for it.

TRANSPORT MODE SELECTION

The goal of mode selection process is to achieve the lowest possible transit cost and maximum service. In selecting carrier, transportation management must be able to address specific shipping issues. These issues are related to choice of mode that best gives its service for the cost, any special needs and instructions of the customer, the rates and services offered by competing carries and likelihood loss, damage or delivery delay.

According to McKinnon, there are 4 general principles for mode of transport selection. In the first place, while comparing available alternatives, transport managers must be able to match

the shipment characteristics with the most appropriate mode of transport. Moreover, a carrier should be selected on the basis of documented performance regarding speed and reliability.

Second principle of carrier selection is for the firm to continually review their carrier selection process, with an eye towards cost cutting and service improvement.

Developing selection criteria constitutes the third principle of carrier selection. Selection criteria can be traffic related, shipper related or service related as depicted in Table.

Table Selection Criteria for Carrier Selection

S.No	Traffic Related	Shipper Related	Service Related
1	Length of haul	Size of firm	Speed (transit time)
2	Consignment weight	Investment priorities	Reliability
3	Dimensions	Marketing strategy	Cost
4	Value	Spatial structure of production and distribution systems	Customer relationship
5	Regularity of shipment	Availability of rail sidings	Availability of special vehicles / equipment
6	Perishability	Stockholding policy	Monitoring of goods
7	Type of packing	Management structure	Unitization
8	Special handling characteristics	System of carrier evaluation	Ancillary service

The last mentioned principle recommends formulation of rigorous selection procedure. In practice, the process for selecting the best mode or combination thereof must be the one that meets the needs of companies, customers and objective. There are essentially four potential selection methods.

METHODS OF SELECTION

Selection procedure for the choice of transport mode could vary from very simple one to complex decisions, where every cost incurred is accounted until an optimum solution is achieved. There are four potential selection methods:

1. Judgement

The transport manager identifies the important factors effecting the transport problems and transport mode from a shortlist of alternatives considered available, in order that the most important feature of the transport requirements is satisfied. The shortcomings of this particular selection method are numerous, some of which are:

a) Factors other than transport matters are ignored.
b) Transport is considered as service rather than a part of the distribution system.
c) A complete list of alternatives transport methods may not be considered.
d) The costs are not important as the decision is made on operational ability.

2. Cost trade off

Under this, the impact of transport is calculated in relation to its immediate activities and the total cost, so that the distribution system can be optimized. This approach acknowledges the existence of trade-offs within the numerous alternative approaches in an attempt to assess the situation to minimize total costs.

3. Distribution models

These identify and explain the inter-relationships between the components of the distribution system at various levels of daily/weekly/monthly demand.

These models could be built to examine the impact of alternative transport models and methods, as wither the demand changes or the components in the system change. The models are logical and mathematical, and attempt to simulate operating practice and conditions. The major shortfall of this model building lies in the ability of the operations research analyst to accurately calculate the operational algorithm as otherwise, errors will be compounded.

4. Systematic selection

In this analysis the most important factors effecting the selection of mode of transport are first identified. Based on these factors the trade-offs between different modes of transportation is undertaken to minimize the total distribution cost. Finally, there is the operations analysis to experiment with a planned system to validate the calculations and check that the interrelationships of transport with other areas have been calculated correctly.

TRANSPORTATION INFRASTRUCTURE

INTRODUCTION

Transportation infrastructure consists of the right-of-way, vehicles, and carrier organizations that offer transportation services on a for-hire or internal basis. The nature of the infrastructure also determines a variety of economic and legal characteristics for each mode or multimodal system. A mode identifies the basic transportation method or form. In

The five basic transportation modes are rail, highway, water, pipeline, and air. The relative importance of each mode can be measured in tens of system mileage, traffic volume, revenue, and the nature of traffic composition. Improved infrastructure has played a significant role in the development of transportation and the resulting growth of trade. The role of the railroads and canals in the economic development is well-documented.

Although there is a large private-sector involvement in transportation, government plays a large regulatory and developmental role. The Central Government has ministries to handle civil aviation, railroads, and surface transportation. Counterpart agencies are found at the state and union territory level. Critical to improving the entire transportation sector in the late 1990s is the ability of the sector to adjust to the Central Government's national reform initiatives, including privatization, deregulation, and reduced subsidies. The sector must also adjust to foreign trade expansion,

demographic pressures and increasing urbanization, technological change and obsolescence, energy availability, and environmental and public safety concerns.

INDIAN TRANSPORT INFRASTRUCTURE

1. Roads and Highways

The country has a road network sparing 41.09 million km. and ranks among the largest in the world. Road infrastructure is of prime importance for the growth of the economy, since around 60% of freight and 85% of passenger traffic moves by road in India. The National Highways only constitute around 1.7% of the road network, but carry 40% of the total road traffic. Yet only 24% of the country's national highways are four-lane and meet the required standards. Total investments in roads and highways sector stood at about Rs.3.4 trillion between 2007-08 and 2011-12, of which state roads accounted for a major share. Government focus has led to greater private sector participation and faster awarding of projects. The highest number of projects was awarded in 2011-12. Of the 7,406km of NH projects awarded, 6,498km projects were awarded by the Nation Highways Authority of India (NHAI), while about 908km were awarded by the Ministry of Road Transport and Highways (MORTH).

2. Railways

India has the world's fourth-largest rail network in the world after the U.S., Russia and China, and the second-largest network under a single management. The growth in rail infrastructure has not matched the demand' with many projects running behind schedule, leading to time and cost overruns of more than 100%. Only 1,750km of new lines was added from 2006-to 2011, as compared to 14,000 in China.

According to Vision 2020, the railways shall on an average to execute "works" covering around 6,300km per annum including new lines, doubling and electrification. In the first year of its initiation, Vision 2020 completed 3,290km of rail tracks. Procurement of rolling stock is also lagging behind, since the railways have. Only procured 20,300 units, which falls far short of the annual target of 34,000 units.

3. Ports

The port sector is a crucial cog in the infrastructure wheel, since it handles 95% of India's external trade by volume and 70% by value, and therefore, directly impacts the country's positioning in global trade, and thereby, its economy as a whole. India has a long coastline of 7,500km, which has 13 major ports and 60 operational non-major ones.

Port traffic has increased at CAGR of 8.1% to reach 884.6 million tones with an average utilization of 90%, as compared to the international average of 70%. The main issues faced by ports include the level of containerization, custom procedures and insufficient connectivity to their hinterlands. In 2011-12, Indian port traffic is estimated to have grown by merely 2 per cent due to a sharp fall in iron ore traffic.

Despite this, the outlook for port traffic remains stable, with an expected 7-9 per cent growth from 2011-12 to 2016-17. Growth in port traffic will be largely driven by strong growth in coal imports and healthy container traffic.

4. Airport

Overall, the total passenger traffic handled at Indian airports in 2011-12 increased to 162 million, rising by 13.3 percent due to rising demand from business and leisure travelers. During 2006-07 to 2011-12, traffic cat Indian airports grew at a CAGR of about 10.9 per cent. The entry of low-cost carriers, coupled with an increase in consumers disposable income resulted in a substantial 11.4 per cent CAGR in domestic passenger traffic during 2006-07 to 2011-12. Domestic passenger traffic at airports grew by 15.4 percent to 121.3 million passengers in 2011- 12. During 2006-07 to 2011-12, international air freight traffic increased to 1,468 thousand tones from 1,021 thousand tones, at a CAGR of 7.5 percent. Pharmaceuticals, auto and electrical equipments, IT hardware, sent and perishable goods were the primary contributors to this growth. However, in 2011-12, it declined by 1.9 per cent to 1,468 thousand tones on account of a slowdown in both the global and domestic economies.

GOVERNMENT STRATEGIES TO IMPROVE TRANSPORT INFRASTRUCTURE IN INDIA

India's Eleventh Five Year Plan identifies various deficits in transport sector which include inadequate roads/highways, old technology, saturated routes and Slow speed on railways, inadequate berths and rail/road connectivity at ports and inadequate runways, aircraft handling capacity, parking space and terminal building at airports. Government aims to modernize, expand, and integrate the country’s transport services. It also seeks to mobilize resources for this purpose and to gradually shift the role of government from that of a producer to an enabler. In recent years, the Government has made substantial efforts to tackle the sector shortcomings and to recon its transport institutions. These include:

1. Increasing public funding for transportation in its Five Year Plans.
2. Launching the ambitious National Highway Development Program which has seven phases and is expected to be completed by 2012. It includes improved connectivity between Delhi, Mumbai, Chennai, and Kolkata, popularly called the Golden Quadrilateral, in the first phase, North-South and East-West corridors in phase two, four lining of more than l2, 000km in phase three, two lining of 20,000km and six lining of 6,500km respectively in phase four and five, development of 1,000km of expressway in phase six and other important highway projects in phase seven. Total expected investment is INR 2.2 trillion.
3. Accelerated Road Development Program for the North East Region to provide road connectivity to all state capitals and district head quarters in the region.
4. Financing the development and maintenance of roads by creating a Central Road Fund (CRF) through an earmarked tax on diesel and petrol.
5. Operational sing the National Highway Authority of India (NHAI) to act as an infrastructure procurer and not just provider.
6. Improving rural access by launching the ‘Pradhan Mantzri Gram Sadak Yojana’ (Prime Minister’s Rural Roads Program).

7. Reducing the congestion on rail corridors along the highly trafficked Golden Quadrilateral and improving port connectivity by launching the National Rail Vikas Yojana (National Railway Development Program).
8. The development of two Dedicated Freight Corridors from Mumbai to Delhi and Ludhiana to Dankuni.

TRANSPORT REGULATIONS

Government transportation regulations can be classified into categories such as

I. Economic regulations of business practices
II. Safety and social regulations.

I Economic Regulations of Business Practices

Control of business practices is one of the oldest forms of government regulation. To provide dependable transportation service and to foster economic development, both central and state governments have actively utilized this tool. Since independence, Government of India is involved in an effort to make transportation equally available to all users without discrimination. To encourage economical and widespread transportation supply, the government of India has invested heavily into public infrastructure over the years, such as highways, airports and deepwater ports.

The government also invests into transport infrastructure, both at central and state level. At state levels, the states are supposed to take independent decisions to excel in the industry. For instance, Haryana is known as the best state road transport undertaking in the country. It has made a provision of Rs 41.85 Crore for acquisition of road transport fleet and Rs 100 lakh for modernization of workshops for the year 2002-03.These are just a small part of the efforts undertaken by the states to promote transport.

A network of also supports the Indian government, in its quest to improve transportation facilities in India, privately owned for hire carriers. Thus, a regulatory system for privately owned carriers also exists. Since transportation has a complimentary role on economic growth, the government tries to give special protection to the carriers to ensure service availability and stability. In some countries, the government, at times, may practice some authority to regulate rate-related practices like ratemaking, rate-changes, rate-subsidies and actual rates. But in Indi, road, air and water transport, set and negotiate terms of tariff or rate privately through unions.

The unions have played a major role in the poor state of small-scale industries. The executives of the unions fix the rates charged by these unions arbitrarily. The base of fixation rates is the paying capacity of industrialist. Secondly, rates are charged for both sides, up and down sides. However, state governments are trying to reduce the hold of truck unions on industry. Punjab has recently taken a step in this direction by abolishing truck unions.

II Safety and social regulations:

This aspect of government regulation refers to government trying to set safety standards for hire/trade carriers, so as to protect the interests of the public. Department of transportation has taken an active role regarding transporting and handling of sensitive, hazardous material, vehicle

safety, vehicle fitness and so on, so as to protect the environment. Various acts like the Motor Vehicles, 1988 were imposed to maintain social safety.

Motor Vehicle Act, 1988: This Act extends to the whole of Indi. It regulates conditions and formalities for issuance of driving license, procedure for registration of motor vehicles, ways in which government can control transport services, for example, through permits limits and restriction on operations and so forth. It also emphasized the need for classification of goods into dangerous and hazardous category, and sets standards of packaging these products as to ensure public safety. Government of India is thus tying to regulate the transport industry by formulating rules and regulations, so as to make it safe industry for the society in general.

INTRA AND INTERSTATE TRANSPORT OF GOODS

Irrespective of the goods to be transported interstate, or intrastate, the vehicle has to acquire permit under the Motor Vehicles Act, 1988.

Application for goods carriage permit (As per Motor Vehicles Act, 1988 and Amendments (section 77).

An application for a permit to use motor vehicle for the carriage of goods on hire or reward, or for the carriage of goods as a part of trade or business carried on by the applicant shall, as far as may be, contain the particulars namely,

1. The area or the route or routes to which the application relates;
2. The types and capacity of the vehicle;
3. The nature of the goods it is proposed to carry;
4. The arrangements intended to be made for housing, maintenance and repair and for the storage and safe custody of goods;
5. Such particulars as the Regional Transport Authority (RTA) may require with respect to any business as the carrier of goods for hire or reward carried on by the applicant at time before the making of the application, and of the rates charged by the applicant;
6. Particulars of any agreement or arrangement, affecting in any material respect the provision within the region of the RTA, of facilities for the transport of goods on hire or reward entered into by the applicant with any other person, by whom such facilities are provided whether within or without the region; and
7. Any other particulars, which may be prescribed.

Consideration of application for goods carriage permit (Section 78)

An RTA shall, in considering an application for a goods carriage permit, have regard to the following matters:

1. The nature of the goods to be carried, with special reference to their dangerous or hazardous nature to the human life;
2. The nature of chemicals or explosive to be carried, with special reference to the safety of human life.

Grant of goods carriage permit (Section 79)

1. An RTA may, on an application made to it, grant a goods carriage permit to be valid throughout the state or in accordance with the application or with such modifications as it deems fit or refuse to grant such a permit, PROVIDED no such permit shall be granted in respect of any area or route not specified in the application;
2. The RTA, if it decides to grant a goods carriage permit, may grant the permit and may subject to any rules that may be made under this Act, attach to the permit any one or other of the conditions namely;
    a. That the vehicle shall be used only in a specified area or on a specified route or routes
    b. That the gross vehicle weight of any vehicle used shall not exceed a specified maximum
    c. That goods of a specified nature shall not be carried
    d. That specified arrangement shall be made for the housing, maintenance and repair of the vehicle and the storage and sage custody of the goods carried
    e. That the holder of the permit shall furnish to the RTA such periodical returns, statistics and other information as the state government may, from time to time, prescribe
    f. That the RTA may, after giving notice of not less than one month:
        i. Vary conditions of the permit, or
        ii. Attach to the permit further conditions;
    g. That the conditions of the permit shall not be departed from, save with other conditions which may be prescribed.
3. The conditions referred to in sub-section (2) may include conditions relating to the packaging and carriage of goods of dangerous or hazardous to human life.

INTRASTATE MOVEMENT OF GOODS

This section provides a brief overview of the nature of intrastate regulations and its impact on transportation performance. In India, we have a regulated intrastate transport. These regulations, try to manage the transport system, for a shipment, within a state. Every state is committed to providing adequate, well co-ordinate, economical and efficient transport services to the people of the state.

The regulations may be based on vehicle size, market entry, rates and routes. For instance , in India, state governments try to regulate the documentation system by dividing the vehicles into categories. For example, for the transportation of goods within Haryana, the state government has divided goods vehicles into following categories:

1. Heavy weight vehicle

This refers to any goods carriage, the gross vehicle weight of which, or a tractor or a road-roller, the unladen weight of either of which exceeds 12,000 kilograms.

2. Light weight vehicle

Any goods carriage the gross vehicle weight of which, or a tractor or a road roller the unladen weight of either of which is less than 7,500 kilograms come under this category.

3. Medium weight vehicle

Any goods carriage the gross vehicle weight of which, or a tractor or a road-roller the unladen weight of either of which is between 7,500-12,000 kilograms are considered medium weight vehicles.

These vehicles, in order to transport goods within a state need to follow the procedures such as:

1. Apply to the district transport office for permit with the vehicle registration card. (Example Form No. 26 provided in the Appendix)
2. Clarify the purpose of transport vehicle
   For hire/reward
   For trade/business
3. The fees for the two categories are the same but the form to be filled is different.
4. Deposit form and fee with the District Transport Office (DTO)
5. DTO issues permit under Form 32/33

Thus, intrastate transport also requires a permit. The permit acquired for this purpose is generally referred to as temporary permit.

Temporary permit (Section 87)

1. The RTA and the STA may, without following the procedure laid down in Section 80, grant permits, to be effective for a limited period which shall not exceed four months, to authorize the use of transport vehicles temporarily

a) For conveyance of passengers on special occasions such as to and from fairs and religious gatherings, or
b) For the purpose of seasonal business, or
c) To meet a particular temporary need, or
d) Pending decision on an application for the renewal of a permit, and may attach to any such permit such conditions as it may think fit.

Provided that the RTA, or the STA, in case of goods carriages under circumstances of exceptional nature, and for reasons to be recorded in writing, grant a permit for a period exceeding four months, but not exceeding one year.

2. Notwithstanding anything contained in sub-section (1), a temporary permit may be granted there under, in respect of any route or area, where

a) No permit could be issued in respect of that route or area by reason of a court order or other competent authority restraining the issue of the same, for a period not exceeding the period for which the issue of the permit has been restrained; or
b) As a result of suspension by a court or other competent authority of the permit of any vehicle in respect of that route or area, there is no transport vehicles of the class with a valid permit in respect of that route or area, or there is no adequate number of such

vehicles in respect of that route or area, for a period not exceeding the period of such suspension

Provided the number of transport vehicles in respect of which temporary permits are so granted shall not exceed the number of vehicles in respect of which the issue of the permits have been restrained or, as the case may be, the permit has been suspended.

INTERSTATE MOVEMENT OF GOODS

The purpose of interstate regulation is to scrutinize the activities of for hire or trade carriers in the public interest. Interstate transportation of goods refers to movement of goods across the boundary of a state.

For transportation of goods within a state or between two states, a transporter requires permit. The permit can be national permit or temporary permit. National permit is used for interstate transport only, while temporary permit is for intrastate transport, but in certain cases, can be used for interstate transport as well.

Validation of temporary permit for use outside region in which granted (Section 88)

1. Except s may be otherwise prescribed, a permit granted by the RTA of any one region shall not be valid in any other region, unless the STA of that other state or the RTA concerned has countersigned the permit.

PROVIDED that a goods carriage permit, granted by the RTA of any one region for any area in any other region or regions within the same state, shall be valid in that area without the countersignature of the RTA of the other region or each of the other regions concerned:

PROVIDED FURTHER that where both starting point and the terminal point of a route are situated within the same state, but a part of that route lies in any other state, and the length of such part does not exceed sixteen kilometer, the permit shall be valid in the other state not withstanding that such a permit has not been countersigned by the STA or by the RTA of that other state

PROVIDED ALSO that:

(a) Where a motor vehicle covered by a permit granted in one state is to be used for the purpose of defense in any other state, such vehicle shall display a certificate, in such form, and issued by such authority, as the central government may, by notification in the official gazette specify, to the effect that the vehicle shall be used for the period specified there is exclusively for the purposes of defence; and

(b) Any such permit shall be valid in that the STA or RTA of that other state has not countersigned other state notwithstanding that such a permit.

2. Notwithstanding anything contained in the sub-section (1), a permit granted or countersigned by an STA shall be valid in the whole states or regions within the state as may be specified in the permit.

3. An RTA when countersigned the permit may attach to the permit any conditions, which it might have imposed if it had granted the permit and may likewise, vary and condition attached to the permit by the authority by which the permit was granted.

4. The provisions relating to grant, revocation and suspension of the permit as laid down by the Motor Vehicles Act 1988, shall apply to countersignatures of permits as well:

PROVIDED that it shall not be necessary to follow the procedures laid down in section 80 for the grant of countersignature of permits where the permits granted in any one state are required to be countersigned by the STA of another state or by the RTA concerned, as a result of the any agreement arrived between the stated after complying with the requirements of sub-section (5).

5. Every proposal to enter into an agreement between the states to fix up the number of permits to be granted or countersigned in respect of each route or area, shall be published by each of the state governments concerned in the official gazette and in any one or more of the regional newspapers in the regional language of the area or the route proposed to be covered by the agreement, together with the notice of the date before which representations therewith may be submitted, and the date not being less than thirty days from the date of publication in the official gazette, on which, and the authority by which, and the time and place at which, the proposal and any representation received in connection therewith will be considered.

6. Every agreement arrived at between the states shall, in so far as it relates to the grant of countersignature of permits, be published by each of the state governments concerned, in the official gazette and in any one or more of the newspapers in the regional language of the area or the route covered by the agreement and the STA of the state and the RTA concerned shall give effect to it.

7. Notwithstanding anything contained in the sub-section (1), the RTA of one region may issue a temporary permit under section 87 to be valid in another region or state with the concurrence, given generally or for particular occasion, of the RTA of the other region or state transport authority of that other state, as the case may be.

But seeing the formalities and all the 'ifs' and 'provided' clauses the manufacture/transporter has to comply with, these parties prefer to opt for a National permit. For the purpose of encouraging long distance interstate road transport, governments have started promoting National permits.

NATIONAL PERMIT

This means a permit granted by the appropriate authority to goods carriage to operate throughout the territory of India or in such contiguous states not being less than four in number, including the states in which the permit is issued as may be specified in such permit in accordance with the choice indicated in the application.

Application for the national permit (Section 86)

An application for the grant of a national permit shall be made in Form 48 to the registering authority.

Form, Contents and duration of authorization (Section 87)

1. An application for the grant of authorization for a national permit shall be made in Form 46 accompanied with a fee of Rs. 500 per annum in the form of a bank draft.

2. Every authorization shall be granted in Form 47 subject to payment of taxed or fees, if any, levied by the concerned states.

The authority, which grants the authorization, shall inform the STAs concerned of the registration number of the motor vehicle, the name and the address of the permit holder and the period for which they said authorization is valid.

3. The period of validity of an authorization shall not exceed one year at a time.

Age of motor vehicle for the purpose of national permit (Section 88)

1. No national permit shall be granted in respect of a goods carriage, other than multi-axle vehicle, which is more than 12 years old at any point of time.
2. No national permit shall be granted for a multi-axel goods carriage, which is more than 15 years old at a point of time.
3. A national permit shall be deemed to be invalid from the date on which the goods carriage covered by the permit completes 15 years, in case of multi-axle goods carriage and 12 years here the vehicle is other than a multi-axle goods carriage, unless such goods carriage is replaced.

Quarterly Return to be filed by a national permit holder (Section 89)

A national permit holder shall file return quarterly in Form 49 in respect of a motor vehicle covered by the national permit to the authority, which granted the national permit.

Additional conditions for national permit (Section 90)

The national permit issued, shall be subject to the following additional conditions:

1. The vehicle plying under national permit shall be painted in dry leaf brown colour with thirty centimeters broad white borders and the words “National permit” be inscribed on both sides of the vehicle in bold letters within a circle of sixty centimeters diameter

PROVIDED that the body of the tanker carrying dangerous or hazardous goods shall be painted in white colour with a dry leaf brown ribbon of 5 centimeters width around in the middle at the exterior and that of the drivers cabin in orange colour.

2. A board with the inscription “National permit valid in the states(s) of … “ With blue letters on white background shall be carried in front of such vehicle.

3. No such vehicle shall carry any goods without a bill of lading.

4. The vehicle shall have a minimum of two drivers to stretch himself and sleep

PROVIDED that this sub-rule shall apply to light motor vehicle and medium goods vehicles only from a date to be notified by the central government.

5. The vehicle shall at times carry the following documents and shall be produced on demand by an officer empowered to demand documents by or under the Act namely;

(a) Certificate of fitness
(b) Certificate of insurance
(c) Certificate of registration
(d) National permit
(e) Taxation certificate
(f) Authorization

6. The vehicle shall be subject to all local rules or restrictions imposed by a state government
7. The vehicle shall not pick up or set down goods between two points situated in the same rate other than the home state.

The concept of interstate and intrastate movement of goods may be explained further with the help of a case study of a company, which engages in both their intrastate movement of goods.

SUMMARY

Transport creates place utility. Geographical and climatic factors force industries to be located in particular places far away from the markets and places where there may not be any demand for the products. Transport bridges the gap between production and consumption centers. Transport creates time utility. Of late transport has started creating the time utility also. It has been made possible by virtue of the improvements in the speed of transport. It helps the product to be distributed in the minimum possible time. Transport helps in stabilization of price. Transport exerts considerable influence upon the stabilization of the prices of several commodities by moving commodities from surplus to deficit areas. This equalizes the supply and demand factor sand makes the price of commodities stable as well as equal. Transport ensures even flow of commodities into the hands of the consumers throughout the period of consumption.

A well–knit and coordinated system of transport plays an important role in the sustained economic growth of a country. The transport system in India comprises a number of distinct modes and services, notably railways, roads, road transport, ports, inland water transport, coastal shipping, airports, and airlines. Railways and roads are the dominant means of transport carrying more than 95% of total traffic generated in the country. Although other modes such as coastal shipping and inland water transport would play a greater role, the railways and roads would continue to dominate the transport landscape in the foreseeable future.

KEY TERMS

Brokers
Certificate of Insurance
Customer Service
Documentation
Equipment scheduling
National Permit
Product Pricing
Rate negotiation
Routing and carrier selection
Shippers Association
Tracing and expediting
Traffic Management
Transport pricing

REVIEW QUESTIONS

1. Explain the relationship with other business functions.
2. Explain the concept of transport management.
3. Write the legal types of transportation?
4. Explain the interstate and intra state movement of goods.

5. Write short notes on
a) National Permit.
b) Motor Vehicle Act, 1988.
6. What are the different Transportation security initiatives and the role of technology?

DISCUSSION QUESTIONS

1. Discuss the selection criteria for mode of transportation.
2. Discuss the transport regulations in India.
3. Discuss the followings,
   a) Government strategies to improve transport infrastructure in India.
   b) Airways as a mode of transport.
4. What are the issues involved in constraints with rail transportation?
5. Explain Transportation as a derived demand.
6. Discuss major factors affecting location decisions

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TRANSPORTATION

INTRODUCTION

Transportation ensures the movement of people as well as goods (produced or manufacture) from one place to another and helps in the growth of trade and commerce. A good transportation system is one of the pillars in the structure of a county's industry and commerce. In the movement of raw materials or products from their place of production to their place of consumption, transportation is the most important component of the logistic system. Transportation serves two purposes, one is product movement and the other is the in-transit product storage.

The movement of product can be achieved through various modes such as road, rail, air and sea subject to the availability of and access to infrastructure. The other functions of transportation, i.e., in-transit storage are not cost effective for a long period. The guiding principle for choosing a mode of transportation is the least cost per unit weight / volume of the product moved over a unit distance. However, selection of a particular mode is dependent on the availability of transportation infrastructure in the region. In India 49 percent of the total cargo movement is by road, followed by rail, which contributes to the extent of 34 percent, the balance is shared by inland water, air and sea.

DEFINITIONS OF TRANSPORTATION

The transportation activity moves products to markets that are geographical disparate (different) and provides added value to the customers when the products arrive on time, undamaged, and in quantities required. The utility provided by the transportation function is called place utility, while the gentility created by the storage function is called time utility. However, without transportation function, the delivery of time utility will not be completed since transportation function decides how fast and how consistently a product moves from one point to another. Transportation can be defined as the action of transporting someone or something or the process of being transported.

According to Burkart and Medlik, "Transportation can be defined as the means to reach the destination and also the means of movement at the destination."

The importance of transportation in physical distribution emanates from a variety of factors. Transportation confers 'time and place utility' to the product; it determines the company's customer service; it also has a crucial bearing on the other elements of physical distribution and marketing, like warehousing, inventory control and channel management. Finally, transportation is also a very important cost element in most businesses.

ROLE / FUNCTIONS OF TRANSPORTATION

Transportation provides following basis role and function,

1. Product Movement

It is a primary transportation function. It moves the product up and down the value chain. Whether the product is in the form of materials, components, assemblies, work-in progress finished

goods transportation is necessary to move it to the next stage of the manufacturing process or physically closer to the ultimate customer.

2. Product Storage

A less common function is that of temporary storage though vehicles make rather expensive storage facilities. In-transit storage requires to be moved again shortly, e.g., in a few days. Temporary storage becomes advantageous as the cost of unloading and reloading the product in a warehouse may exceed the daily charge of storage in transportation vehicles. Where the warehouse space is limited, utilizing transportation vehicles may become a viable option.

3. Creating Economic Utility

In economic theory terms, transportation function is to create place utility (right place) for the goods produced or, distributed by the firm. Place utility exists when goods are in the place where they can be consumed. Along with the necessity to have goods in the right place, the goods must be there at the right time (time utility) and in the right form (form utility) and in the possession or ownership of the person(s) who wants to consume them (possession utility).

4. Geographic Specialization

It takes place when a nation, region, state, or city produces those products and services for which it is best suited in terms of its capital, labor, raw materials and other resources and talents. If geographic specialization does not occur, then a nation, region, state, or city will be forced to devote some of its resources and energies to production of goods and/or services for which it is not well suited thus resulting in economic inefficiency and slower standard of living for all concerned.

5. Large-Scale Production

The role of transportation in large-scale production is similar to that discussed in connection with geographic specialization in that the availability of an adequate transportation system is a requirement to sustain large-scale production. But large-scale production by a firm requires that raw materials, parts, and supplies be collected from a variety of sources and a large geographic market for the product(s) produced is accessible at reasonable cost.

PARTICIPANT IN TRANSPORTATION

In transportation there are several participants:

1. Shipper and Consignee

The shipper and the consignee have the common objective of moving goods from origin to destination within a prescribed time at the lowest cost.

2. Carrier

The carrier is an intermediary between the shipper and the consignee. The main objective of the carrier is to maximize the revenue associated with the transaction while minimizing the cost associated to complete the transaction.

3. Government

Transportation is an important factor as' it affects the economy to a large extent and hence it maintains a high level of interest in it. Many Movements are more involved with carrier activities and practices.

4. Public

The public is concerned with the transportation accessibility 'expenses' effectiveness and environmental and safety standards. The public ultimately determines the need for transportation by demanding goods at reasonable price.

MODES OF TRANSPORTATION

There are five basic transportation modes are rail, highway' Water air but package carrier and inter-modal system are the recent y emerge. in transportation forms. The relative importance of each mode can be terms of system mileage, traffic volume, revenue, and the nature composition. Each mode is discussed with respect to these measures.

1. Roadways.
2. Railways.
3. Waterways.
4. Airways.
5. Pipelines.
6. Ropeways.

1. ROADWAYS

Road transport is the most promising means of agricultural and industrial advancement of a country. It is suitable for short and medium distances where other means are unable to reach. It provides door-to-door service, which is not possible with other means. Road transportation provides the basic infrastructure to bring trade and commerce from the remote rural areas to urban areas or vice- versa, and brings far-off villages into the mainstreams of national life, ensuring connectivity. With the manifold growth in industrial and agricultural activities and output, road transport has assumed greater importance due to the growing demand for making available the right product at the right place of consumption.

As regards the trucking industry in the country, it is predominantly in the unorganized private sector and the bulk of truck operators are single truck owners. Operations face may irritants like levy of multipoint actor and traffic police checks, which impede the free flow of goods traffic. Currently 25 million trucks are plying on Indian roads. Indian road conditions not being good, the average distance travelled by a truck is 250-300 kilometers per day as against 550-600 kilometers by their counterparts in developed countries.

Advantages of road transport

1. Door to Door Services

The outstanding advantage of road transport is that it provides door to door or warehouse to warehouse services.

2. Flexible Services

Road transport a great advantage over other modes of transport for its flexible services. Its routes and timings can be adjusted and changed to individual requirements without much inconvenience.

3. Suitable for Short Distance

Delays in transit of gods on account of intermediate loading and handling are avoided. Goods can be loaded direct into a road vehicle and transported straight to their place of destination.

4. Lesser Risk of Damage in Transmit

Road transport is most suited fool transporting delicate goods like chinaware and glassware, which are likely to be damaged in the process of loading and unloading.

5. Rapid Speed

If the goods are to be sent immediately are quickly, motor transport is more suited than the railways or water transport. Water transport is very slow.

Disadvantages of Road Transport

1. Seasonal Nature

Motor transport is not as reliable as rail transport. During rainy or flood season, roads become unfit and unsafe for use.

2. Accidents and Breakdown

There are more chances of accidents and breakdown in case of motor transport. Thus, motor transport is not as safe as rail transport.

3. Unsuitable for Long Distance and Bulky Traffic

This mode of transport is unsuitable and costly for transporting cheap and bulky gods over long distances.

4. Slow Speed

The speed of motor transport is comparatively slow and limited.

5. Lack of Organization

The road transport is comparatively less organized. More often, it is irregular and undependable. The rates charged for transportation are also unsuitable and unequal.

2. RAILWAYS

Railways are the principal carriers of man and material, and play a major role in the country's trade and commerce activities. It is the main source of supply of essential commodities, which are transported across the length and breadth of the country. Across the world, railways have played an important role in the industrialization and development of nations. It was the primary means available. for movement of raw material and finished goods and all countries have heavily relied on railways in the initial stages of industrialization and development.

Railways commonly haul heavy materials, chemical, and farm products, as well as low-value products and an increasing number of automobiles. They are especially efficient for fill carloads, which can be shipped\at lower rates than smaller modes of transportation. They require

less handling. Many companies locate factories near major rail lines for convenient loading and unloading.

Advantages of Rail Transport

1. It is a convenient mode of transport for travelling long distances;
2. It is relatively faster than road transport;
3. The major advantage that the railway offers is the efficient transportation of larger loads over long distances.
4. The variable cost per tone-mile of transportation is lower for railways as comparison to other modes of transportation.
5. The physical capability of transporting all types of commodities.
6. Railways also offer more reliable and safe transportation under all weather, a condition as compare to other modes of transportation.
7. Traditionally railways have been the most economical mode of transportation for low value, high density and high volume products and hence the preferred mode for inbound logistical movements for firms.

Disadvantages of Railway Transport

1. It is relatively expensive for carrying goods and passengers over short distances.
2. It provides service according to fixed time schedule and is not flexible for loading or unloading of goods at any place.
3. It involves heavy losses of life as well as goods in case of accident.
4. Low accessibility, the inability to service areas where railway tracks are not present.
5. Slow speed due to consolidation of rail wagon into complete train units for a particular route, and breaking out of rail wagons that have reached their destination or have to be transferred to other trains for different routes.
6. The packaging costs for railway are higher due to the impact experienced by goods during the coupling of the rail cars.
7. In India, meter-gauge routes remain underutilized due to the problem of transfer of loads from broad- to meter-gauge routes being uneconomical.

3. WATERWAYS / SEA

Waterways transportations are cheapest method of shipping heavy, low-value, non-perishable goods such as ore, coal, grain, and petroleum products. Water carriers offer considerable capacity. Powered by tugboats and towboats, barges that travel along inter-coastal canals, inland rivers, and navigation systems can haul at least ten times the weight of one rail car, and ocean-going vessels can haul thousands of containers. However, many markets are inaccessible by water transportation unless supplemented by rail or truck.

The shipping industry is divided into several parts such as:

a) Liner service
b) Tramp shipping

c) Industrial services
d) Tanker operations

India has 55 shipping companies, of which 19 are exclusively involved in trade, 29 in overseas trade, and remaining engage in both. Indian shipping highly dependent on international trade with 95 percent of the total cargo being volume overseas cargo and 77 percent of the value moving by sea. Crude oil, oil products, iron ore, coal, food grain and steel are some major products transported through sea.

Advantage of Water Transport

1. Important for Foreign Trade

Expansion of foreign trade has taken place mainly due to water transport. Current volume of India's foreign trade is quite large and it is likely to further expand for the sake of economic development of the country.

2. Defense

Development of shipping is essential for defense of the country also. It is regarded as second line of defense.

3. Cheapest and Slowest of Transport

Water transport-is the cheapest and slowest means of transport. Waterways are provided by nature. Oceans and rivers are the free gifts of nature. No cost is involved in their construction unlike that of railways and roads.

4. Transport of Heavy Goods

Heavy and bulky goods can be transported at little cost through water transport. A boat has a large carrying capacity which can carry goods weighting about six times its own weight.

Disadvantages of Water Transport

1. Limited Area

Area of water transport is restricted. Rivers and oceans are free gifts of nature. Accordingly the operational area remains fixed.

2. Slow Speed

It is a slow means of transport. Water transport suffers from relatively slow speed as compared to other means of transport. Failure of monsoon results into fall in the water level of rivers making navigation difficult.

3. Less Safety

As against other means of transport, it is less safe. The loss of ships due to perils of the sea involves considerable loss of life as well as goods. There are all possibilities that ships or boats may sink:

4. Deterioration of Goods

Sea water is corrosive which may affects the quality and the flavor of goods. The element of pilferage also seems to be very high in water transport. Because of these reasons goods transported by water transport requires special packing.

4. AIRWAYS

Compared to all other modes of transport, air transport is the least hazardous in nature. Air transportation is costly and is used only for highly perishable commodities and items, whose life is short and value is high.

Air transportation is the fastest but most expensive form of shipping. It is used most often for perishable goods; for high-value, low-bulk items; and for products requiring quick delivery over long distances, such as emergency shipments. Some air carriers transport combinations of passengers, freight, and mail. Despite its expense, air transit can reduce warehousing and packaging costs and losses from theft and damage, thus helping to lower total costs (but truck transportation need. end for pick-up and final delivery adds to cost and transit time). Although air transport accounts for less than 1 per cent of total ton-miles carried, its importance as a mode of transportation is growing. In fact, the success of many businesses is now based on the availability of overnight air delivery service provided by such organizations.

Advantages of Air Transport

1. High Speed

Air transport is the fastest mode of transport and therefore suitable carriage of goods over a long distance requiring less time.

2. Quick Service

Air transport provides comfortable, efficient, and quick transport service. It. is regarded as best mode of transport for transporting perishable goods.

3. No Infrastructure Investment

Air transport does not give emphasis on construction of tracks like railways. As no capital investment in surface track is needed, it is a less costly mode of transport.

4. Easy Access

Air transport is regarded as the only means of transport in those areas which are not easily accessible to other modes of transport. It is therefore accessible to all areas regardless the obstruction of land.

5. No Physical Barrier

Air transport is free from physical barriers because it follows the shortest and direct routes where seas, mountains, and forests do not obstruct.

6. National Defense

It plays a significant role in the national defense of the country because modern wars are conducted with the help of airplanes. An airway has an upper hand a destroying the enemy in a short period.

Disadvantages of Air Transport

1. Risky

Air transport is the most risky form of transport because a minor accident may put a substantial loss to the goods, passengers, and the crew.

2. Very Costly

Air transport is regarded as the costliest mode of transport. The operating cost of airplanes is higher and it involves a great deal of expenditure on the construction of aerodromes and aircraft.

3. Small Carrying Capacity

The aircrafts have small carrying capacity and therefore these are not suitable for carrying bulky and cheaper goods.

4. Unreliable

Most of the air transports are uncertain and the unreliable because these are controlled by weather condition. It is seriously affected by adverse weather conditions.

5. Huge Investment

Air transport requires huge investment for construction and maintenance of aerodromes. It also requires trained, experienced, and skilled personnel which involves a substantial investment.

5. PIPELINES TRANSPORT

A pipeline as a mode of transportation was first developed by Samuel Van Syckel in 1870 to transport Petroleum at Pithole, Pennsylvania. Twenty Years the Stand is of Northwestern Pennsylvania chanced the grace of transportation. Through petroleum was the first product transported In this manner, pipelines became useful for several other commodities such as coal in slurry form, iron ore fines in slurry form, chemicals, natural gas etc. the basic advantage of pipelines is that they reduce operational costs, though initial investment is high In India pipelines are used for oil transportation by all public and private sector petroleum refineries. For iron Ore transportation it is used by the Kudremukh Iron Ore project to transport-to-transport iron ore over a distance of 67 kilometers, along the hilly terrain of the Western Ghats.

Pipelines are the most automated transportation mode, usually belong to the shipper and carry the sh1pper's products. Most pipelines carry petroleum products or chemicals. Pipelines have also been a concern to environmentalists, who fear installation and leaks could harm plants and animals.

Advantages of Pipeline Transportation

1. Loss in Transit

It is less in pipeline transportation as compared to other modes.

2. Economies of Scale

Pipeline offers large-scale economies of scale in transportation liquid petroleum products.

3. Eco-Friendly

Environmental impact during construction, operation and maintenance is negligible and reversible which is environment friendly.

4. Flexibility

Pipeline transportation is flexible, as the volume transported can be increased / decreased quickly and at negligible cost.

5. Low Operational Cost

Operation/maintenance costs are relatively lower.

Disadvantages of Pipeline Transportation

1. High Initial Cost

Initial cost of laying pipelines is high and very expensive.

2. Long Transit Time

It takes long transit time of about ten miles per hour per liquid.

3. Leakage in Pipelines

Some solids, liquids and gases when transported cause damage to the pipeline. Leakage in pipes might pollute soil and contaminate ground water. If pipelines are damaged, it can cause soil pollution and damage to environment.

6. ROPEWAYS

In India more than 16 percent of the geographical area is hilly. Transportation in such areas is a problem because of the long circuitous routes. The transportation of goods and essential commodities to the hilly areas is sometimes crucial because of their strategic importance in the country's defense program. In view of oil shortages, ropeways can prove more economical and faster than road transport, particularly in hilly areas.

This mode of transport is used in hilly areas in Sikkim, Meghalaya, Mizoram Himanchal Pradesh, and Uttar Pradesh. Currently 175 Kilometers of ropeway; are in use in the above territory for transportation of both materials and human beings.

Advantages of Ropeways Transportation

1. Eco-Friendly

They cause least damage to the ecology.

2. High Accessibility

Inaccessible hilly areas can be reached using the shortest routes

3. Economical

This route is very cheap and has low initial and variable cost other modes of transportation are uneconomical.

4. Faster Coverage

Bulk material can be moved faster over short distances.

Disadvantages of Ropeways Transportation

1. Limited Scope of Operation

The major drawback of ropeways is lack of space due to which only a limited amount of quantity can be transported.

2. Slowest Mode of Transport

It moves from one end to another with a consistent speed according to the weight of the goods loaded for transit.

TRANSPORTATION COST

INTRODUCTION

Transport costs are a monetary measure of what the transport provider must pay to produce transportation services. They come as fixed (infrastructure) and variable (operating) costs, depending on a variety of conditions related to geography, infrastructure, administrative barriers, energy, and on how passengers and freight are carried. Three major components, related to transactions, shipments and the friction of distance, impact on transport costs.

Rates are the price of transportation services paid by their users. They are the negotiated monetary cost of moving a passenger or a unit of freight between a specific origin and destination. Rates are often visible to the consumers since transport providers must provide this information to secure transactions. They may not necessarily express the real transport costs.

The difference between costs and rates either results in a loss or a profit from the service provider. Considering the components of transport costs, rate setting is a complex undertaking subject to constant change. For freight transportation and many forms of passenger transportation (e.g., air transportation) rates are subject to a competitive pressure. This means that the rate will be adjusted according to the demand and the supply. They either reflect costs directly involved with shipping (cost-of-service) or are determined by the value of the commodity (value-of-service). Since many actors involved in freight transportation are private, rates tend to vary, often significantly, but profitability is paramount.

TOTAL TRANSPORTATION COST STRUCTURES

Transportation costs are classified into a combination of categories:

1. Variable Costs

These are those costs that change in a predictable, direct manner in relation to some level of activity during a time period. Variable costs can be avoided only by not operating the vehicle. Aside from exceptional circumstances transport rates must at least cover variable cost. The variable category includes direct carrier costs associated with movement of each load. Expenses generally measured as a cost per mile or per unit of weight. Typical components in this category include labor, fuel and maintenance.

2. Fixed Costs

These are those costs that do not change in the short run and must be covered even if the company is c1osed-down (e.g., during a holiday Strike). The fixed category includes carrier costs not directly influenced by Shipment volume. For transportation firms, fixed components include terminals, rights-of-way, information systems, and vehicles. In the short-term, expenses associated with fixed assets must be covered by contributions above variable cost on a per shipment basis. In the long-term, the fixed cost burden can be realized somewhat by the sale of fixed assets; however, it is often very difficult to sell rights-of-way or technologies.

3. Joint Costs

These are expenses unavoidably created by the decision to provide a particular service. For example, when a carrier elects to haul a truckload from point A to point B, there is an implicit decision to incur a "joint" cost for the back-haul from point B to point A. Either the joint cost must be covered by the original shipper from A to B, or a back-haul shipper must be found. Joint costs have significant impact on transportation charges because carrier quotations must include implied joint costs based on considerations regarding an appropriate back-haul shipper and/or back-haul charges against the original shipper.

4. Common Costs

This category includes carrier costs that are incurred on behalf of all shippers or a segment of shippers. Common costs, such as terminal or management expenses, are characterized as overhead. These are often allocated to a shipper according to a level of activity like the number of shipments handled (e.g., delivery appointments). However, allocating overhead in this manner may incorrectly assign costs.

FACTORS INFLUENCING TRANSPORT COST

The various factors affecting transportation cost are as follows:

1. Geography

Its impact mainly involves distance and accessibility. Distance is commonly the most basic condition affecting transport costs. The more it is difficult to trade space for a cost, the more the friction of distance is important. It can be expressed in times of length, time, economic costs or the amount of energy used. It varies greatly according to the type of transportation mode involved and the efficiency of specific transport routes.

2. Type of Product

Many products require packaging, special handling, and they are bulky or perishable. Coal is obviously a commodity that is easier to transport than fruits or fresh flowers as it requires rudimentary storage facilities and can be trans-shipped using rudimentary equipment. Insurance Costs are also to be considered and are commonly a function of the Value to Weight ratio and the risk associated with the movement,

3. Economies of Scale

Another condition affecting transport costs is related to economies of scale or the possibilities to apply them as the larger the quantities transported, the lower the unit cost. Bulk commodities such as energy (coal, oil), minerals and grains are highly suitable to obtain lower unit transport costs if they are transported in large quantities.

4. Energy

Transport activities are large consumers of energy, especially oil. About 60% of all the global oil consumption is attributed to transport activities. Transport typically account for about 25% of all the energy consumption of an economy. The costs of several energy intensive transport modes, such as air transport, are particularly susceptible to fluctuations in energy prices.

5. Trade Imbalances

Imbalances between imports and exports have impacts on transport costs. This is especially the case for container transportation since trade imbalances imply the re-positioning of empty containers that have to be taken into account in the total transport costs. Consequently, if a trade balance is strongly negative (more imports than exports), transport costs for imports tend to be higher than for exports.

6. Infrastructures

The efficiency and capacity of transport modes and terminals has a direct impact on transport costs. Poor infrastructures imply higher transport costs, delays and negative economic consequences. More developed transport systems tend to have lower transport costs since they are more reliable and can handle ‘more movements.

7. Mode

Different modes are characterized by different transport costs, since each has its own capacity limitations and operational conditions. When two or more modes are directly competing for the same market, the outcome often results in lower transport costs. Containerized transportation permitted a significant reduction in freight transport rates around the world.

8. Competition and Regulation

Transport services taking place over highly competitive segments tend to be of lower cost than on segments with limited competition (oligopoly or monopoly). International competition has favored concentration in many segments of the transport industry, namely maritime and air modes.

9. Surcharges

Refer to an array of fees, often set in an arbitrary fashion, to reflect temporary conditions that may impact on costs assumed by the transporters. The most common are fuel surcharges, security fees, geopolitical risk premiums and additional baggage fees. The passenger transport industry, particularly airlines, has become dependent on a wide array of surcharges as a source of revenue.

INDIAN SHIPPING

Learning Objectives

After reading this chapter, you should be able to:

- Understand the history of Indian shipping.
- Know the shipping industry in India.
- Understand the various ports in India.
- Know the ports infrastructure development in India.
- Understand the Indian ports during pre and post Independence Era.
- Know the shipping associations in India.

INTRODUCTION

It is now recognized that a very well developed and fully coordinated transport system is vital to the prosperity and planned development of any country. Strictly speaking, this is not a newly discovered truth. Before the development of air transport, railways, and shipping constituted the twin forms of a transport system and it is well known that during the last century the most advanced nations have had either one or both of these modes highly developed.

India has been a significant maritime nation that has recognized and encouraged "shipping" as a critical contributor to the country's trade and overall economic development. In fact, shipping is the lifeline of a nation, necessary for transportation of essential goods, men, and materials during peace as well as emergencies like war and famine.

HISTORY OF INDIAN SHIPPING

1. India had a maritime history dating back around 5,000 years. During the Indus Valley Civilization, ships from the harbor at the ancient port city of Lothal established trade with Mesopotamia.
2. After the decline of the Indus civilization however, maritime trade on the Indian sub-continent came to an end for over 1,000 years. Due to the monopoly of the Persians and later the Arabs over land—based caravan routes, the impetus to later re-develop maritime links was trade (primarily in cotton, pepper, and other spices).
3. The later maritime journeys spread the influence of ancient and medieval Indian civilization as far as the islands of Indonesia in the East, the islands of Japan in the North, and the East Coast of Africa in the West.
4. During the early days of British rule, Indian shipping was in a flourishing stage. A shipbuilding yard was maintained at Surat up to I735 AD. In that year, most of the work was transferred to Bombay. The Bombay built ships were very strong and durable. What is more, they were 25 per cent cheaper as compared with ships built in the dock of England.

5. On 15th August I947, the Indian shipping place comprised 59 vessels totaling 1,92,000 GRT (Gross Registered Tonnage), of which 48 vessels with 1,19,000 GRT were engaged in exclusively coastal trade and 11 vessels with 73,000 GRT were employed to a limited extent in the adjacent country trades, namely, India-Burma, India-Ceylon (Sri Lanka) and India-Pakistan.
6. The shipping and shipbuilding industries of India once had a proud record of achievement as witnessed by the single fact that Nelson's flagship at the battle of Trafalgar was an Indian built vessel. Events of the mid nineteenth century, however, dealt blows from which these industries have never recovered; and the establishment of the merchant marine adequate to aid India's economic survival remains a thorny problem.
7. Up to 1923, the Government of the country took little interest in the question of building a mercantile marine for India. It was in that year that the Indian mercantile committee was established. The committee recommended the building-up of mercantile marine, the maintenance by government of a training ship to train personnel, and a far reaching scheme of Indian of coastal marine in a period of 25 years.
8. The Indian fleet consisted of 420 ships of 9,973,277 DWT (Dead Weight Tonnage) or 6,05,69,50 GRT at the end of Dec, 1990. Although the Indian fleet added up to 6.06 million GRT, it was only marginally over the 6 million GRT achieved at the end of the sixth plan period.
9. The working group of the Ministry of surface transport made a proposal in 1989 based on requirements of cargo projection. It suggested increasing the shipping tonnage to 14.89 million DWT. It involved gross acquisition of 427 ships of 7,732 million Dead Weight Tonnage (DWT) in the existing fleet which would have outlived their life.

SHIPPING INDUSTRY OF INDIA

Indian shipping industry is one of those few sectors of economy which has constantly engaged the attention of the policy-makers ever since the attainment of Independence in I947. Government owned the responsibility of developing national shipping industry so that it may get its due share in world trade. Government committed to take necessary steps to get 100 per cent share in the coastal trade and 50 per cent share in foreign trade to Indian shipping industry:

1. Growth of Indian Shipping Industry

Growth of Indian shipping industry since independence is studied in four periods as under:

a) Period from 1948 to 1960

This period has been characterized by slow but steady growth. In I950, government adopted the policy of reserving coastal shipping service for Indian ships. On August 15, 1947, we had only 59 ships with a capacity of 1.92 lacs Gross Registered Tonnage (GRT).

b) Period from [960 to 1980

Shipping industry made rapid progress in these two decades. This period is called the golden period of the development of GRT, which increased eightfold.

c) Period from 1980 to 1995

New policies, initiated in the wake of the process of liberalization in 1991, emphasized that shipping industry be given free hand to procure new ships and to increase GRT of the country. On account of record increase in GRT in 1985 in India, shipping industry achieved Eighth Plan target of 70ylacs GRT much before the deadline in 1995.

d) Period from 1995 to 2002

This period showed tendency of fall in GRT. Record GRT of 71 lacs as achieved in I996 could not be sustained. In 2002, GRT declined to 69 lacs. Several facts on account for this fall in growth rate of Indian shipping industry, viz.

- Fall in the tempo of measures taken to provide relief to shipping industries over the years.
- Shortage of funds to buy new ships.

2. Shipping Companies

On March 31, 2003, in all 122 shipping companies were operating in the country. Shipping companies are operating in both public sector and private sector. These are discussed below:

a) Private Sector

Of the total shipping industry capacity, private sector companies constitute 55 per cent capacity. Some main private shipping companies having 2 laces or more GRT capacity are:

- Great Eastern Shipping Co. Ltd. (8.86 lack GRT).
- Essar Group's SISCI Shipping Co. Ltd. (6.75 lack CRT) and
- Chogle Steam Ships Ltd. (2.29 lack GRT).

b) Public Sector

Shipping Corporation of India is the largest public sector company in the country. It has a fleet of 79 merchant ships. Their total capacity is 27 lacks GRT. It operates on almost all sea-routes. Of the total shipping industry capacity, share of this corporation comes to 31 per cent.

3. Shipping Corporation of India Limited

In 1961, government merged Eastern Shipping Corporation and V Western Shipping Corporation and established 'Shipping Corporation of India'. It is a profit earning public sector unit. It has been getting handsome return on its investment. Company has 79 ships with 27 lacks GRT. Company's fleet comprises of variety of ships, viz., general cargo ship, crude oil tankers, cellular container ships, passenger liners: timber cargo vessels, etc. On behalf of government organizations and departments, the company also manages another 53 ships. The company has collaborated with Iran in a joint venture shipping company named 'Irano-Hind Shipping Company'.

4. Ship Building

India has nine shipyards in public sector, and about 21 small shipyards in private and State Governments sectors, which cater to the needs of small and medium ships. Cochin shipyard in Cochin and Hindustan shipyard in Vishakhapatnam are main Indian shipyard companies. Small shipyards also cater to the needs of small/medium sized fishing vessels and ships. Ship building industry has also been opened to private sector. Private sector is allowed to build ships of different capacities.

Although Indian shipping industry is one of those few sectors of the economy which has constantly engaged the attention of the policy-makers since independence, yet the progress of this industry has not been encouraging over the past few years. Since liberalization, Indian shipping industry has been facing lot of competition and challenges. Because of several factors Indian shipping industry is not in a position to face these challenges on equal footing. Cyclical freight market, large capital requirement, inadequate essential basic facilities and services, limited capacity of ports, etc., are some of the factors on account of which shipping industry of the country has to encounter several problems.

PROBLEMS OF SHIPPING INDUSTRY

1. Inadequate Capacity

Carrying capacity of Indian shipping is inadequate. In I95], major ports had a carrying capacity of20 millions tonnes. In 2008 carrying capacity of ports increased to 519 million tonnes. But still this capacity is inadequate keeping in view increased foreign trade. There is a need to upgrade the cargo handling capacity of ports to 1,273 million tones.

2. Inefficient Performance of Indian Ports

As compared to other ports in the Asian region performance of Indian ports is inefficient. Many factors account for it, viz., obsolete equipments, defective training, and wastage time in the docks, etc.

3. Paucity of Funds

Shipping industry requires huge capital investment. Main problem faced by Indian shipping industry today is shortage of funds to execute its plans of extension. Replacement of ships is a very expensive proposition. As per the estimate made by Planning Commission, shipping industry will require huge amount to buy new ships and replace obsolete ones.

4. Working Age of Indian Ships

Working age of Indian ships is relatively short. It is 14 years on an average as against 18 years of world ships. Another problem of Indian shipping industry is that 44 per cent of its ships are more than 15 years old. Of these 24 per cent are more than 20 years old. Consequently, huge expenditure is incurred on their repair.

5. Under-Developed Container System

Container system has been introduced by developed countries in their quest for labor-saving device in carrying goods. In container system, goods are packed in large containers. These containers are loaded and unloaded with the help of cranes. India has made very little progress in this matter. Backwardness of Indian container system is detrimental to national shipping industry.

On international ports, one laborer is required per container as against 59 laborers on Indian ports. As a result, cost of loading and unloading increases.

6. More Operating Cost

Operating cost of water transport has increased tremendously mainly due to inefficiency of Indian ports, shortage of cranes, delay in loading the cargo, etc. Consequently, profits earned by shipping companies are very little.

7. Competition with Foreign Companies

Indian Shipping Companies have to compete with foreign companies. It adversely affects their profitability.

REFORMS IN SHIPPING SECTOR

Along with economic reforms introduced in 1990-91, a new policy was chalked-out to reform the shipping sector. Its main features are as follows:

1. Except for some special categories of ships, there is no need on the part of Indian Shipping Companies to get sanction from the Central Government for the purchase or sale of ships.
2. Ship can be got repaired from any shipyard. Previously, there was a committee to decide the place of repair of ships; the same has now been abolished.
3. Reserve Bank of India will make available foreign exchange for the import of parts and components required for repairing ships.
4. Indian Shipping Companies have been allowed to keep the sale proceeds of Indian ships abroad for the purchase of new ships.
5. By amending Merchant Shipping Act, shipping companies have been authorized to get loans in foreign currency by pledging their ships.
6. To attract foreign ships to Indian Ports, laws were relaxed for foreign cargo ships and boats.
7. In order to buy old ships, norms regarding their condition were further relaxed.
8. Any ship engaged in coastal trade will be free to determine passenger fare or tariff for cargo.
9. Sanction has been accorded to the private sector to participate in the construction of ports.
10. Railroad connectivity with ports will be improved to facilitate movement of goods and reduce congestion at the ports.
11. Foreign direct investment up to 100% has been permitted for the construction and maintenance of ports and harbors.

In short, shipping has an important role in India's foreign trade. Special efforts should be made .to develop and enhance its efficiency.

PORTS IN INDIA

A port is essentially an economic concept, an economic infrastructure that serves coastal and overseas traffic. Port is a subsystem of total transport network and meeting place of other modes of transport. It is a gateway for the entrance from surface water to land and vice versa, Port is also construed as the major crossroad of traffic in ideas, people, and goods over the centuries.

Ports play an important economic role in maritime nations by facilitating foreign trade. They are thus, catalysts for social and economic development in any man time nation. Maritime transportation has been the life-blood of world trade since time immemorial. Though new modes of transportation have grown enormously, shipping shall continue to occupy the lion‘s share in world trade in the foreseeable future because of economies inherent in sea transportation.

India has a long coastline spanning 7600 kilometers forming one of the biggest peninsulas in the world. It is serviced by 12 major ports and 185 notified minor and intermediate ports. Major ports handled over 80% of all cargo traffic in 2007. However, the words “major”, “intermediate” and “minor” do not have a strict association with the traffic volumes served by these ports. For example, Mundra Port, a newly developed minor port in the State of Gujarat registered cargo traffic of around 28.8 million tons per annum during the financial year of 2008, which is higher than that of many major ports.

Ports play a crucial role in the transportation sector of India. They provide an interface between the ocean transport and land-based transport. India has extensive port infrastructure covering major ports and 187 minor/intermediate ports (non-major ports).

The total non-major ports are in the following States: Gujarat (42); Maharashtra (48); Tamil Nadu (15); Karnataka (10); Kerala (17); Andhra Pradesh (12); Orissa (13); Goa (5); West Bengal (1); Daman and Diu (2); Lakshadweep (10); Pondicherry (2); and Andaman and Nicobar (23).

While, the major ports along the coastline include:

1. East Coast

This includes,

a) Kolkata Port (including Dock Complex at Halide)

It is the oldest and the only reverie major port of India. It has a vast hinterland comprising the entire eastern India and two landlocked neighboring countries of Nepal and Bhutan.

b) Para dip Port

It is the only major sea port in Orissa.

c) Visakhapatnam Port

It is a natural harbor. It is the only port to possess the three international accreditations of ISO 14001; 2004 (EMS)/OHSAS 18001 and 150 9001:2000 (QMS).

d) Chennai Port

It is an all weather artificial harbor with one outer harbor and an inner harbor with a wet dock and a boat basin with round the clock navigation facilities.

e) Ennore Port

It is situated on the coromandal coast about 24km North of Chennai Port, along the coastline, in the state of Tamil Nadu. it is the first corporatized major port in India. It is an all weather port with State-of-the art infrastructure and modern navigational facilities.

f) Tuticorin Port

It is also situated on the East coast of Tamil Nadu. it has two operational wings, namely, Zone 'A' comprising the new major port and Zone '8' representing the old anchorage port, which were merged together on 1st April 1979, when Tuticorin Port Trust was constituted.

2. West Coast: This includes:

a) Cochin Port

It was formally opened for vessels by 1930-31 and was given the status of a major port in 1936.

b) New Mangalore Port

It has a modern all weather artificial lagoons situated at Panambur, Mangalore in the State of Karnataka on the West coast of India.

c) Mormugao Port

It is one of the century old ports with modern infrastructural facility and is one of the finest natural harbors in the world.

d) Jawaharlal Nehru Port

It is India's largest container port and also has become a world class international container handling port.

e) Mumbai Port

It is a natural deepwater port. The second oldest major part of-India after Kolkata. Originally a general cargo port, today it is a multi-purpose port handling all types of cargo-liquid bulk. Dry bulk, break bulk and containers.

f) Kandla Port

A natural harbor situated in the State of Gujarat, is a gateway to the North-Western States.

All the Major Ports are administered by the 'Port Trusts' governed by the provisions of Major Port Trust Act, I963 which are autonomous bodies, except the newly 'Ennore Port' which is run by 'Ennore Port Limited' (registered under the Companies Act, I956). Besides, there is a subordinate office namely, the 'Andaman Lakshadweep Harbor Works (ALHW)'. It is under the control of the Department of Shipping and is entrusted with the responsibility of formulating and implementing the program for providing port and harbor facilities in A&N Islands and Lakshadweep Islands. The remaining ports are under the administrative control of the respective maritime State Government,

POLICY INITIATIVES

1. The government has studied various strategies adopted by ports worldwide to address similar issues facing ports In India. The government envisages commercialization/privatization/modernization of major existing ports. These are

expected to result in technological upgrades and overall improvement of performance levels of the ports.

2. Measures to strengthen the regulatory structures your major ports have also been initiated. These pertain to tariff rationalization and the establishment in a phased manner, of a corporate structure for the existing ports.
3. The government has announced a series of measures to promote foreign investment in the port sector:
    a. Issued guidelines for private/foreign participation that permits formation of joint ventures or foreign collaboration for setting-up port facilities.
    b. 100 per cent foreign investment is permitted for construction and maintenance of ports and harbors and in projects providing support services to water transport.
    c. Private sector is allowed to set-up captive facilities.
4. The government is offering various fiscal incentives to private investors; e.g., a 10~year tax holiday in the port development, operation, and maintenance. Investors in inland waterways and inland ports are also entitled to these incentives.
5. The Tariff Authority for Major Ports (TAMP), an independent authority, is now responsible for determining and revising tariffs on major ports. The existing legal framework is being amended to facilitate changes consistent with structural re-organization of major ports. The government has announced guidelines for revised bidding and bid evaluation process.

Maritime states have also formulated their port development plans through private investment as well as by user agencies. An integrated port development strategy comprises creation of port facilities, industrialization and development of infrastructure facilities like roads and railways linking the hinterland. Suitable sites have been identified, techno-economic pro-feasibility studies have been carried-out and guidelines formulated.

POLICY INITIATIVES IN PORTS SECTOR

1. FDI up to 5l% is allowed on automatic basis in support services like operation and maintenance of piers and loading and discharging of vessels.
2. Up to 100% FDI under automatic route is permitted in projects for vehicular tunnels, ports, and harbors.
3. The BOT model will generally be used for private sector participation with the assets reverting back to the port after the concession period.
4. Major ports have been permitted to form joint ventures with foreign ports, minor ports and other companies to attract new technology, better management practices, and implementation of development schemes and creation of optimal port infrastructure.
5. Inputs and concessional import duty allowed with liberalized trade policy.

6. The process of phased corporatization has been initiated for the major ports.
7. An independent Tariff Authority for major ports set-up to fix and revise ceiling tariff.
8. Ten year tax holiday that may be availed in a block of fifteen years for infrastructure facility relating to port, inland port and inland waterways, i.e. 1 April 2002 under section 80-1A of it Act.

PORT INFRASTRUCTURE DEVELOPMENT

MEANING OF INFRASTRUCTURE

It would be useful to determine what is meant by "infrastructure" in the context of international logistics. A couple of dictionary definitions would be a good start:

a) The basic facilities, services, and installations needed for the functioning of a community or society, such as transportation and communication.
b) A collective term for the subordinate parts of an undertaking; substructure, foundation.
c) The permanent installations forming a basis for military operations, as air fields, naval bases, training establishments, etc".

In the field of logistics, the definition can be very broad: infrastructure is a collective term that refers to all of the elements in place (publicly- or privately-owned goods) to facilitate transportation, communication, and business exchanges. It would therefore include not only transportation and communication elements, but also the existence and quality of public utilities, banking services, and retail distribution channels.

INTRODUCTION TO PORT INFRASTRUCTURE

A port is a key component or a vital system of infrastructure, indispensable to the smooth operations of international trade. Efficient, economical, and expeditious infrastructure is a necessary pro-condition for the growth of exports. In its role as a transfer point between land and sea transport networks, a port serves as a (pivotal element in the circulation of trade flows.

Ports reflect the country's commitment to trade and earnings that stem from trade. Ports serve as a funnel through which the impulses of modernization and development are channeled along with imports. A port is a visible manifestation of economic activity, a symbol of integrated growth center and an agent of balanced regional development. Establishing a strategic network of ports is essential to promote regional economic development. The up gradation and expansion of ports is vital to strengthen India's position in world trade and to handle the growing volume of international trade.

The Indian subcontinent is blessed with a long navigable coastline with 12 major ports and several minor ports. The 2,965km long Western coastline extending from Kandla to Kanyakumari has 6 major ports, 17 intermediate and 127 minor ports.

RAKESH MOHAN COMMITTEE REPORT AND PORT INFRASTRUCTURE

Even though most of the Indian ports have completed forty years of their operation, still they are costlier than other ports in the region for handling containers. The additional cost burden due to use of second and third generation vessels has been estimated at US$ 250 million a year, container delays at Indian ports cost USS 70 million a year. The Government of India constituted a committee under the Chairmanship of Rakesh Mohan to identify the bottlenecks in the sector and to suggest measures for augmenting resources and removing constraints to make the sector a catalyst for economic growth in the country. Rakesh Mohan Committee Report on India Infrastructure, which was submitted to Government of India on June 22, I996 recommended the following measures for the development of ports:

1. Port authorities should be permitted to raise resources from the primary market by way of debt and equity and from financial institutions.
2. Major port trust boards should be delegated powers to incur capital expenditure up to Rs. 500 million to facilitate speedy creation and operation of assets.
3. The Major Port Trust Act, 1963 should be amended to permit expansion of projects to be [taken-up on a Build-Operate-Transfer (BOT) basis.
4. Options of leasing-out berths or other assets, wherever it is cost effective, should be considered.
5. A moving away from a monolithic pay structure for workers to a piece-based wage structure.
6. At least two ports. One each and the West and East Coasts. be developed as mega ports, the warehouse for the Indian sub-continent.
7. While commercial operations of the ports be entrusted to existing port authorities with adequate autonomy, a separate regulatory authority to be set-up to consider issues relating to pricing and conditions which would govern private sector participation.
8. Ports with private participation be continued to be exempted from corporate taxes to augment internal resource base and increase eligibility for raising resources from the market.

A scheme for joint venture formation between major port and minor foreign ports has been finalized. Also similar schemes between major port and minor ports, without tender, and between major port and company following tender route have been considered. The measure aims at facilitating port trust to:

- Attract new technology,
- Introduce better managerial practices,
- Expedite implementation of schemes,
- Foster strategic alliances with minor ports for creation of optimal port infrastructure and
- Enhance confidence of private sector in funding ports.

In a sequel to the above recommendations of the committee, the government has declared some liberalized measures and incentives for privatization of the ports, such as:

Port will regulate as per the major Port Trust Act.

1. An independent Tariff Authority has been established to induce confidence to the investor and have a level playing field.
2. Tender with the transparent conditions shall be put on BOT basis.
3. Proper feasibility report shall be made at the expense of the port but subsequently recovered from the successful tender.
4. Evaluation is made transparent as per the criteria indicated in the tender.
5. Automatic approval of an Indian company with 74 per cent foreign equity is permitted.
6. Even up to 100 per cent foreign equity shall be considered by the F IPB (Foreign Investment Promotion Board).
7. Tax holiday for 5 years and tax concession for another 5 years within 12 years for infrastructure project are granted.
8. Custom duties for imported equipment are drastically reduced.
9. Subscription to debentures or equity of companies engaged in infrastructure would qualify for income tax relief for the investor.
10. Infrastructure Development Financing Company (IDFC) has been set-up for financing infrastructure project.
11. HUDCO also proposed to finance such projects.
12. Model tender document has been prepared.
13. Bankable agreement has also been prepared so that the entrepreneurs would be able to raise funds.

They special committee for infrastructure paved the way for removing the bottlenecks and ensured smooth progress under BOT concept:

1. Leasing-out existing assets of the port has been allowed.
2. Construction and creation of additional assets, such as:
    a. Construction and operation of container terminals.
    b. Construction and operation of bulk, break-bulk, multi-purpose, and specialized cargo berths.
    c. Warehousing container freight stations storage facilities and tank farms.
    d. Carnage and handling equipment.
    e. Setting—up of captive power plants.
    f. Dry -docking and ship repair facilities.
3. Leasing of equipment for port handling and leasing of floating crafts from the private sector.
4. Pilotage.

ISSUES IN PORT INFRASTRUCTURE

Port infrastructure is made up of several items, most of which are interconnected, and obviously affect the type of ships that can call on a given port, as well as the type of merchandise that can transit through it. With the advent of the much larger post-Panamax containerships, ports have been faced with many challenges. Specifically, the size of those ships is stretching the

capabilities of the ports: they are wider, longer, higher above the water, and have a much deeper draft.

1. Depth of Water

The first issue is undoubtedly the depth of the water. In many ports, the depth of the channels and of the berths is not sufficient to accommodate such large ships. Therefore, in a great number of ports, the port authorities have had to engage in dredging activities in order to allow ships with drafts exceeding forty feet (13.5 meters) to access the port. Only a few ports with naturally deep channels have been exempt from this activity.

This dredging of channels and ports can be exceedingly expensive, but ports have little alternative but to undertake this improvement of their capabilities. Competition is such that ships would call on a different port if one were not expanding.

In a parallel fashion, longer ships require longer turning circles, and therefore a redesign - and dredging « of different access channels. As ships become yet longer, wider and heavier, the challenge for the ports will be to keep up with these ships' requirements and adapt.

2. Cranes

Ports have found out that the width of those post-Panamax ships can also be a challenge for their cranes. Traditional Panamax ships can load up to thirteen containers in the width of a ship.

Some of the post-Panamax ships can load sixteen or more containers side-by-side. This presents a problem for ports in which the cranes cannot reach. One alternative is to load the ship from one side, turn the ship around, and then load the rest of the container. The problem is one of balance, as the ship will seriously list is heavier on one side.

3. Bridge Clearance

Another factor of great importance in ports is the clearance under its bridges. In many older ports, such as New York-New Jersey, the bridges are too close to the water, leaving very little clearance for tall ships or ships for which the cargo is outsized. In some cases, the cargo has to be partially dismantled or repositioned, or the ship lowered with ballast water and the delivery made at low tide so that it clears the bridge span.

All of these factors (depth of channels and berths, crane capacities, and bridge clearance) are likely to affect how ports are used in the future. There is a strong likelihood that all ports will not be able to accommodate the largest ships and that there will be the creation of large port hubs, to and from which mega-containerships will travel.

Smaller "feeder" ships that would not tax the ports infrastructure beyond their capacity would then serve smaller ports. Such ports have already been created in the Mediterranean Sea: Marsaxlokk, in Malta, and Cagliari, in Sardinia (Italy) serve as trans-shipment ports, loading and unloading large containerships in a deep- water port and using feeder services to serve the local markets of France, Italy and Spain.

DEVELOPMENT OF SHIPPING IN INDIA

INTRODUCTION TO DEVELOPMENT OF SHIPPING

India is a developing maritime nation with an extensive coastline of about 6,000km. There are nine coastal States namely:

1. Gujarat
2. Maharashtra
3. Karnataka
4. Goa
5. Kerala (West Coast)
6. Tamil Nadu
7. Andhra Pradesh
8. Orissa, and
9. West Bengal (East Coast)

These nine States have in all 12 major and 179 minor ports. Among these 12 major ports, six are located on the west coast (Kandla, Mumbai, Jawaharlal Nehru, Mormugao, Cochin, and New Mangalore) and six on the east coast (Chennai, Tuticorin, Paradeep Vizag, Kolkata, and Haldia). Four of the major ports, viz, Kolkata, Mumbai, Chennai, and Mormugao are more than 100 years old. Cochin and Vizag ports have recently celebrated their golden jubilee. The ports of Kandla, Tuticorin, New Mangalore, and Para deep came into existence after independence. INPT became operational after 1989.

Out of total declared 179 minor ports including 13 non-working ports. 120 ports (67 per cent) belong to West Coast States, 24 ports to East Coast States, and the rest (22) belong to the Island Union Territories. Maharashtra is the only State having the highest numbers of both major (Two) and minor (Fifty-three) ports in India. Next is Gujarat where one major and 40 minor ports are situated. West Bengal is the only maritime State, which, even with adequate waterfront has no declared minor port. The development of ports in the country can be described broadly in two phases, i.e., in the pre-independence era and post-independence era.

DEVELOPMENT OF INDIAN PORTS DURING PRE-INDEPENDENCE ERA

Realizing the growing significance of ports in the Indian economy in general and transportation infrastructure in particular, the Government of India has laid adequate emphasis on the development of ports in the country. During the pre-independence era the port facilities, which existed in India till about the middle of the 19th century, were meager. The ships used to lie in anchorage and small lighters transshipped cargo passengers. The great technological changes like advent of steam power for ship population and replacement of wooden hull by steel and also the major event of opening of Suez Canal in 1869 gave boost to world maritime commerce and triggered the first wave of port development in India.

A beginning was made to construct the first of the dock systems that exists today in the ports of Bombay, Chennai and Kolkata, This phase of development lasted till about the beginning of the First World War. During the period of l9l5-50, there was lull in the port development activity at these ports. But the work on the development of two new major ports at Visakhapatnam and

Cochin were taken-up. Visakhapatnam was declared as a major port in 1933 and the new major port Cochin was born in 1936. During World War II and thereafter there was no significant port development activity in the country.

DEVELOPMENT OF INDIAN PORTS DURING POST-INDEPENDENCE ERA

1. First Five-Year Plan Period (1951-56)

After the attainment of independence in 1947, India adopted the strategy of planned development through various Five Year Plans. The beginning of the First Five Year Plan from 1951 onwards saw another period of hectic activity in port development. The significant development in the First Five Year Plan was the development of a major port at Kandla as a replacement to Karachi, which became a part of Pakistan. A modern maritime oil terminal to receive the crude requirements of the two refineries set-up at Trombay was commissioned in Bombay in 1955-56. In the other major ports, the developmental activities were mainly in the nature of making good the deficiencies, which were thrown-up during the war and the post war period.

2. Second and Third Five Year Plan Period (1956-61 to 1961 to 66)

With the increased tempo of industrialization after independence, the volume of cargo traffic handled at Indian ports almost doubled from about 19 million tons to about 40 million tons during the 10-year period from 1950-51 To 1960-61. More significantly, the composition of traffic, which is a very important parameter in planning the port facilities also, changed from break bulk to predominantly bulk cargoes. The percentage of bulk cargo increased from about 50 per cent to 68 per cent. These factors mainly governed the port development during the Second and Third Five Year Plan Period. Some of the important developments witnessed during this period are as follows:

1. There was the creation of 4 alongside berths at Visakhapatnam with a fully mechanized berth exclusively for loading of iron ore. This was the first installation of its kind in Indian ports for fast loading of iron ore. An oil wharf with two berths to cater to tankers bringing crude to the Caltex refineries was also constructed.
2. Virtually the berth capacity of Madras port was doubled by construction of Jawaharlal Dock with three berths: Earmarked for handling bulk cargoes of iron ore and coal.
3. Four additional cargo berths and coal berth were constructed.
4. The port of Mormugao joined the list of Indian ports after its liberation from Portuguese control in I961 and formal declaration as major port in I963.
5. Paradeep was developed as an outlet for iron ore export with draught of 12m, which was the deepest in the country at that time.
6. A beginning was made for the development of two all weather major harbors at Mangalore and Tuticorin with complete indigenous know-how.

In addition to these, considerable improvement in the handling capacity at various ports was made by modernization of handling equipments. Despite port development activities carried-out during the first three Five Year Plans, port conditions remained deficient and imposed costly delays in shipping and high port costs. Draft limitations and lack of mechanized handling facilities especially for bulk cargoes proved to be a major constraint in receiving large tankers and bulk

carriers. These factors along with pressing need to be competitive in international market influenced the port development programs undertaken subsequent to the third Five Year Plan.

3. Fourth and Fifth Year Plan Period (1969-74 to l980-8l)

Some of the major port development works undertaken during this period are enumerated as follows:

1. Creation of a Satellite Port at Haldia, 56 nautical miles downstream of Kolkata was undertaken in 1966 to overcome the limitations of Kolkata port in respect of permissible drafts and lengths of vessels which could be received and cater to modern bulk carriers and tankers. This was commissioned in 1977.
2. An outer Harbor at Visakhapatnam port with an iron ore berth was designed to receive 1,00,000 DWT vessels initially and 2,00,000 DWT size vessels ultimately and with modern ore handling system with the rated capacity of 8,000 TPH initially and capable of increasing to 16,000 TPH was commissioned in December.
3. An outer harbor! at Madras known as Bharathi Dock was created to receive 1,00,000 DWT oil tankers and ore carriers,
4. At Mormugao, the premier ore loading port of India,‘ modernization of ore handling facility which was primitive and very slow was launched in l970 to enable the port to receive 60,000 DWT ore carriers initially and 1,00,000 DWT size vessels ultimately with the rated loading capacity of the system as 8,000 TPH.
5. At Bombay, t1; Dock Expansion Scheme providing additional berthing facility in the Alexandra Dock (Present lndira Dock) and Ballard Pier Extension to cater to passenger traffic were completed and commissioned.
6. At Mangalore, a new lagoon type major harbor with 30m draft was developed and a new major port called new Mangalore port was born in May, 1974.
7. Similarly, at Tuticorin, an artificial new harbor with 30m draft was developed and Tuticorin was declared as a major port in July, 1974.
8. The agreement to export Kudremukh Iron Ore to Iran necessitated the creation of bulk ore handling facilities at Mangalore capable of receiving 60,000 DWT ore carriers. This project was taken-up with the financial assistance of Iran and was commissioned in August 1980. This scheme marked a technological advancement in the export of iron ore in the sense that the ore is transported in slurry form from the mines to the port where it is converted to pellet form and then were exported.

4. Sixth Five-Year Plan Period (1980-85)

The notable development in the Sixth Five Year Plan was the sanctioning of construction of new port at Nava Sheave, near Bombay, which was under consideration for a long time, with the objective of decongestion and mitigation of environmental degradation of Bombay. Under this project, it was envisaged to provide the state- of-the-art handling facilities for bulk cargoes (fertilizer/food grains) and containers. POL handling berths and other facilities were added at Kandla, Bombay, Cochin, and Visakhapatnam. A new fertilizer berth with fertilizer unloading equipment was completed at Cochin. The first modern container terminal was commissioned in

December 1982 at Madras port. Additional general cargo berths were commissioned at Kandla, Mormugao, New Mangalore, Tuticorin, Visakhapatnam, and Paradeep.

5. Seventh Five-Year Plan Period (1985-90) and Annual Plans (1990-92)

The remarkable development during the Seventh Plan period was commissioning of Nhava Sheva Port (now known as Jawaharlal Nehru Port).

The other important developments were:

- Deepening of outer harbor at Madras from 14m to 16.2m to cater to 1,30,000 DWT size ore carriers.
- Commissioning of second oil berth at Madras port.
- Commissioning of a captive berth for handling fertilizers at Paradeep port.
- Acquisition of container handling equipment at the ports of Bombay and Cochin.

The main developments during the Annual Plans 1990-92 were:

- Commissioning of second oil jetty at Haldia.
- Completion of construction of barge unloading jetty at Mormugao for handling iron ore.
- Completion of extension of container terminal at Madras port.

6. Eighth Five-Year Plan (1992-97) and Ninth Five Year Plan (1997-2002)

In the return of economic reform launched in mid-l99l, the stress on infrastructure assumed added significance during two Five Year Plan periods. The concept of privatization of major ports and corporatization of some ports took its shape in this decade. The Rakesh Mohan Committee appointed by the Government of India submitted its report on present state of infrastructure and the future trend for ten years 0996-2006). Some of the major developments in this decade are:

a) Container terminal was commissioned at JNPT and Tuticorin port with private participation. The JNPT container terminal has been completed with the collaboration of the Pacific and Orient ports (P&0 ports) of Australia at a cost of $1 70 million. The Tuticorin terminal, costing $21 million, was executed under a tie-up with the port of Singapore Authority.

b) The fifth oil jetty in Kandla has been completed with an investment of $5 million with Indian Farmers and Fertilizer Cooperative (IFFCO) while the work for another facility was completed in the same port having tieing with Indian Oil Corporation at a cost of $20 million.

c) Two mega projects, namely, construction of a new port at Ennore and establishment of mechanized coal handling facility at Paradeep port costing a total of Rs. 19 billion with financial assistance from Asian Development Bank (ADB) were completed.

SHIPPING ASSOCIATIONS

Shippers associations are operationally similar to freight forwarders in that they consolidate individual small loads into bulk shipments to gain cost economics. However, shipper associations are voluntary non-profit entities where members operating in a specific industry, join

to manage small shipment purchases. Typically, members purchase products from common sources or the sources of supply are located in one area.

A group of shippers form an association, and establish an administrative office at a point of frequent merchandise purchase. The office arranges for individual purchase orders to be placed with a local supplier. When sufficient volume is accumulated, a consolidated shipment is made to each member's place of operation. Each member is billed for their portion of the shipment plus a prorated share of the association's fixed costs.

1. Shippers organizations and consultancy machinery

The shippers making use of liner shipping services were in a very weak bargaining position, because while they were unorganized, the liner shipping companies with whom they dealt with were organized into powerful conferences.

The reports prepared by the UNCTAD such as Consultation in Shipping Establishment of National and Regional Shippers Bodies, for consultation and negotiation between shippers and ship-owners and its further Report on Consultation in Shipping gave an impetus to the formation of Shippers Councils.

UNCTAD has prepared a number of reports on Consultation Machinery Protection of Shippers interests, under headings such as Shippers councils, Shipping Investigation Units, etc. The prime purpose of shippers council is to unite shippers and to gain bargaining power to obtain adequate and efficient shipping services at the minimum cost. They also provide ship-owners, government agencies, and port authorities with a means of communicating with shippers and of obtaining an authoritative shipper viewpoint. By encouraging and giving guidance on the function of shippers councils and setting-up of consultation machinery the UNCTAD had helped to reduce imbalance between the powers of organized liner conference and a very large number of shippers who were unorganized previously.

2. Directorate General of Shipping

Till 1949 the Ministry of Commerce was responsible for all matters relating to policy and administration of merchant shipping. Subsequently. it was felt that there should be a separate organization to deal with all executive matters relative to merchant shipping. Accordingly, the Directorate General of Shipping was set-up in September 1949. Its function, inter alia, include:

1. Matters affecting merchant shipping, navigation, administration of merchant shipping.
2. Development of Indian shipping; and
3. Regulation of ocean freight rates in overseas trade.

It is headquartered in Mumbai and deals with all matters (including ocean freight rates) that pertain to merchant shipping. It has a Freight Investigation Bureau (FIB), which looks specifically into the question of freight rates. The FIB organizes shippers on the basis of regional shippers associations and there is an apex body known as the All India Shippers Council.

3. Freight Investigation Bureau (FIB)

FIB was set-up in the Directorate General of Shipping in 1959. This has branch offices at Calcutta, Cochin, Kandla, Chennai, and Vishakhapatnam. The main functions of FIB are:

a) To investigate into the representations of shippers/shippers councils relating to high/anomalous/discriminatory freight rates and to secure necessary adjustment;
b) To critically examine proposals of conference lines on periodic increases in freight rates and to provide guidance to shippers councils with a view to enable them to have an effective dialogue with conferences on such proposals;
c) To provide spot assistance to shippers all over the country in procuring timely and adequate shipping space;
d) To collect, maintain, and examine freight rates of conferences/shipping lines operating in India's overseas trade and also in international cross trade;
e) To analyze the impact of changes in freight rates and to keep shippers; councils and other organizations concerned posted on such amendments;
f) To investigate into complaints regarding lack of shipping facilities; and
g) To serve as a liaison organization between shippers and shipping companies to solve shipping and freight problems through mutual consultation.

4. All India Shippers Council

The Freight Investigation Bureau has organized shippers in all parts of India in five regional shippers associations namely, Eastern, Western, Southern, Northern and Southwestern. There is an apex body on all India basis, namely, All India Shippers Council, New Delhi.

SUMMARY

India's shipping sector continues to expand at a frenetic pace, with a combination of increasing consumer demand and overseas trade volumes boosting shipping volumes to and from the country's ports. The country's growing trade relationship with China is expected to provide fuel for growth over the next few years, while the country is also establishing itself as an important transshipment hub on Europe-Asia trade routes.

Government is responsible for creation of the trained manpower required for the merchant navy fleet of the country. This national obligation is being met through the Government training institutes and number of other approved training institutes in private sector. The training institutes established by the Government are: Training Ship 'Chanakya'; Marine Engineering and Research Institute (MERI), Kolkata; Marine Engineering & Research Institute (MERI), Mumbai; LBS College of Advance Maritime Studies & Research, Mumbai. In addition to the above, there are about 128 training institutes in the private sector approved by the Director General of Shipping, imparting pre-sea and post-sea training in various disciplines.

KEY TERMS

All India Shipping Council
Air Cargo
Charting Wing
Directorate General Shipping
Freight Investigation Bureau (FIB)
Port Infrastructure
Shipping Arrangements
Shippers Associations
Ship Building
Shippers Organizations

International Air Transport Association (IATA)

REVIEW QUESTIONS

1. Explain the history of Indian shipping industry.
2. What are the problems faces by the shipping industry?
3. Explain about various ports in India.
4. What are the development activities in shipping industry in India?
5. Write short notes on:
   a) Shippers Organizations.
   b) Directorate General of Shipping.
   c) Freight Investigation Bureau (FIB).
   d) All India Shippers Council.

DISCUSSION QUESTIONS

1. Discuss about past and present shipping industry in India.
2. Discuss ports infrastructure development in India.
3. Discuss the Indian ports during pre-independence and post-independence Era.

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GLOBAL SHIPPING OPTIONS

Learning Objectives

After reading this chapter, you should be able to:

- Understand the international shipping industry.
- Know the regulation of International shipping.

- Understand the different types of ships.
- Know the international shipping routes.
- Understand the types of operating ships
- Know the organization of a shipping company.

INTRODUCTION

Shipping, or carriage of goods by water, has played a significant role in the development of human society over the centuries. Shipping has been a crucial link by which commercial relationships have been established between widely separated parts of the world. There are two major types of shipping services:

a) Shipload Services

These move goods in bulk for one or a few shippers.

b) Liner Services

These carry relatively small shipments of general cargo on a regular schedule for many shippers.

Some ships are owned by firms engaged in the production or processing of goods in bulk. For example, tankers owned by petroleum companies, and bulk carriers owned by steel companies. Most ships, however, are owned by firms whose prime business is shipping. These owners make their vessels available to importers/exporters through a highly efficient international network of shipping brokers.

Shipping is not just about vessels and tonnage. Shipping capabilities of a nation are not solely measured in terms of the quantum of tonnage under control. [n a knowledge based economy, soft intangible parameters like human capital, information technology and expertise are becoming increasingly important giving rise to innumerable opportunities. India has already become the source of quality Seafarers to global ship-owners. Indian shipping firms can rely on the inherent it skills of Indians to play a pivotal role in its activity involved in international shipping and the country may come-up as a business center for information processing requirements of the international ship-owners.

HISTORY OF SHIPPING INDUSTRY

Commercial shipping can be traced back to the Phoenician merchants who transported goods across the ’Mediterranean. The Venetians-owned huge merchant fleets from 1300 AD to 1500 AD. The Dutch held the largest shipping fleet from 1600 AD to 1650 AD. Till the nineteenth century, merchants dominated the shipping industry. The common freight carrier service started in 1818.

The shipping industry boomed after the opening of the Suez Canal (1869), which facilitated faster trade between Europe and Asia. In 1960, the first nuclear powered cargo and

passenger ship 'Savannah' was launched. From the 19705 to the 1980s, the container-shipping sector grew exponentially. The industry continued its growth, especially in Asia, with Hong Kong inaugurating the world's largest container port in 1989.

INTRODUCTION TO INTERNATIONAL SHIPPING INDUSTRY

Shipping is the safest and most environmentally benign form of commercial transport. Perhaps uniquely amongst industries involving physical risk, commitment to safety has long pervaded virtually all-deep sea shipping operations. Shipping was amongst the very first industries to adopt widely implemented international safety standards. Because of its inherently international nature, the safety of shipping is regulated by various United Nations agencies, in particular the international Maritime Organization (IMO) which has developed a comprehensive framework of global maritime safety regulations.

Around 90% of world trade is carried by the international shipping industry. Without shipping the import and export of goods on the scale necessary for the modern world would not be possible. Sea-borne trade continues to expand, bringing benefits for consumers across the world through competitive freight costs. Thanks to the growing efficiency of shipping as a mode of transport and increased economic liberalization, the prospects for the industry's further growth continue to be strong.

There are around 50,000 merchant ships trading internationally, transporting every kind of cargo. The world fleet is registered in over 150 nations, and manned by over a million seafarers of virtually every nationality. Shipping is the least environmentally damaging form of commercial transport and, compared with land-based industry, is a comparatively minor contributor to marine pollution from human activities. There has been a substantial reduction in marine pollution over the last 15 years, especially with regard to the amount of oil spilled into the sea, despite a massive increase in world sea-home trade.

GLOBAL NATURE OF SHIPPING

1. Shipping by its very nature has always been global. Long before the term globalizations became fire buzzword it is today, ships plied the world's oceans playing the dual role of serving crucial links in support of other industries and making maritime transport an industry in its own right.
2. The European Commission in the Green Paper towards a future maritime policy has recognized the international dimension of shipping. As oceans are global by nature, cooperation with third countries within multilateral is a must; and that governance of the oceans must be developed within an international context, based on universally applicable rules that concurrently cater for the specific needs and circumstances of different maritime regions.
3. Even within a commercial context, the European Commission recognizes that European maritime activity mostly competes in a global market and that such competitiveness necessitates an international level playing field for maritime sectors. For Malta this clear

acknowledgement by the European Commission's document is very significant and positive.

4. It is of paramount importance that the global nature of shipping is never far from the collective consciousness of regulators. We need to keep in mind the fact that a global level playing field is vital for the smooth flow of world trade.
5. The vast majority of entrepreneurs in any business of a trans boundary nature perceives "the disparate sets of laws and regulations governing trade from one country or region to another" as the biggest obstacle to trade. One can clearly see that a level playing field in the regulatory regime is of even more vital importance to the shipping industry, which is international by its very nature.
6. Every effort has to be made to ensure such uniformity and that the premier institution governing maritime affairs, the international Maritime Organization, should not be undermined in its role as the UN. Agency tasked with setting global safety standards safeguarding the environment and human life at sea.
7. The shipping industry is transporting 90% of the world's trade. Most ships move from country to country as part of their normal trading pattern.
8. The safety record of shipping is the envy of shore-based transport sectors, and shipping is the most environmentally friendly form of transport.
9. The 50,000 ships that make-up the international trading fleet are technically sophisticated, high value assets. The cost of a new ship can easily exceed U.S $100 million. Merchant ships generate an estimated annual income of over U.S$200 billion in freight rates within the global economy.
10. The shipping industry is a truly global community. It is intrinsically international; indeed it was the very first global industry.
11. In addition to moving cargo around the world, ships may be owned in one country, managed from another, registered in a third country and manned by seafarers from one or more others. Multinational crews are very common, and the international language in shipping is English.
12. Approximately one and a quarter million seafarers are employed worldwide in shipping, which relies heavily on the expertise of officers, especially senior officers, from countries in the developed world. But most ratings, and a growing number of officers, are employed from countries in the developing world.
13. Modern shipping comprises companies from across the globe with virtually every nationality represented as seafarers and in the industry's shore-based workforce.
14. Because of the international nature of shipping, its safe operation is regulated by United Nations agencies. Ships' officers, regardless of their country, are trained and qualified in accordance with precise global standards. There are also thirty international conventions directly governing the employment standards of seafarers, more than any other industry. They cover issues such as living and working conditions, which are generally very high, as well as health and safety, the payment of wages and repatriation.

REGULATION OF INTERNATIONAL SHIPPING

Most shipping is by nature international the carriage of goods between countries and across international waters. A ship may be owned, financed, registered, insured, and managed, each in a different country. When a ship is registered in any given country, it becomes subject to the laws of that country at all times. Each country has the right to establish its own shipping laws. International shipping conventions have been reached by a number of intergovernmental organizations, such as the international Labor Organization (ILO) and the international Maritime Organization (IMO).

Indian shipping has the 14th largest fleet in the world as per deadweight tonnage. Indian shipping fleet consists of around 515 vessels with an art of 7.06 million and dwt of 11.5 million ones. The industry is governed by three separate Acts, via:

a) The Merchant Shipping Act, 1958,
b) The Inland Vessels Act, 1917, and
c) The Coasting Vessels Act, 1838.

While most of the vessels are registered under Merchant Shipping Act and dealt with in the sector, smaller barges and coastal vessels are governed by the latter two Acts, and lighter age or barging industry comes under port sector. The industry has historically catered to only domestic shipping requirements, although the private sector companies like Great Eastern Shipping Company and Essar shipping are increasingly getting involved in international cross trade. Not with standing this focus on Indian trade. Ship-owners' share in the country's overseas trade is hardly thirty per cent in volume terms.

Regulations concerning shipping are developed at the global level. Because shipping is inherently international, it is vital that shipping is subject to uniform regulations on matters such as construction standards, navigational rules and standards of crew competence. The alternative would be a plethora of conflicting national regulations resulting in commercial distortion and administrative confusion, which would compromise the efficiency of world trade.

The shipping industry is principally regulated by the international Maritime Organization (IMO), which is the London-based United Nations agency responsible for the safety of life at sea and the protection of the marine environment. The international Labor Organization (ILO) is also responsible for the development of labor standards applicable to seafarers worldwide.

IMO has adopted a comprehensive framework of detailed technical regulations, in the form of international diplomatic conventions, which govern the safety of ships and protection of the marine environment. National Governments, which form the membership of IMO, are required to implement and enforce these international rules, and ensure that the ships, which are registered under their national flags, comply.

The level of ratification and enforcement of IMO Conventions is generally very high in comparison with international rules adopted for shore-based industries. The principal responsibility for enforcing IMO regulations concerning ship safety and environmental protection rests with the

flag states (i.e., the countries in which merchant ships are registered which may be different to the country in which they are owned).

Flag states enforce IMO requirements through inspections of ships conducted by a network of international surveyors. Much of this work is delegated to bodies called classification societies. However, flag state enforcement is supplemented by what is known as Port State Control, whereby officials in any country which a ship may visit can inspect foreign flag ships to ensure that they comply with international.

DIFFERENT TYPES OF SHIPS

Ships are categorized into various types based on the goods they carry as given below:

1. Mail and Supply Ships

The mail and supply ships carry mail, supplies, and cargo to out-of-the-way locations. Typical of this type of ship is the RMS St. Helena that sails from the U.K. to Smith Africa, the Ascension, Canary, and St. Helena Islands.

2. General or Dry Cargo Ships

Prior to the days of containerization, all cargoes were carried on what is known as general or dry cargo ships. The cargo was known as break-bulk cargo. With the advent of containerization there are fewer general cargo ships. Generally, they carry cargo that is too large to be carried in a container, e.g., steel, rolls of wire, and machinery. However, they also carry boxed goods that are too small to justify use of a full container. They are much slower to load/unload and thus have longer port times. Whereas a dry cargo vessel may take 3-4 days to load or discharge. A container ship can achieve the same in a matter of hours. Although general/dry cargo vessels remain as the largest (in pure numbers) of cargo carrying vessels, they are often smaller (rarely above 50,000 gross tone) than the specialized vessels that are slowly replacing them.

3. Expedition Ships

Expedition ships are built for adventurous purposes. They can carry between 52 and 117 passengers on voyages to Antarctica. South Georgia and the Falkland islands, Patagonia islands of the South Atlantic, the Amazon, Iceland, Greenland, the North-West Passage, Hudson Bay, plus Lost Islands of the South Pacific, Polynesia, Melanesia, and the Russian Far East. The services are available via the Cruise People Ltd. Maris Freighter Cruises, and Travel Tips as well as traditional freighter voyages.

4. Coasters

Coasters are small cargo ships, often containerships that run on feeder routes. They carry relatively small number of containers from small ports to major ports. They travel along the coast, making many stops in a short period of time, hence the name.

5. Bulk Carriers

Bulk carriers or Bulkers are the great work horses of the shipping world, carrying raw dry cargoes in huge cavernous holds, such as coal, iron ore, grain, sculpture, scrap metal, and other loose cargo. They come in various sizes and configurations depending on the nature of the cargo and the area served.

Bulkers are of three sizes as given below:

a) Handy Size Bulkers: of about 25,000 Dead Weight Tons (DWT)
b) Medium Size Bulkers (Panama): of about 75,000DWT
c) Giant Bulkers (Cape Size): over 2, 00,000DWT

Due to the heavy use that these vessels are put to, their life expectancy is less than it would be for, say, a container ship. A problem facing the industry is that with freight rates so high, there is little incentive for ship-owners to scrap the, older tonnage when it is still able to earn very good money. Also, the dangers of running old bulk carriers can be catastrophic not only for the vessels and their insurers, but also for the crews who have to man them.

Some of these vessels are too large for the Panama Canal (the same can be true of containerships) known as post Panama ships. These ships are more than 32.2 meters long in the beam. They are almost always employed in tramp service, i.e., they often work on contract basis rather than carrying cargo on a regular, established route. A special type of bulk carrier is found on the Great Lakes between the U.S., Canada, and, adjoining waterways.

Currently, there is a huge demand for these vessels, driven by the extraordinary expansion of the Chinese economy. Imports of iron ore into China have boosted the earnings of bulk carrier owners as freight rates Eve gone thr0ugh the roof into uncharted territory. A rare beast, Geared Bulk Carrier carries its own cargo cranes very useful when visiting ports lacking portside cargo handling equipment.

6. Tankers

Tankers are vessels used to carry liquid cargoes such as crude oil and acids, chemicals. Normally, the cargoes carried in tankers are hazardous items and, therefore, passengers do not travel on such ships.

a) Types of Tankers

Tankers are designed to carry all liquid cargoes. The carriage of crude oil has brought it unwelcome attention and largely unjustified criticism. Oil tankers come in two basic flavors:

- The crude carrier, which carries crude oil, and
- The clean products tanker, which carries the refined products such as petrol, gasoline, aviation fuel, kerosene, and paraffin.

b) Size of Tankers

Range in all sizes, from the small bunkering tanker (used for re-fueling larger vessels) of 1OOODWT to the real giants - the Very Large Crude Carrier (VLCC) of between 2, 00,000-3,00,000DWT and the Ultra Large Crude Carrier (ULCC) of over 3,00,000DWT.

Over 60 per cent of the world's oil is transported by these tankers, and over 99 per cent of them arrive safely without causing pollution. Indeed most oil pollution on beaches comes from the engine rooms of vessels (of all types) rather than from the cargo tanks of tankers.

To ensure that oil tankers transport to all parts of the world, and not just to the affluent western countries where freight rates might be expected to be higher (and therefore tankers congregate), there is a system called World scale, which ensures that net freight earnings for tanker owners are the same regardless of length of voyage and varying costs (such as bunkers and port

dues), so that a tanker will earn (net) as much per day on a route that is long and arduous as they will on a short and low cost voyage. This ensures that tankers take their goods to all parts of the world and do not just concentrate on the lucrative Middle East to Western Europe or U.S.A. routes.

Increased regulation on the construction, maintenance, and crew competence of tankers, such as under the Maritime Pollution (MARPOL) Convention and the U.S. Oil Pollution Act of 1990 (OPA90), has required double hull tankers to be built (of questionable value, but of undoubted high cost), the phasing out of single skinned tankers and the Certificates of Financial Responsibility (COFR), which demonstrate that the operator has the financial backing, should a pollution event occur, to pay for the necessary clean-up.

In the "real world" (i.e., outside U.S.A.) international agreements such as the CLC and Fund Conventions have provided clear levels of compensation that the oil industry are committed to provide to protect the environment. The MARPOL convention is a major force in ensuring a professional attitude is required of any vessel owner/operator carrying persistent oil as cargo.

7. Refrigerated Vessels and Reefers

Refrigerated cargo carrying vessels or reefers is purpose built to carry fruit, meat, and other food products across the sea fresh and clean. Perhaps the most famous of these types of vessels are the banana carriers, trading between the Caribbean and Europe. They are sleek and fast, as their trade demands, with cooling (refrigeration) equipment to keep their cargoes fresh.

Historically, the months of February and March see the greatest use of these vessels. This gives a clue to their current weakness, from a commercial point of view. Any vessel that is governed by highly seasonal trades will inevitably have high and low activity periods during a year. At some points, there is very little work for dedicated refrigerated cargo vessels, making them rather inefficient compared to container ships (now with reefer box capabilities) which can switch trades in low season to carry other goods. The dedicated reefer vessel (probably the most loved type of cargo ship to work on as a seaman) is becoming less common, with fewer being built each year, due to the flexibility of usage of reefer containers.

8. Livestock Carriers

The carriage of live animals around the world is performed by specialist vessels, designed (or adapted) to transport a large number of cattle and sheep in secure but humane conditions. The trade is largely 'from Australia to the Middle East and/or South-East Asia. One modern vessel may can't up to 1, 25,000 sheep.

This transport of live animals requires experienced and specialized operators. The dangers of disease, rejection, injury, and death to and from the animals are particularly high, coupled with the current activities of animal rights activists. When a livestock carrier catches fire or sinks, the loss of life of the animals can be appalling. (Remember, sheep and cattle cannot swim and they are not very good, when it comes to Lifeboat Drills.). In 2003, Australia exported over 4,500,000 sheep and nearly 8, 00,000 cattle. These vessels have to sail—in for trade in the hot temperatures for various reasons.

9. LNG Carriers

The Liquefied Natural Gas (LNG) carrier and the Liquefied Petroleum Gas (LPG) carrier are products of the late twentieth century. LNG and LPG are the preferred fuel types of certain countries for their industrial power needs, e.g., Japan. But is not easy to transport these cargoes. LNG is a gas, in its natural state, it needs to be either pressurized into a liquefied form, or kept as a liquid by reducing the temperature to transport it, (by simple application of Boyle's law in physics).

The shape of an LNG carrier is quite unmistakable, with the shape of the Moss tanks (which are like enormous spherical thermos flasks) visible along the deck, which has led to the nickname Dinosaur Eggs Carriers. An alternative design is known as the membrane type, which allows for a more standard shape of vessel without the "eggs". Obviously, the carriage of an explosive gas kept at below freezing temperatures as an unstable liquid presents a very dangerous cargo, yet it is for this very fact that LNG carriers have about the best safety record of all maritime vessels. Only the best officers and crews are employed on these vessels, and the vessels themselves are maintained meticulously, and renewed frequently. There have been accidents involving LNG/LPG carriers, but where such events have occurred, the crews or salver have so far successfully managed to vent-off the cargo into the atmosphere, thus rendering the lethal cargo harmless.

10. Car Carriers

The car carrier or more correctly the PCC (Pure Car Carrier) or Pacific (Pure Car/Truck Carrier) could never be described as a beauty of the seas, yet in its rectangular design, is purpose built to carry a large number of cars. The manufacturers of cars, mainly in Japan and Europe, use these vessels to ship large quantities of their products around the world. Every Japanese Korean, or European car seen on the roads may have been brought across on one of these car carrier vessels.

A few years ago, a car carrier called the Reijin ran aground and partially sub-merged off Spain full of Japanese cars. The manufacturers were so scared of people stealing w hole cars or pans from the wrecked vessel (which was easy to reach from land) that the} ordered the vessel. And its cargo of brand new vehicles, to be towed-out into deep water and sunk.

More recently, the Tricolor sank after a collision with another vessel in the English Channel proving, yet again the inherent weakness in the design of vessels with a single massive hold (at least until water gets in). With no watertight bulkheads, the free surface effect of that water rapidly destabilizes the whole vessel, and in a matter of seconds they can capsize.

11. Container Ships

General cargo was historically carried in dry cargo vessels. without any particular specialization. Cargo loading and unloading was always a slow, laborious task due to the varying shapes, sizes, weights, and fragility of the numerous cargoes being carried on any one vessel. The idea of standardizing the carrying box, or container at 20 feet long was a breakthrough that allowed for vessels to be designed to carry these standard sized boxes and for dockside equipment also to be designed to stack and store these specific shapes.

The boxes they carry are containers that are generally found in twenty and forty-foot lengths. They can be filled with any type of cargo e from television sets to fruit or meat. The capacity of a containership is measured in Twenty-foot Equivalent Units (TEU). Thus, a freighter

that can carry 1,600TEUs is a bit small by the standards of some of the giant container ships that pi)' the seas of the world today. This is a sister ship of the vessel l sailed on from Los Angeles, CA. to Australia (boomerang route). The Regina M is one of the largest containerships in the world, capable of carrying about 6,600TEUs.

Initially, container ship were small vessels of up to [0,0000WT, carrying no more than a few hundred Twenty-foot Equivalent Units (TEU). but have grown in size as the success and economies of these vessels have become more obvious. Today‘s container ships are being built to take 8,000TEU, with plans to build 10,000-l 2,000TEU ships.

Many goods need larger boxes, so there is a larger standard sized container the Forty-foot Equivalent Unit (FEU). On board a modern containership, the complex method of loading the TEU and FEU in an order that will facilitate off loading at the other end is now largely computerized. As if specially designed to give Chief Officers a headache, a modern development is the high box, a standard container in length and width, but 9' 6" high instead of the standard 8'. Refrigerated containers (reefer containers) have become very popular for the carriage of meat and fruit.

The containers are anchored by twist locks, which simply twist round to lock the eight comers of the container to the neighboring container. The outside of the whole stack is then further fixed with lashing poles diagonally from one comer to another. These vessels are built for speed, and can reach upwards of 28 knots, moving cargoes around the world. Globally, storing and returning empty boxes has become an industry in itself.

These containers can be off loaded from a ship, and rapidly loaded onto trains or onto container lorries for onward transport to the place of delivery. This is known as through-transport or inter-modal transport. Recently this has led to serious security concerns.

12. Heavy Lift Vessels

Heavy lift vessels are quite amazing vessels built to load, carry. And discharge large, unusual shaped cargoes (or even smaller vessels) that will simply not fit inside the holds of conventional vessels.

Shaped quite unlike other cargo vessels, some heavy lift vessels can be ballasted down, so that cargoes (often on pontoons) can be floated on. The vessel is then raised, lifting the cargo (now resting on the deck of the heavy lift vessel), out of the water, ready for ocean transit, often to the other side of the world. These specialist vessels are often used in the oil industry for the carriage of jack-up rigs. Other unusual cargoes include power plants, desalination units generators and yachts.

13. Tugs

The tugboats have a number of functions ranging from towing vessels into berths, to fire fighting, salvage and anchor handling/positioning. They range in size (and power) depending on the tasks they are required to perform. A small harbor tug may only have 300-4OOBHP (Brake Horse Power), whereas a large ocean-going salvage tug may have up to 10,0008HP engines capable of towing large cargo vessels off sandbanks and rocks.

The European tug is very different in shape and size from the US. Push boat which plies up and down the Mississippi river system. Modern European style tugs are being built with 'their

engines amidships, propellers on the two sides of the vessel, to allow 360 degree maneuverability, i.e., turning on a Sixpence, and sailing in any direction with equal ease — forwards, backwards, and sideways.

14. Ro-Ro Vessels

The Ro-Ro or the Roll-on Roll-off vessel comes in a number of shapes and sizes. But generally they are in two types:

a) Passenger Ro-Ro

These are the common passenger ships. It is probably the only type of cargo vessel that most people have traveled on. Usually, a rear door (but sometimes a bow door) allows for vehicles to be driven on and off, stored on the car deck below the passenger accommodation areas.

b) Cargo Ro-Ro

The cargo Ro-Ro is less "plush" than the passenger type, as these vessels are designed for the carriage of commercial vehicles where luxurious passenger accommodation is not a primary consideration. Considerable concerns have been expressed over the bow-door type of Ro-Ro design.

The indication of free enterprise was one such vessel, where a practice of sailing, before the bow door was fully closed, had been allowed to develop. Tragically, on leaving Zeebrugge, the folly of this practice led to the disaster that claimed nearly 200 lives. If water is allowed to enter the car deck the stability of the whole vessel can be rapidly affected. It is estimated that it only needs one centimeter of water over the whole car deck, for the vessel to become so unstable that it can overturn. The Estonia was another such vessel where, in a storm, the shield over the bow door was ripped-off. Once water penetrated the car deck the vessel began to turn over and sink. In the bitter waters of the Baltic Sea, the loss of life was terrible. Another earlier accident was that of the Sea speed Dora which led many to call these vessels Ro-Ro, Roll-on, Roll-off and Roll-over – ships.

Advantages of Ro-Ro

- It is the safest and most inexpensive way to handle and transport oversized or special project cargo.
- Shipments often move as one piece using specialized trailers from origin to port to destination.
- It involves less physical handling and there is no need for costly dismantling and reassembly.
- There is no exposure to water or the elements because the cargo is always secured in ship's RORO/Containership's garage decks for the entire voyage.

SHIPPING ROUTES

Shipping routes reflect world trade flows. Sailings are most numerous and most frequent on routes where trade Volumes are largest and demand is therefore greatest. Sea routes are confined to a stipulated voyage and distance. They normally are inclined to approximate to a curve of the

great circle since it guides the ships to undertake the shortest journey. They connect various ports or destinations or trade points. They determine the freight.

There are many shipping routes and the following are the busiest and important ones:

1. Northern Sea Route

The Northern sea route connects Asia and Europe. It is the shortest waterway between the North Europe the Asia-Pacific region and the Western Coast of the U.S. and Canada. It plays an important role in Russia's national transportation policy. In l993-l997, the volume of see cargo along this route was already ISO-200 thousand tons a year.

In 1993 during the Antic's summer shipping season when the cargo traffic peaked 15 Russian ships with 2l0 thousand tons of transit goods passed along the route. Also, 8 ships carrying metals, fertilizers, and timber traveled from ports in Russia. Latvia. Sweden, and Finland to China, Japan and Thailand.

2. North Atlantic Route

The North Atlantic route connects the Western Europe, Eastern Canada and the Unites States. It is one of the busiest shipping routes. The traded items are mostly industrial goods foods and manufactured products.

3. Mediterranean Route

The Mediterranean route connects the North-Western Europe with the Mediterranean Eastern Europe, Southern and Eastern Asia, Australia, and New Zealand. It is also an important and busiest sea route in the world. It relies mainly on the Suez Canal.

This route is famous for the exchange of commodities such as cotton, dairy products, meat, rubber, and tea as well as for tourism. This route touches the most developed economies and the Petroleum Oil Lubricant (POL) rich countries. The POL products are carried from the Middle East and European countries through this route.

4. Panama Route

The Panama Canal links the Pacific Ocean, the Caribbean Sea, and the Atlantic Ocean. it lies in Central America. This route connects the Eastern North and South America to their western pans. Before the existence of Panama Canal, the ships were en route through Cape Horn to reach these destinations and it would take considerable time. The canal saved time and cost.

5. Baltic Sea Route

The Baltic Sea is located in the Northern Europe, bounded by the Scandinavian Peninsula and the main lands of Northern Europe, Eastern Europe, Central Europe, and the Danish islands. lt drains into the Kattegat by way of the Oresund the Great Belt and the Little Belt. Kattegat then continues in the Skagerrak into the North Sea and the Atlantic Ocean. The Baltic Sea is linked to the White Sea by the White Sea Canal and directly to the North Sea by the Kiel Canal.

In I999 the huge bridge over the Sound limited the Baltic Sea to the middle-sized vessels. In the meantime, the Baltic Sea is the main trade route for export of Russian oil. Many of the neighboring countries are rather concerned admit this since a major oil leak would be disastrous in

the Baltic given the slow exchange of water and the many unique species. The tourism industries, especially in economics dependent on tourism, e.g. in North Eastern Germany, are naturally very concerned.

6. Indian Ocean Route

The Indian Ocean is the third largest water body covering about 20 per cent of the earth‘s water surface. It is bounded on the North by the Southern Asia (the Indian sub-continent); on the West by the Arabian Peninsula and Africa; on the East by the Malay Peninsula, the Sunda Islands and Australia; and on the South by the Antarctica.

The Indian Ocean is separated from the Atlantic Ocean by the 20° East meridian South of Africa and from the Pacific by the 147° cast meridian. The Northern most extent of the Indian Ocean is approximately 30° North latitude in the Persian Gulf.

This ocean is nearly l0,000km (6200 mile) wide at the Southern tips of Africa and Australia; its area is 73,556,000ka (28,400,000 mile) including the Red Sea and the Persian Gulf. Its volume is estimated to be 2,92,l31.000Km (70,086,000 mile). Small islands dot the continental rims. The island nations within the ocean are Madagascar (formerly Malagasy Republic), Comoros, Seychelles Maldives Mauritius and Sri Lanka Indonesia borders it.

The ocean‘s importance as a transit route between Asia and Africa has made it a scene of conflict. Because of its size however, no other nation had successfully dominated it until the early 18005 when the Great Britain controlled much of the surrounding land.

7. Cape Route

The Cape route connects the Western Africa and Australia by the way of Cape of Good Hope. The traffic and the movement of cargo through this route have been now reduced by the competition with the Suez route. The exporters can reduce their movement charges if they choose the Suez route. It is tedious for the traders of Europe and Australia if they are connected through Cape route.

8. South Atlantic Route

The South Atlantic route connects the South America the North-Western Europe and the Mediterranean. Agricultural products, meat, raw materials, coffee, and fruits are mow through this route.

9. North Pacific Route

About 2000 ships a year passes through Aleutians on a major shipping route between the West Coast of North America and the Far East.

ROLE OF PANAMA CANAL IN OCEAN SHIPPING

The Panama Canal is approximately 80km long between the Atlantic Ocean and the Pacific Ocean (Panama Canal Authority 2009). This waterway cuts through one of the narrowest saddles of the isthmus that joins North America and South America. The Panama Canal uses a system of locks -compartments with entrance and exit gates. The locks function as water lifts, they

raise ships from sea level (the Pacific or the Atlantic) to the level of Lake Gatun (26m above sea level), ships then sail the channel through the Continental Divide. The Panama Canal -handles about 12% of American international Seabee trade. In December 1999, the Panama Canal became the property of Panama under the jurisdiction of the Panama Canal Authority.

The same year, Hong Kong port operator Hutchison-Whampoa took operational control of the ports at both the Atlantic (Port of Colon) and the Pacific (Port of Panama City) sides of the Panama Canal with a 25-year lease. The company also became involved in the improvement of the rail line between the two ports to handle the growing amount of containerized traffic. This rail line is important as it offers an alternative to the size limitations of the Panama Canal, which prevents large - Post-Panamas- container ships from going through.

ROLE OF SUEZ CANAL IN OCEAN SHIPPING

The Suez Canal is an artificial waterway in Egypt, connecting the Mediterranean sea to the Gulf of Suez, and then to the Red Sea. The Suez Canal is 163km long with a canal width of a minimum of 60m (Suez Canal Authority 2008). Modern ships extensively use the Suez Canal, as it is the fastest crossing from the Atlantic Ocean to the Indian Ocean. Taxes paid by the vessels represent an important source of income to the Egyptian Government. The Suez Canal has no locks because the Mediterranean Sea and the Gulf of Suez have roughly the same water level. It acts as a short cut for ships between both European and American ports and ports located in Southern Asia, Eastern Africa, and Oceania.

OPERATING SHIPS

Operating ships can be categorized as:

1. Liner Ships vessels
2. Tramp Ships vessels

I LINER SHIPS

Liner ships represent the organized sector of the shipping industry due to their fixed schedules of arrival and departure, predetermined voyages and trade routes, and published ocean freight rates.

TYPES OF LINER SHIPS

1. Conference Lines

A “conference” is a group of shipping companies that have agreed amongst themselves to levy the same freight rates and to observe the same shipping conditions (e.g., liability for damage to the exporter’s goods). Such conference lines operate on the entire observed regular shipping schedules, an important consideration for the exporter, which enables him to quote reliable shipping, and delivery dates. The conference lines charge two different shipping rates for the goods carried. The lower rate is available to exporters who sign an exclusive patronage contract, whereby

they agree to use only those ships served by the conference lines for their goods. Exporters who have not signed such a contract can still ship by conference line ships but at a higher freight rate.

2. Non-Conference Lines/Independent Lines

Non-conference lines are individual lines that operate on the traditional shipping routes with regular sailing schedules and in competition with the conference lines. They have their own individual freight rates as much as 10% lower than the conference ships. Backings are accepted from any shipper so long as space is available. No exclusive patronage contracts are required.

Advantages of Liner Ships

Liner shipping is governed by shipping conference and offers the following advantages to shippers:

a) Regular sailings to scheduled ports of call.
b) Stable freight rates for a long period of time, which helps the shipper to quote C&F prices with confidence.
c) Uniform rates for all shippers.
d) Coverage of wide range of ports. Rebates of height rates based on loyalty agreements.

II TRAMP SHIPS

A ship engaged in the tramp trade is one, which does not have a fixed schedule or published ports of call. Tramp ships have following characteristics.

1. They are tree to move anywhere on the high seas at their will.
2. Their voyage routes and schedules are flexible.
3. They travel from one port to another port on various trade routes looking for the cargo and carrying the same to various destinations around the world.
4. They arrive or depart without a fixed route or schedule.
5. They fix their voyages according to availability of cargo and as per the requirement of the shippers of these cargoes.
6. The freight rates of tramp ships depend upon the demand and supply conditions in the shipping industry. If there is a glut of shipping space, the tramp freight rates plummet. Whereas in case of shortage of shipping space, the tramp freight rates shoot-up.
7. The cargo space on the tramps is booked by the brokers located in major port cities like New York, London, Rotterdam, Hamburg, Singapore, Hong Kong, etc. They work as a link between tramp operators and shippers.
8. Tramps normally carry homogeneous cargo like grains, ores, coal, timber, fertilizers, etc.

TYPES OF TRAMP VESSELS

a) Tramp Lines

Tramp lines are companies, which do not have. a fixed route or schedule of sailing. They operate on a per-Voyage basis. Usually, their freight rates are extremely competitive.

b) National or Flag Lines

The exporter may receive a lower rate of import duty and other benefits if he uses the national shipping line of the importing country.

c) Steamship Agent

Steamship agent is a person employed by a shipping company to arrange for the berthing and clearance of vessels, including Customs formalities, loading and unloading of cargo, preparation of bills of lading, and the booking of shipping space.

d) Ship-Broker

A shipbroker acts as an intermediary between the shipping companies on the one hand and the exporters on the other. He helps to arrange charters or booking for tramp vessels and advises the ship- owner on competitive rates.

CARGOES SUITED FOR TRAMP OPERATIONS

- Bulky and heavy cargo.
- The cargo with low value cargo, which has no urgency to reach the destination port and hence is ideally suited for carriage by tramps.
- Cargoes, which do not require protective covers and any special material handling facilities.
- Cargoes, which are normally shipped in large quantities.

LINERS VERSUS TRAMP SHIPS

1. Nature

Liner ships represent the organized sector of the shipping industry due to their fixed schedules. 0n the other hand, tramps being the masters of their own will have flexible schedules of arrival and departure.

2. Weight Rates

Liner ships have fixed freight rates policy. Whereas freight rates of tramp ships-every according to the situation.

3. Operation

The Liners are mainly interested in carrying the general cargo, which requires punctual, regular, and continuous service. Also the freight rates of the Liners are fixed and the shippers know them in advance in order to plan their shipments. All these requirements do not confirm to the style of operations of Tramps.

4. Situation Favoring Cooperation

On the other hand, they cooperate with each other in a situation where the business is booming and there is a dearth of shipping space on some routes. in such cases. Liner Companies charter tramp vessels in order to meet the increasing demand from the market.

5. Situation Favoring Competition

However, under the following circumstances Liners and Tramps compete with each other:

a) As the volume of world trade is growing, the shipping companies are buying larger ships in order to carry more and more cargo in a single voyage. But sometimes when the market is depressed, large ships do not find general cargo and there is a possibility of the shipping space remaining empty. In such a situation, Liner companies may undercut Tramps to carry bulk cargo.
b) Tramps are normally hired for one-way movement of cargo. They do not often get any suitable cargo for their return voyage. In such case, they compete with Liners by under cutting the rates and try to get any cargo for their return voyage. .
c) In case of specialized ships, such as refrigerated vessels and oil carriers they may not get suitable return cargo and may carry any cargo during return voyage, just to meet operational overheads.

SHIPPING PORT OPERATIONS

The sequences of various operations right from arrival of a ship at the port till its departure from the port are as follows:

1. Shipping line submits its Import General Manifest (IGM) which is a document giving the description of the ship's cargo to the customs authorities at the port of destination of the cargo. IGM is submitted in advance of the arrival of the ship.
2. Ship announces its arrival to the port authorities.
3. Port authorities request the ship to wait on high seas until a vacant berth is available at the port to receive the ship.
4. A pilot vessel is sent by the port authorities to high seas to escort the visiting ship to the vacant berth.
5. Shipping or cargo agents complete the paper formalities for the unloading of import cargo and loading of export cargo.
6. Unloading and loading operations begin. Import cargo is unloaded and stacked at the designated area in the port premises for custom's examination and further formalities.
7. Export cargo, which is customs cleared, is loaded on board the ship.
8. After the port operations are complete, the pilot vessel escorts the ship out of the port back on the high seas.

ORGANIZATION OF SHIPPING COMPANY

A bigger organization will normally consist of the head office, the ships, offices abroad and a network of business connections such as shipbrokers, agents, stevedores, etc. who frequently is independent companies with whom the big company has fixed contracts.

Bigger liner companies, for instance, will normally operate with a great number of agents, the same number of stevedore connections as ports called at, and a number of special firms such as

shipbrokers, lawyers, tax consultants and auditors who have standing orders according to agreement.

Most shipping companies prefer to use independent firms familiar with the local conditions in each country. Partly because such firms may already be tied up with a competitor, or difficult to find, and partly because the companies have sizeable spheres of interest in the areas concerned, they have established their own offices abroad. This is particularly true of the liner companies. In this way the head office has direct Control of the Went: and will, for instance, be able to expand the offices abroad under its own supervision. In towns of molar importance one may find Owners' Representatives who have assignments of different kinds.

DEPARTMENT OF SHIPPING COMPANY

A shipping company has following departments, responsible for various tasks to be performed by the shipping company.

1. The Superintendents

Other kinds of representatives abroad are the superintendents at shipyards which build new ships on contracts, inspectors who manage repair and maintenance work, etc, Cargo Superintendents for the liner trade are also quite usual. in general there are not many of these representatives sent out from head office, and consequently they seldom create any organizational problems.

2. Organization Chart

The usual aid to showing the pattern of different functions in a shipping company is like organization chart. Such a chart is an important means of defining the areas of authority and responsibility and thus of evaluating if the organization satisfies the demands placed on it.

3. Operating Department

It is a very crucial department for a shipping company. The main concern of this department is to maximize the economic employment of the ship. As part of this end it has a major function of coordination, not only amongst the other departments of the company but also with the ship, the ship-broker and the agents who will have to attend to the various and many problems that may arise at the ports of call. Where to bunker and the amount to take one of its continuing problems is as the varying price of bunkers in different ports of the world means that careful planning can make substantial savings in this direction.

Another specific activity of this department is scheduling. Scheduling concerns where to send the ships and when. For general cargo liners this is a twofold problem:

a) The decision to open-up a new route or line. This is fundament and will require a special research team. Not only has the supply and demand on the new possible routes to be considered but also the strength of the competition, the conference involved, and whether they will accept, reject or tolerate someone joining the trade. If it is decided to join the c0nference many of the questions conceding scheduling such as sailing dates, ports, and frequency of service will be determined by the conference.

b) Once the run is established the line manager must maintain schedules and combat the problems of delays, strikes, breakdowns, and the annual dry dock. For Tramp ships the problem is to charter the vessel out to obtain a maximum profit to the company.

4. Marketing Department

Marketing in shipping as in any other business involves:

a) Product Analysis

It means assessing,

- The supply and demand for the particular shipping service in question,
- The costs of such a service.

b) Traffic and Market Development

It is concerned with how to deploy ones resources, e.g., in cruising for instance this would be to consider:

- How long should the cruises be?
- Where should the cruise go?
- For which age group, class and nationality is it designed to cater for?

c) Promotions and Product Development

It deals with

- Analyzing mistakes (passengers' complaints),
- Shore excursions.
- Food and entertainment,
- Advertising.

5. Technical Department

This department will be responsible for the maintenance and efficient functioning of the ships as carrying vehicles and the safety of the ship and crew. This requires not only a comprehensive technical knowledge but also an awareness of the precise details regarding national and international regulations concerning pollution, port health, safety, and the like.

This department must be aware of and be frequently involved in research so that the company's ships and future ship buildings does not fall behind in technical efficiency and safety.

6. Husbandry

It keeps the ships seaworthy. So, it has to look into Maintenance Research and Technical Repairs. It also looks into the personnel affairs of the ship.

7. The Legal Department

It covers a large field. Among the main functions can be mentioned Board matters, contracts for new buildings, company law, national/international taxes, and purchase/sale of ships. The company lawyer is also often on the Managing Director's staff, and/or secretary to the Board of Directors. Frequently, this department wills otter: also handle marine and other insurance matters that may arise of a legal nature.

8. The Marine insurance and Claims Department

It handles ship insurance, statistics of damage and makes preparations for adjustments of claims. All claims relating to damage of the cargo are handled in this department. These claim adjustments cause considerable work, particularly within the liner trade.

10. The Finance and Accounting Departments

It covers budgets, accounting, EDP and finance. Regular reports are prepared for each operational department, each ship and for the company as a whole. The management needs to keep the Accounting Department up to date with operational proceedings at all times, with changes and corrections when necessary. Computers have been used for a long time in modern shipping companies and for different purposes. Computers are indispensable as an advanced technical aid in the office, also for complicated techno/ economic calculations and for budgeting in the operational and liquidity area.

During the last few years long range planning has also been taken up as a subject of business administration. The reporting of the economic results should follow the organization chart and will, if correctly done, be of prime importance to the management, for the organization's mode of operations and for the control of the results. Financial questions will always be vital to any shipping company. To arrange finance when building new vessels and to find the highest interest for liquid working capital are main activities together with follow—up of the unstable currency markets and protection against currency losses.

11. The Shipping and Chartering Departments

a) The Liner Department

From an organizational point of view it is of interest to observe that the liner trade seldom operates under the supervision of one company alone. There are many different types of cooperation. In some cases the owners put their ships at the disposal of a separate joint venture company which manages the operations. In other cases they may participate as partners in a co-sailing agreement, possibly also with a pool agreement. Usually this takes place within the framework of a liner conference, where all the shipping companies serving the trade in question participate. No matter how the cooperation is arranged, it includes marketing and booking of cargo, allocation of space, responsibility for routing, documents, port facilities. Loading/discharging, control of port expenses, stevedore contracts, etc.

b) Tanker and Dry Cargo Department

It arranges employment for the vessels and is also responsible for their operation. These departments are frequently referred to as chartering Departments. The tanker department will usually handle bunker contracts for the company fleet. In the organization the lines of authority show chart the interaction between the main department and the sub departments. This authority goes down from the ship owners, through the main departments and down into the organization. The lines show how the instructions for the company are transmitted from the ship owners through the different links. All department heads are fully responsible for their own activities and they must therefore be fully familiar with all matters concerning their departments so as to be capable of

managing and making decisions. Information, messages, tasks, proposals, etc. follow the formal lines of communication. This is the main principle, but modifications occur.

c) Other Departments

Department for other activities reflects the fact that many shipping companies have assumed other shipping related activities within their organization, such as agencies activities, forwarding, travel agency, and in recent years, offshore activities. Shipping companies will frequently build their organization around key personnel and the dividing lines between the departments might therefore be very different from those mentioned above. Shipping companies should also be aiming at flexibility in their organizational development, as changing conditions in the world market might make it necessary to reorganize or set up new departments to cater for new trades or ventures.

WORLD SEABORNE TRADE

INTRODUCTION

The volume of exchanged goods and services between nations is taking a growing share of the generation of wealth. Mainly by offering economic growth opportunities in new regions and by reducing the costs of a wide array of manufacturing goods. By 2007, international trade surpassed for the first time 50% of the global GDP, a two-fold increase in its share since I950. The facilitation of trade involves how the procedures regulating the international movements of goods can be improved. It depends on the reduction of the general costs of trade, which considers transaction, tariff, transport and time costs. Thus the ability to compete in a global economy is dependent on the transport system as well as a vast array of supporting service activities. These activities include:

a) Distribution-Based

A multi-modal and inter-modal-freight transport system composed of modes, infrastructures and terminals that spans across the globe. It insures a physical capacity to support trade.

b) Regulation-Based

Customs procedures, tariffs, regulations and handling of documentation. They insure that trade flows abide to the rules and regulations of the jurisdictions they cross.

c) Transaction-Based

Banking, finance, legal and insurance activities where accounts can be settled. They insure that the sellers of goods and services are receiving an agreed upon compensation and that the purchasers have a legal recourse if the outcome of the transaction is judged unsatisfactory or is insured if a partial or full loss incurs.

Ocean transport or shipping through waterways plays an important role in the trade and economic development of nations. In fact, transport, trade, and economic development are mutually supportive. The overwhelming share of shipping in the carriage of about 95 per cent in terms of

volume and almost two-thirds of the total value of international trade establishes its predominance and importance as a mode of international transportation system. The importance of shipping, over the period, has also increased due to the technological developments in transport. Especially in terms of containerization culminating in multi-modal transportation on door-to-door basis, since majority of the containers move by this mode of transport.

GLOBAL TRADE PATTERNS

International trade, both in terms of value and tonnage, has been a growing trend in the global economy. The emergence of global trade patterns can mainly be articulated within three major phases:

1. First Phase

Concerns a conventional perspective on international trade that prevailed until the 19705 where factors of production were much less mobile. Particularly, there was a limited level of mobility of raw materials, parts and finished products in a setting, which was fairly, regulated with impediments such tariffs, quotas and limitations to foreign ownership.

Trade mainly concerned a range of specific products, namely commodities, (and very few services) that were not readily available in regional economies. Due to regulations, protectionism and fairly high transportation costs, trade remained limited and delayed by inefficient freight distribution. In this context, trade was more an exercise to cope with scarcity than to promote economic efficiency.

2. Second Phase

From the 19805, the mobility of factors of production, particularly capital, became possible. The legal and physical environment in which international trade was taking place lead to a better realization of the comparative advantages of specific locations. Concomitantly, regional trade agreements emerged and the global trade framework was strengthened from a legal and transactional standpoint (GATT/WTO). In addition, containerization provided the capabilities to support more complex and long distance trade flows, as did the growing air traffic. Due to high production (legacy) costs in old industrial regions, activities that were labor intensive were gradually re-located to lower costs locations.

The process began as a national one, then went to nearby countries when possible and afterwards became a truly global phenomenon. Thus, foreign direct investments surged, particularly towards new manufacturing regions as multinational corporations became increasingly flexible in the global positioning of their assets.

3. Third Phase

There is a growth in international trade, now including a wide variety of services that were previously fixed to regional markets and a surge in the mobility of the factors of production. Since these trends are well-established, the priority is now shifting to the geographical and functional integration of production, distribution and consumption with the emergence of global

production networks. Complex networks involving flows of information, commodities, parts and finished goods have been set, which in turn demands a high level of command of logistics and freight distribution. In such an environment powerful actors have emerged which are not directly involved in the function of production and retailing, but mainly taking the responsibility of managing the web of flows.

The global economic system is thus one characterized by a growing level of integrated services, finance, retail. manufacturing and nonetheless distribution, which in turn is mainly the outcome of improved transport and logistics, a more efficient exploitation of regional comparative advantages and a transactional environment supportive of the legal and financial complexities of global trade. The outcome has been a shift in global trade flows with many developing countries having a growing participation to international trade.

The nature of what can be considered international trade has also changed, particularly with the emergence of global commodity chains. This trend obviously reflects the strategies of multinational corporations positioning their manufacturing assets in order to lower costs, maximize new market opportunities while maintaining the cohesion of their freight distribution systems. In addition, another important trade has been growing imports of resources from developing countries, namely energy, commodities and agricultural products.

Events Conducive to World Trade

Many events have been conducive to trade facilitation in recent decades:

1. Integration Processes

Integration processes such as the emergence of economic blocks and the decrease of tariffs at a global scale through agreements, promoted trade as regulatory regents were harmonized. One straightforward measure of integration relates to custom delays, which can be a significant trade impediment. The higher the level of economic integration, the more likely the concerned elements are to trade. International trade has consequently been facilitated by a set of factors linked with growing levels of economic integration, the outcome of processes such as the European Union or the North American Free Trade Agreement. The transactional capacity is consequently facilitated with the development of transportation networks and the adjustment of trade flows that follows increased integration. Integration processes have also taken place at the local scale with the creation of free trade zones where an area is given a different governance structure in order to promote trade, particularly export oriented activities. In this case, the integration process is not uniform as only a portion of a territory is involved.

2. Production Systems

Production systems are more flexible and embedded, which favors exchanges of commodities, parts and services. Information technologies have played a role by facilitating transactions and the management of complex business operations. Foreign direct investments are commonly linked with the globalization of production as corporations invest abroad in search of

lower production costs and new markets. China is a leading example of such a process. There is consequently a growing availability of goods and services that can be traded on the global market.

3. Transport Efficiency

Transport efficiency has increased significantly because of innovations and improvements in the modes and infrastructures. Ports are particularly important in such a context since they are gateways to international trade through maritime shipping networks. As a result, the transferability of commodities has improved. Decreasing transport costs does more than increasing trade; it also helps to change the location of economic activities. Yet, transporter transportation issues remain to be better addressed in terms of capacity and security.

4. Transactional Efficiency

The financial sector also played a significant role in integrating global trade, namely by providing credit for international commercial transactions. For example, a letter of credit may be issued based upon an export contract. An exporter cart thus receives a payment guarantee from a bank until its customer finalizes the transaction upon delivery. This is particularly important since the delivery of international trade transactions can take several weeks due to the long distances involved. During the transfer, it is also common that the cargo is insured in the event of damages, theft or delays, a function supported by insurance companies. Also, global financial systems permit to convert currencies according to exchange rates that are commonly set by market forces, while some currencies, such as the Chinese Yuan, are set by policy. Monetary policy can thus be a tool, albeit contentious, used to influence trade.

VOLUME AND VALUE OF WORLD TRADE

The nature of global seaborne trade has continued to evolve as a truly internationalized activity. Manufacturer's products may be assembled from elements produced in several countries and then transported to national seaports under multinational control. It is then loaded aboard a vessel that is built in one country, owned by another, insured by a third registered in a fourth and crewed by a fifth country. This 'globalization' of sea- borne trade raises some complex issues relating to the nature of the strategic concern and the extent and responsibility of international regulation for trade and shipping. The short-term future of sea-home trade in the Asia Pacific may be well-shaped by the emergence of new markets the reduction of trade barriers under the ASEAN Free Trade Area (AFTA) regime by 2003, and the forthcoming World Trade Organization negotiation.

Volume and value of sea borne world trade may be summarized as given below:

1. It is difficult to quantify the value of volume of world seaborne trade in monetary terms, as figures for trade estimates are traditionally in terms of tons or tons-miles, and are therefore not comparable with monetary-based statistics for the value of the world economy. The United Nations Conference on Trade and Development (UNCTAD) estimates that the operation of merchant ships contributes about US$ 380 billion in freight rates within the global economy, equivalent to about 5% of total world trade.

2. Shipping trade estimates are often calculated in tons-miles, as a way of measuring the volume of trade (or "transportation work", as it is sometimes referred). In 2008, e.g., it is estimated that the industry transported over 7.7 thousand million tons of cargo, equivalent to a total volume of world trade by sea of over 32 thousand billion tons-miles.
3. Throughout the last century, the shipping industry has seen a general trend of increases in total trade volume. Increasing industrialization and the liberalization of national economies have fuelled free trade and a growing demand for consumer products. Advances in technology have also made shipping an increasingly efficient and swift method of transportation. Over the last four decades total seaborne trade estimates have quadrupled, from just over 8 thousand billion tons-miles in 1968 to over 32 thousand billion tons-miles in 2008.
4. As with all industrial sectors, however, shipping can be susceptible to economic downturns. Indeed, following several years of incredibly buoyant shipping markets, for many trades the best in living memory, much of the international shipping industry has fallen prey to the worldwide economic downturn. Shipping is inherently the servant of the economy, so the contraction in trade, following the beginning of the 'credit crunch' in late 2008, has translated into a dramatic and abrupt reduction in demand for shipping.
5. Notwithstanding the current gloom and doom, the longer-term outlook for the industry remains very good. The world's population continues to expand, and emerging economies will continue to increase their requirements for the goods and raw materials that shipping transports so safely and efficiently. In the longer-term, the fact that shipping is the most fuel-efficient and carbon friendly form of commercial transport should work in favor of an even greater proportion of world trade being carried by sea.

From 1990-95, world trade grew at a rate of 6.5%. Exports from the developing economies in Northeast and South-East Asia increased by 13% at a time when GDP in Asia increased by 6.1%. By 1997, the world GDP has slowed to 3.4% and by 1998 it was just 1.8%. The pick-up in world GDP in 1999 was 2.7%, of which Asian GDP increased by 5.1%. Global seaborne trade is expected to grow by 2%. The development of international seaborne trade could be summarized as follows:

a) World sea-borne trade in 1999 reached a record high of 5.23 billion tons highlighting the enlarging ship borne trade and its impact on the security of sealanes of communication particularly in the Indian Ocean.
b) Socialist countries of Asia substantially increased their dry cargo imports in 1999.
c) Asian exports growth improved as Japan's exports recovered and the five developing Asian economies (Indonesia, Malaysia, Philippines, Thailand, ROK) affected by 1997-98 financial crises, achieved a significant growth in combined exports of dry cargo in 1999.
d) Developing Asia, including China, India and Indonesia, accounts for approximately 10% of world GDP, 54% of world population, and 18.6% of world energy consumption (of which 40% of energy is oil). Their share of energy consumption by fuel is oil (18%), natural (7%) and coal (36%).

e) The US and Japan are the major importers of crude 0“ng petroleum products, with 23.7% and 16.4% respectively of total world imports.
f) The developing countries share in total world exports in 1999 remained at 50.5%, and share of imports was 27.8%. 7) In 1999, Asia maintained 26.1% of world exports and 18.8% of world imports.

WORLD TONNAGE

The international carriage of goods, shipping has a supreme position. The growth and development of world’s merchant fleets is the result of interplay of many constraints:

1. Economic Constraint

The demand for shipping space is obviously the basic consideration but this is dependent on the speed and efficiency with which the vehicles can operate. In assessing demand, it is important to recognize world patterns, as it is vital to establish where it really lies. The supply of shipping follows demand. But shipbuilding can take several years. There can therefore be considerable lag between supply and demand and often ships ordered when the market conditions were favorable, will not be ready for service until the market conditions are reversed and shipping is going through a depressed period.

2. Technical Constraint

British were the first to develop large non-steam driven' ship and hence the growth of the British fleet at the beginning of the century. In the last ten years, the advent of container ship has been a noteworthy accomplishment. Bulk carriers and very large tankers could only become a reality after changes and improvements in ship building technology. Technical development has also taken place in propulsion units and in much of the ancillary equipments that are now essential for the safe and efficient operation of ships.

3. Political Constraint

a) Wars cause losses of ships and the need for rapid re-building and produce a considerable stimulus to research. Artificial demands may be created. Routes may be lengthened (as in the closure of the Suez Canal). Blacklisting can occur and ships trading in one country may not be allowed to visit another, i.e., ships visiting Cuba are not allowed to trade in the USA. Because of war, nations do not like to be dependent on the ships of other nations for the delivery of essential supplies (withdrawal of British ships from liner trade routes during the wars made countries aware of their dependence on foreign ships and thus determine to develop their own fleets).
b) Changes in trading pattern are brought about by changes in government, political ideology, political alignments, and the like.
c) Forms of taxation and government restrictions can have their effect as with the growth of flags of convenience.
d) Government can protect her own flags and thus exert influent.

SUMMARY

Shipping is the safest and most environmentally benign form of commercial transport. Perhaps uniquely amongst industries involving physical risk, commitment to safety has long pervaded virtually all deep sea shipping operations. Each country has the right to establish its own shipping laws. International shipping conventions have been reached by a number of intergovernmental organizations, such as the international Labor Organization (ILO) and the international Maritime Organization (IMO).

Shipping routes reflect world trade flows. Sailings are most numerous and most frequent on routes where trade Volumes are largest and demand is therefore greatest. Sea routes are confined to a stipulated voyage and distance. They normally are inclined to approximate to a curve of the great circle since it guides the ships to undertake the shortest journey. The volume of exchanged goods and services between nations is taking a growing share of the generation of wealth. Mainly by offering economic growth opportunities in new regions and by reducing the costs of a wide array of manufacturing goods.

The nature of global seaborne trade has continued to evolve as a truly internationalized activity. Manufacturer's products may be assembled from elements produced in several countries and then transported to national seaports under multinational control. It is then loaded aboard a vessel that is built in one country, owned by another, insured by a third registered in a fourth and crewed by a fifth country.

KEY TERMS

Bulk Carriers
Cargo Department
Flag Identification
Freight Rates
Global Trade Patterns
Liner Ships
LNG Carriers
Operating Department
Organizational Chart
Product Analysis
Ro-Ro Vessels
Ship Broker
Shipping Industry
Tramp Ships
Tankers
World Tonnage

REVIEW QUESTIONS

1. Explain the basic functions of shipping.
2. Write notes about history of shipping industry?
3. Explain the regulations of international shipping industry.
4. Explain the different types of ships.
5. Explain the organization of a shipping company.
6. Briefly explain any three of the following.
   a) Container ships
   b) Roll on/Roll off ships
   c) Break bulk ships
   d) Combination ships

DISCUSSION QUESTIONS

1. Discuss the International Shipping Industry.
2. Discuss the shipping routes.
3. Discuss the different departments in shipping organization.
4. Discuss the various types of ships available for international transportation of goods? Classify the ships based on their sizes and type of cargo they carry. Also discuss the differences between liner and tramp vessels.

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